

Department of Computer Science & Engineering

**Curriculum
For
Computer Science & Technology**

1st Semester to 8th Semester

(Effective from 2021-22 Admission Batch)

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
 L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

<u>1ST Year 1st Semester: 1st Semester</u>								
Sl. No.	Category	Course Code	Course Title	Hours per week				Credits
	THEORY			L	T	P	Total	
1	Basic Science course	PH101	Physics-I	3	0	0	3	3
2	Basic Science course	M101	Mathematics –I	4	0	0	4	4
3	Humanities and Social Sciences including Management courses	HSMC 101	Professional Communication	2	0	0	2	2
	PRACTICAL							
4	Basic Science course	PH191	Physics-I Lab	0	0	3	3	1.5
5	Engineering Science Courses	ME 191	Workshop & Manufacturing Practices Lab	0	0	3	3	1.5
6	PROJECT	PR191	Theme based Project I	0	0	1	1	0.5
7	PROJECT	PR192	Skill Development I: Soft Skill	0	0	1	1	0.5
	MANDATORY ACTIVITIES / COURSES							
8	Mandatory Course	MC181	Induction Program	0	0	0	0	2Units
	TOTAL CREDIT							13.0

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<u>1ST Year 2nd Semester: 2nd Semester</u>								
Sl. No	Category	Course Code	Course Title	Hours per week				Credits
	THEORY			L	T	P	Total	
1	Basic Science courses	CH 201	Chemistry-I	3	0	0	3	3
2	Basic Science courses	M 201	Mathematics –II	4	0	0	4	4
3	Engineering Science Courses	EE 201	Basic Electrical Engineering	3	0	0	3	3
4	Engineering Science Courses	CS 201	Programming for Problem Solving	3	0	0	3	3
	PRACTICAL							
5	Basic Science course	CH 291	Chemistry-I Lab	0	0	3	3	1.5
6	Humanities and Social Sciences including Management courses	HSMC 291	Professional Communication LAB	0	0	2	2	1.0
7	Engineering Science Courses	EE 291	Basic Electrical Engineering Lab	0	0	3	3	1.5
8	Engineering Science Courses	ME 292	Engineering Graphics & Design Lab	0	0	3	3	1.5
9	Engineering Science Courses	CS 291	Programming for Problem Solving Lab	0	0	3	3	1.5
10	PROJECT	PR291	Theme based Project II	0	0	1	1	0.5
11	PROJECT	PR292	Skill Development II: Life Skill	1	0	0	1	0.5
	MANDATORY ACTIVITIES / COURSES							
12	Mandatory Course	MC281	NSS/Physical Activities / Meditation & Yoga / Photography	0	0	3	3	3 Units
	TOTAL CREDIT							21

2nd Year 1st Semester: 3rd Semester								
Sl. No.	Category	Course Code	Course Title	Hours per week				Credits
				L	T	P	Total	
A. THEORY								
1	Basic Science course	M301	Discrete Mathematics	3	0	0	3	3
2	Engineering Science Courses	ESC301	Analog Electronics	3	0	0	3	3
3	Engineering Science Courses	ESC302	Digital Electronics	3	0	0	3	3
4	Program Core Course	PCC-CS301	Programming in Python	3	0	0	3	3
5	Program Core Course	PCC-CS302	Data Structures	3	0	0	3	3
6	Humanities and Social Sciences including Management courses	HSMC 303	Universal Human Values 2: Understanding Harmony	3	0	0	3	3
B. PRACTICAL								
7	Engineering Science Courses	M (CS)391	Numerical Methods Lab	1	0	3	3	2.5
8	Engineering Science Courses	ESC391	Digital Electronics Lab	0	0	3	3	1.5
9	Program Core Course	PCC-CS391	Programming in Python Lab	0	0	3	3	1.5
10	Program Core Course	PCC-CS392	Data Structures Lab	0	0	3	3	1.5
11	PROJECT	PR391	Theme based Project III	0	0	1	1	0.5
12	PROJECT	PR392	Skill Development III: Technical Seminar Presentation	1	0	0	1	0.5
C. MANDATORY ACTIVITIES / COURSES								
15	MC	MC 301	Environmental Science	0	0	3	3	3 Units
	TOTAL CREDIT WITHOUT MOOCS COURSES							26.0

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D.MOOCs COURSES**								
16	MOOCs COURSES	HM301	MOOCs COURSE-I	1	3	1	4	4
TOTAL CREDIT WITH MOOCs COURSES								30

2nd Year 2 nd Semester: 4 th Semester								
Sl. No.	Category	Course Code	Course Title	Hours per week				Credits
				L	T	P	Total	
A. THEORY								
1	Program Core Course	PCC-CS401	Computer Organization and Architecture	3	0	0	3	3
2	Program Core Course	PCC-CS402	Design and Analysis of Algorithms	3	0	0	3	3
3	Program Core Course	PCC-CS403	Operating Systems	3	0	0	3	3
4	Program Core Course	PCC-CS404	Formal Language and Automata Theory	3	0	0	3	3
5	Humanities and Social Sciences including Management courses	HSMC 402	Gender Culture and Development	2	0	0	2	2
6	Basic Science course	BSC401	Probability and Statistics	3	0	0	3	3
B. PRACTICAL								
6	Program Core Course	PCC-CS491	Computer Organization and Architecture Lab	0	0	3	3	1.5
7	Program Core Course	PCC-CS492	Design and Analysis of Algorithms Lab	0	0	3	3	1.5
8	Program Core Course	PCC-CS493	Operating Systems Lab	0	0	3	3	1.5
9	Engineering Science Courses	ESC491	IT Workshop Lab (C++/python)	0	0	3	3	1.5
10	PROJECT	PR 491	Theme based Project IV	0	0	1	1	0.5
11	PROJECT	PR492	Skill Development IV: Soft Skill & Aptitude-I	1	0	0	1	0.5
C. MANDATORY ACTIVITIES / COURSES								
15	MC	MC 481	Learning an Art Form [vocal or instrumental, dance, painting, clay modeling, etc.] OR Environmental Protection Initiatives	0	0	0	3	3Units
	TOTAL CREDIT WITHOUT MOOCS COURSES							24

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D.MOOCs COURSES								
16	MOOCs COURSES	HM401	MOOCs COURSE-II	3	1	0	4	4
TOTAL CREDIT WITH MOOCs COURSES								28

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3 rd Year 1 st Semester: 5 th Semester								
Sl. No.	Category	Course Code	Course Title	Hours per week				Credits
				L	T	P	Total	
A. THEORY								
1	Humanities and Social Sciences including Management courses	HSMC 505	Principles of Management	2	0	0	2	2
2	Program Core Course	PCC-CS501	Compiler Design	3	0	0	3	3
3	Program Core Course	PCC-CS502	Database Management Systems	3	0	0	3	3
4	Program Core Course	PCC-CS503	Object Oriented Programming using Java	3	0	0	3	3
5	Professional Elective courses	PEC-CS-T-501	Microprocessors and Microcontrollers	3	0		3	3
		PEC-CS-S-501	Advanced Computer Architecture			0		
		PEC-CS-D-501	Artificial Intelligence					
		PEC-CS-A-501	Computer Graphics					
B. PRACTICAL								
7	Program Core Course	PCC-CS591	Compiler Design Lab	0	0	3	3	1.5
8	Program Core Course	PCC-CS592	Database Management Systems Lab	0	0	3	3	1

								5
9	Program Core Course	PCC-CS593	Object Oriented Programming using Java Lab	0	0	3	3	1 .5
10	Professional Elective courses	PEC-CS-T-591	Microprocessors and Microcontrollers Lab	0	0	3	3	1 .5
		PEC-CS-S-591	Advanced Computer Architecture Lab					
		PEC-CS-D-591	Artificial Intelligence Lab					
		PEC-CS-A-591	Computer Graphics Lab					
11	PROJECT	PR 591	Minor Project I	0	0	3	2	1
12	PROJECT	PR 592	Skill Development V: Soft Skill & Aptitude-II	1	0	0	1	0 .5
C. MANDATORY ACTIVITIES / COURSES								
14	MC	MC 501	Constitution of India	3	0	0	3	3Units
	TOTAL CREDIT WITHOUT MOOCS COURSES							21.5
D. MOOCS COURSES**								
15	MOOCS COURSES	HM501	MOOCS COURSE-III	3	1	0	4	4
TOTAL CREDIT WITH MOOCS COURSES								25.5

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3 rd Year2 nd Semester: 6 th Semester								
Sl. No.	Category	Course Code	Course Title	Hours per week				Credits
				L	T	P	Total	
A. THEORY								
1	Humanities and Social Sciences including Management courses	HSMC 604	Economics for Engineers	2	0	0	2	2
2	Program Core Course	PCC-CS601	Computer Networks	3	0	0	3	3
3	Program Core Course	PCC-CS602	Software Engineering	3	0	0	3	3
4	Professional Elective courses	PEC-CS-T-601	Advanced Algorithms	3	0	0	3	3
		PEC-CS-S-601	Advanced Operating Systems					
		PEC-CS-D-601	Machine Learning					
		PEC-CS-A-601	Web and Internet Technology					
5	Professional Elective courses	PEC-CS-T-602	Parallel and Distributed Algorithms	3	0	0	3	3
		PEC-CS-S-602	Embedded Systems					
		PEC-CS-D-602	Soft Computing					
		PEC-CS-A-602	Human Computer Interaction					
6	Open Elective courses	OEC-CS-601A	Introduction to Internet of Things	3	0	0	3	3
		OEC-CS-601B	Bio-informatics					

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		OEC-CS-601C	Robotics					
B. PRACTICAL								
7	Program Core Course	PCC-CS691	Computer Networks Lab	0	0	3	3	1.5
9	Program Core Course	PCC-CS692	Software Engineering Lab	0	0	3	3	1.5
10	Professional Elective courses	PEC-CS-T-691	Advanced Algorithms Lab	0	0	3	3	1.5
		PEC-CS-S-691	Advanced Operating Systems Lab					
		PEC-CS-D-691	Machine Learning Lab					
		PEC-CS-A-691	Web and Internet Technology Lab					
11	PROJECT	PR 691	Minor Project II	0	0	3	2	1
12	PROJECT	PR 692	Skill Development VI: Soft Skill & Aptitude-III	1	0	0	1	0.5
C. MANDATORY ACTIVITIES / COURSES								
13	MC	MC 601	Intellectual Property Right	3	0	0	3	3Units
	TOTAL CREDIT WITHOUT MOOCS COURSES							23.0
D.MOOCS COURSES**								
1	MOOCS	HM601	MOOCS COURSE-IV	3	1	0	4	4

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4	COURSES							
TOTAL CREDIT WITH MOOCS COURSES								27

4 th Year 1 st Semester: 7 th Semester								
Sl No	Course Code	Paper Code	Theory	Contact Hours /Week			Credit Points	
				L	T	P		Total
A. THEORY								
1	Professional Elective courses	PEC-CS-T-701	Information Theory and Coding	3	0	0	3	3
		PEC-CS-S-701	Ad-Hoc and Sensor Networks					
		PEC-CS-D-701	Data Mining and Data Warehousing					
		PEC-CS-A-701	Cloud Computing					
2	Professional Elective courses	PEC-CS-T-702	Quantum Computing	3	0	0	3	3
		PEC-CS-S-702	Mobile Computing					
		PEC-CS-D-702	Natural Language Processing					
		PEC-CS-A-702	Cryptography and Network Security					
3	Open Elective courses	OEC-CS-701A	High Performance Computing	3	0	0	3	3
		OEC-CS-701B	VLSI					
		OEC-CS-701C	Optimization Techniques					

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4	Open Elective courses	OEC-CS-702A	Cyber Law and Ethics	3	0	0	3	3
		OEC-CS-702B	Soft Skills and Interpersonal Communication					
		OEC-CS-702C	Foreign Language					
B. PRACTICAL								
5	Professional Elective courses	PEC-CS-T-791	Information Theory and Coding Lab	0	0	0	3	1.5
		PEC-CS-S-791	Ad-Hoc and Sensor Networks Lab					
		PEC-CS-D-791	Data Mining and Data Warehousing Lab					
		PEC-CS-A-791	Cloud Computing Lab					
6	Open Elective courses	OEC-CS-791A	High Performance Computing Lab	0	0	3	3	1.5
		OEC-CS-791B	VLSI Lab					
		OEC-CS-791C	Optimization Techniques Lab					
7	PROJECT	PR 791	Major Project-I	0	0	0	4	2
8	PROJECT	PR 792*	Industrial Training / Internship	0	0	0	0	1
9	PROJECT	PR 793	Skill Development VII: Seminar & Group Discussion	1	0	0	1	0.5
C. MANDATORY ACTIVITIES / COURSES								
10	MC	MC 781	Entrepreneurship & Innovation Skill	3	0	0	3	3 Units
TOTAL CREDIT WITHOUT MOOCS COURSES								18.5
D.MOOCs COURSES**								
11	MOOCS COURSES	HM701	MOOCS COURSE-V	3	1	0	4	4

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TOTAL CREDIT WITH MOOCS COURSES	22.5
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4th Year 2nd Semester: 8th Semester

Sl No	Course Code	Paper Code	Theory	Contact Hours /Week			Credit Points	
				L	T P	Total		
A. THEORY								
1	Professional Elective courses	PEC-CS-T-801	Advance Graph Algorithms	3	0	0	3	3
		PEC-CS-S-801	Real Time Systems					
		PEC-CS-D-801	Data Analytics					
		PEC-CS-A-801	Image Processing					
2	Open Elective courses	OEC-CS-801A	Human Resource Development and Organizational Behavior	3	0	0	3	3
		OEC-CS-801B	Block Chain					
		OEC-CS-801C	Simulation and Modeling					
3	Open Elective courses	OEC-CS-802A	Values and Ethics in Profession	3	0	0	3	3
		OEC-CS-802B	History of Science					
		OEC-CS-802C	Economic Policies in India					
B. PRACTICAL								
4	PROJECT	PR 891	Major Project-II	0	0	0	12	6
5	PROJECT	PR 892	Grand Viva	0	0	0	0	1

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C. MANDATORY ACTIVITIES / COURSES								
6	MC	MC 801	Essence of Indian Knowledge Tradition	0	0	3	3	3 Units
TOTAL CREDIT								16

PROGRAM OUTCOMES (POs)

Engineering Graduates will be able to:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to

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comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs) of CST

Program Specific Outcome:

PSO1	The ability to understand, illustrate, discuss, explain the fundamental design and working principle of any existing or new computing technology with a focus on acquiring the knowledge of cutting-edge technical skills in the field of Computer Science.
PSO2	The ability to identify and formulate an engineering problem within the scope of Computer Science & Technology domain with a special focus on the probable technical challenges while proposing any software application-based or research-based technical solution with adequate justification
PSO3	The ability to design, implement and analyse a solution proposal with proper documentation and tools by demonstrating adequate software engineering management skill along with the necessary technical skill for driving the intellect towards technological innovation boosted with research as well as entrepreneurial aptitude for producing globally competent engineering professionals capable of making meaningful contributions in the field of computer science and technology with a special focus on its technological aspects.

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Department of Computer Science & Engineering

Curriculum Structure & Syllabus
For
Computer Science & Technology

1stSemester to 8th Semester

(Effective from 2021-22 Admission Batch)

1 ST Year 1 st Semester: 1 st Semester									
Sl. No	Category	Course Code	Course Title	Hours per week				Credits	
	THEORY			L	T	P	Total		
1	Basic Science course	PH101	Physics-I	3	0	0	3	3	
2	Basic Science course	M101	Mathematics –I	4	0	0	4	4	
3	Humanities and Social Sciences including Management courses	HSMC 101	Professional Communication	2	0	0	2	2	
	PRACTICAL								
4	Basic Science course	PH191	Physics-I Lab	0	0	3	3	1.5	
5	Engineering Science Courses	ME 191	Workshop & Manufacturing Practices Lab	0	0	3	3	1.5	
6	PROJECT	PR191	Theme based Project I	0	0	1	1	0.5	
7	PROJECT	PR192	Skill Development I: Soft Skill	0	0	1	1	0.5	
	MANDATORY ACTIVITIES / COURSES								
8	Mandatory Course	MC181	Induction Program	0	0	0	0	2Units	
	TOTAL CREDIT								13.0

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COURSE NAME: PHYSICS –I

Course Code: PH 101

Contact: 3:0:0

Total Contact Hours: 36

Credit: 3

Prerequisites: Knowledge of Physics up to 12th standard.

Course Outcomes (COs):

After attending the course students' should be able to

CO1: describe various types of mechanical resonance and its electrical equivalence

CO2: explain basic principles of Laser, Optical fibers and Polarization of light

CO3: apply superposition principle to explain interference and diffraction

CO4: analyze different crystallographic structures according to their co-ordination number and packing factors

CO5: justify the need of a quantum mechanics as remedy to overcome limitations imposed by classical physics

Course Content:

Module 1 (5L):-

Waves & Oscillations:-

Simple Harmonic Motion (Recap), superposition of waves, damped harmonic motion-over damped, critically damped and under damped motion, energy decay, logarithmic decrement, force vibration and resonance (amplitude, velocity resonance), sharpness of resonance, quality factor, related numerical problems. **5L**

Module 2 (12L):-

Classical Optics:

2.01- Interference of light: Huygens's principle, conditions of sustained interference, classification of interference, Newton's ring (qualitative descriptions of working principles and procedures-no deduction required). Engineering applications, related numerical problems. **4L**

2.02-Diffraction of light: Fresnel and Fraunhofer class, Fraunhofer diffraction of a single slit, double slit, multiple slits, intensity distributions, missing order, Rayleigh criterion (no deduction) and resolving power of grating and microscope (no deduction), related numerical problems. **4L**

2.03-Polarization: Definition, Plane of polarization, Plane of vibration, Malus Law, Fundamental concepts of plane, circular & elliptical polarizations (only qualitative idea) with

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examples, Brewster's law, Double refraction: Ordinary & Extra ordinary rays, positive and negative crystal, Nicol's prism, Numerical problems

4L

Module 3 (8L):-

Quantum Mechanics-I

3.01 Quantum Theory: Inadequacy of classical physics-concept of quantization of energy, particle concept of electromagnetic wave (example: photoelectric and Compton Effect; no derivation required, origin of modified and unmodified lines), wave particle duality; phase velocity and group velocity; de Broglie hypothesis; Davisson and Germer experiment. **4L**

3.02 Quantum Mechanics 1: Concept of wave function, physical significance of wave function, probability interpretation; normalization of wave functions; uncertainty principle, relevant numerical problems. Introduction of Schrödinger wave equation (only statement).

4L

Module 4 (3L):-

Solid State Physics-I:

4.01 Crystal Structure: Structure of solids, amorphous and crystalline solids (definition and examples), lattice, basis, unit cell, Fundamental types of lattices –Bravais lattice, simple cubic, fcc and bcc lattices, Miller indices and miller planes, co-ordination number and atomic packing factor, Bragg's equation, applications, numerical problems. **3L**

Module 5 (8L):

Modern Optics-I:

5.01- Laser: Concepts of various emission and absorption processes, Einstein A and B coefficients and equations, working principle of laser, metastable state, population inversion, condition necessary for active laser action, optical resonator, illustrations of Ruby laser, He-Ne laser, Semiconductor laser, applications of laser, related numerical problems. **5L**

5.02-Fibre optics-Principle and propagation of light in optical fibers (Step index, Graded index, single and multiple modes) - Numerical aperture and Acceptance angle, Basic concept of losses in optical fiber, related numerical problems. **3L**

Recommended Text Books for Physics I (PH 101):

Waves & Oscillations:

1. Sound-N. K. Bajaj (TMH)
2. Advanced Acoustics-D. P. Roy Chowdhury (Chayan Publisher)
3. Principles of Acoustics-B.Ghosh (Sridhar Publisher)
4. A text book of sound-M. Ghosh (S. Chand publishers)
5. A text book of Light- K.G. Mazumder &B.Ghoshs, (Book & Allied Publisher)
6. Physics of Oscillations and Waves- R.P. Singh

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7. College Physics Vol. II - A.B. Gupta
8. Vibration, Waves and Acoustics- Chattopadhyay and Rakshit

Classical & Modern Optics:

1. A text book of Light- K.G. Mazumder & B. Ghosh (Book & Allied Publisher)
2. A text book of Light-Brijlal & Subhramaniam, (S. Chand publishers)
3. Modern Optics-A. B. Gupta (Book & Allied Publisher)
4. Optics-Ajay Ghatak (TMH)
5. Optics-Hecht
6. Optics-R. Kar, Books Applied Publishers
7. Physical Optics Möler
8. Optics -F.A. Jenkins and H.E White

Quantum Mechanics-I

1. Introduction to Quantum Mechanics-S. N. Ghoshal (Calcutta Book House)
2. Quantum Mechanics-Bagde and Singh (S. Chand Publishers)
3. Perspective of Quantum Mechanics-S. P. Kuilla (New Central Book Agency)
4. Quantum Mechanics-Binayak Datta Roy (S. Chand Publishers)
5. Quantum Mechanics-Bransden (Pearson Education Ltd.)
6. Perspective of Modern Physics-A. Beiser (TMH)
7. Quantum mechanics -A.K. Ghatak and S Lokenathan
8. Modern Physics -E.E. Anderson
9. Physics Volume 2 -Haliday, Resnick & Krane, Published by Wiley India

Solid State Physics-I:

1. Solid state physics-Puri & Babbar (S. Chand publishers)
2. Materials Science & Engineering-Kakani Kakani
3. Solid state physics- S. O. Pillai
4. Introduction to solid state physics-Kittel (TMH)
5. Solid State Physics and Electronics-A. B. Gupta and Nurul Islam (Book & Allied Publisher)
6. Problem in Solid state physics -S.O. Pillai (a. b.)

Text Books:

1. Refresher courses in physics (Vol. 1, Vol. 2 & Vol. 3)-C. L. Arora (S. Chand Publishers)
2. Basic Engineering Physics-Amal Chakraborty (Chaya Prakashani Pvt. Ltd.)
3. Perspective & Concept of Modern Physics -Arthur Baiser
4. Principles of engineering physics – Md. N Khan and S Panigrahi.
5. Basic Engineering Physics-Sujoy Bhattacharya, Saumen Pal (MG)
6. Engineering Physics (Vol. 1, Vol. 2)-S.P. Kuila(S. Chand Publishers)
7. Engineering Physics-A. S. Vasudeva

Project Domains

- CO-PO Mapping:**

[illegible]

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Course Name: Mathematics-I

Course Code: M 101

Contact: 3:1:0

Total Contact Hours: 48

Credits: 4

Prerequisite: The students to whom this course will be offered must have the concept of (10+2) standard matrix algebra, calculus, and vector algebra.

Course Outcomes (COs):

On successful completion of the learning sessions of the course, the learner will be able to

CO1: Recall the properties and formula related to matrix algebra, differential calculus, multivariable calculus, vector calculus and infinite series.

CO2: Determine the solutions of the problems related to matrix algebra, differential calculus, multivariable calculus, vector calculus and infinite series.

CO3: Apply the appropriate mathematical tools of matrix algebra, differential calculus, multivariable calculus, vector calculus and infinite series for the solutions of the problems.

CO4: Analyze different engineering problems linked with matrix algebra, differential calculus, multivariable calculus, vector calculus.

Course Content:

Module I: Matrix Algebra

11L

Echelon form and Normal (Canonical) form of a matrix; Inverse and rank of a matrix; Consistency and inconsistency of system of linear equations, Solution of system of linear equations; Eigenvalues and eigenvectors; Diagonalization of matrices; Cayley-Hamilton theorem.

Module II: Differential Calculus and Infinite Series

10L

Rolle's Theorem, Mean value theorems, Taylor's and Maclaurin theorems with remainders; Concept of sequence and series, Tests for convergence of infinite series: Comparison test, D'Alembert's ratio test, Raabe's test, Cauchy's root test, Leibnitz's Test, Power series; Taylor's series, Series for exponential, trigonometric and logarithm functions.

Module III: Multivariable Calculus (Differentiation)

13L

Function of several variables, Concept of limit, continuity and differentiability; Partial derivatives, Total derivative and its application; Chain rules, Derivatives of implicit functions Euler's theorem on homogeneous function, Jacobian. Maxima and minima of functions of two variables, Method of Lagrange multipliers.

Module IV: Multivariable Calculus (Integration)

6L

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Line Integral, Double Integral, Triple Integral, Change of order in multiple integrals, Change of variables in multiple integrals.

Module V: Vector Calculus

8L

Gradient, Directional derivatives, Divergence, Curl, vector line integrals, vector surface integrals, vector volume integrals, Green's theorem, Gauss divergence theorem and Stokes' theorem.

Project Domain:

1. Study on eigenvalues and eigenvectors.
2. Study on convergence of infinite series.
3. Application of partial derivatives.
4. Application of vector calculus
5. Application of integral calculus.

Text Books:

1. Kreyszig, E., Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. Ramana, B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11th Reprint, 2010.
3. Veerarajan, T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2008.
4. Grewal, B.S., Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.
5. Bali, N.P. and Goyal, M., A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2008.
6. Samanta Guruprasad, A text book of Engineering Mathematics-I, New age International Publishers

Reference Books:

1. Thomas, G.B. and Finney, R.L., Calculus and Analytic geometry, 9th Edition, Pearson, Reprint, 2002.
2. Apostol, M., Calculus, Volumes 1 and 2 (2nd Edition), Wiley Eastern, 1980.
3. Kumaresan, S., Linear Algebra - A Geometric approach, Prentice Hall of India, 2000.
4. Poole, D., Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2005.
5. Bronson, R., Schaum's Outline of Matrix Operations. 1988.
6. Piskunov, N., Differential and Integral Calculus, Vol. I & Vol. II, Mir Publishers, 1969

CO-PO Mapping:

Curriculum for B.Tech Under Autonomy
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PO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	-	-	-	-	-	-	2
CO2	3	2	-	-	-	-	-	-	-	-	-	2
CO3	3	2	2	-	-	-	-	-	-	-	-	2
CO4	2	3	2	2	-	-	-	-	-	-	-	2

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Course Name: Professional Communication

Course Code: HSMC 101

Contact: 2:0:0

Total Contact Hours: 24

Credits: 2

Pre-requisites: Basic (10+2) level of knowledge of English grammar, vocabulary reading and writing skills.

Course Outcomes (COs):

After attending the course students' should be able to

CO1: apply the modalities and nuances of communication in a workplace context.

CO2: analyze communication across cultures and societies.

CO3: apply the basic formats, templates of business and official communication.

CO4: employ formal communication modes in meetings and reports.

CO5: justify importance of culturally neutral language in interpersonal and business communication.

Course Content:

Module- 1: Verbal and Non-verbal communication

4L

1.1: Definition, Relevance and Effective Usage

1.2: Components of Verbal Communication: Written and Oral Communication

1.3: Components of Non-verbal Communication: Kinesics, Proxemics, Chronemics, Haptics
Paralanguage

1.4: Barriers to Effective Communication

Module- 2: Social Communication Essentials and Cross-Cultural Communication

6L

2.1: Communication in Society and the Workplace

2.2: Greetings, Courtesies and Socially Useful Language

2.3: Cultural Contexts: High Context and Low Context Cultures

2.4: Understanding Cultural Nuances and Stereotyping

2.5: Achieving Culturally Neutral Communication in Speech and Writing

Module- 3: Meetings

4L

3.1: Meetings: Nature and Types

3.2: Conducting Meetings: Organization and Procedures

3.3: Meeting Coordination: Roles of Chairpersons and Members

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3.4: Notice and Agenda for a Meeting

3.5: Preparing the Minutes of a Meeting (MOM)

Module- 4: Report Writing

4L

4.1: Nature and Function of Reports

4.2: Types of Reports

4.3: Researching for a Business Report

4.4: Format, Language and Style

4.5: Report Documentation

Module 5: Employment Communication

6L

5.1: Writing Business Letters- (Enquiry, Order, Sales, Complaint, Adjustment, Job Application, Offer)

5.2: Preparing a CV or Résumé

5.3: Creating a Digital/Online Profile – LinkedIn (Résumé/Video Profile)

5.4: Writing E-mails: types, convention, and etiquette

5.5: Memo, Notices and Circulars

5.6: Writing Technicalities—Paragraphing, Sentence Structure and Punctuation

Text Books & Reference Books:

1. Meenakshi Raman and Sangeetha Sharma. *Technical Communication*. 3rd edition. New Delhi: Oxford University Press, 2015.
2. Mark Ibbotson. *Cambridge English for Engineering*. Cambridge: Cambridge University Press, 2008.
3. Mark Ibbotson. *Professional English in Use: Engineering*. Cambridge: Cambridge UP, 2009.
4. Lesikar et al. *Business Communication: Connecting in a Digital World*. New Delhi: Tata McGraw-Hill, 2014.
5. John Seeley. *Writing Reports*. Oxford: Oxford University Press, 2002.
6. Judith Leigh. *CVs and JobApplications*. Oxford: Oxford University Press, 2002.
7. Judith Leigh. *Organizing and Participating in Meetings*. Oxford: Oxford University Press, 2002.
8. Michael Swan. *Practical English Usage*. Oxford: OUP, 1980.
9. Pickett, Laster and Staples. *Technical English: Writing, Reading & Speaking*. 8th ed. London: Longman, 2001.
10. Diana Booher. *E-writing: 21st Century Tools for Effective Communication*.

Links:

1. Purdue University's Online Writing Lab (OWL)-<https://owl.purdue.edu/>
2. Business English Pod-<https://www.businessenglishpod.com/>

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CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	-	-	-	-	-	2	-	-	3	-	2
CO2	2	3	2	-	-	2	2	2	-	3	-	3
CO3	1	3	-	-	-	3	3	3	-	3	-	3
CO4	-	-	-	-	-	3	3	3	-	3	-	3
CO5	-	-	-	-	-	-	3	3	-	3	-	3

Curriculum for B.Tech Under Autonomy
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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Paper Name: Physics I Lab

Paper Code: PH 191

Contact Hours: 0:0:3

Credit: 1.5

Prerequisites: Knowledge of Physics up to 12th standard.

Course Outcomes (COs):

After attending the course students' will be able to

CO1: demonstrate experiments allied to their theoretical concepts

CO2 : conduct experiments using LASER, Optical fiber, Torsional pendulum, Spectrometer

CO3: participates an individual, and as a member or leader in groups in laboratory sessions actively

CO4: analyze experimental data from graphical representations , and to communicate effectively them in Laboratory reports including innovative experiments

General idea about Measurements and Errors (One Mandatory):

- i) Error estimation using Slide calipers/ Screw-gauge/travelling microscope for one experiment.
- ii) Proportional error calculation using Carrey Foster Bridge.

Any 6 to be performed from the following experiments

Experiments on Waves & Oscillations:

1. Study of Torsional oscillation of Torsional pendulum & determination of time using various load of the oscillator.
2. Determination of elastic moduli of different materials (Young's modulus /Rigidity modulus)
3. Determination of Q factor using LCR Circuit.
4. Calibration of an oscillator using Lissajous Figure.

Experiments on Classical Optics:

5. Determination of wavelength of light by Newton's ring method.
6. Determination of wavelength of light by Laser diffraction method.
7. To determine the angle of optical rotation of a polar solution using polarimeter

Experiments on Quantum Physics-I:

8. Determination of Planck's constant using photoelectric cell.
9. Verification of Bohr's atomic orbital theory through Frank-Hertz experiment.
10. Determination of Stefan's Constant

Probable experiments beyond the syllabus:

- ### Recommended Text Books for Physics I Lab (PH 291):

1. Vibration, Waves and Acoustics- Chattopadhyay and Rakshit

1. A text book of Light- K.G. Mazumder & B. Ghosh (Book & Allied Publisher)

1. Introduction to Quantum Mechanics-S. N. Ghoshal (Calcutta Book House)

1. Solid State Physics and Electronics-A. B. Gupta and Nurul Islam (Book & Allied Publisher)

1. Practical Physics by Chatterjee & Rakshit (Book & Allied Publisher)
2. Practical Physics by K.G. Mazumder (New Central Publishing)
3. Practical Physics by R. K. Kar (Book & Allied Publisher)

[illegible]

Curriculum for B.Tech Under Autonomy
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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Workshop/Manufacturing Practices

Course Code: ME 191

Contact: 0:0:3

Credits: 1.5

Prerequisite: Higher Secondary with Mathematics, Physics and Chemistry.

Course Outcomes (COs):

After completion of this course students will be able to

CO1: Identify and operate various hand tools related to variety of manufacturing operations

CO2: Safely fabricate simple components with their own hands.

CO3: Get practical knowledge of the dimensional accuracies and tolerances applicable for different manufacturing processes.

CO4: Produce small devices of their interest in project or research purpose.

Course Content:

(i) Theoretical discussion & videos:

3P

1. Manufacturing Methods- casting, forming, machining, joining, advanced manufacturing methods
2. Fitting operations & power tools
3. Carpentry
4. Welding (arc welding & gas welding), brazing
5. Electrical & Electronics
6. Metal casting
7. CNC machining, Additive manufacturing
8. Plastic moulding& Glass Cutting

(ii) Workshop Practice:

Module 1 - Machine shop

6P

Typical jobs that may be made in this practice module:

- i. To make a pin from a mild steel rod in a lathe.
- ii. To make rectangular and vee slot in a block of cast iron or mild steel in a shaping and / or milling machine.

Module 2 - Fitting shop

6P

Typical jobs that may be made in this practice module:

- i. To make a Gauge from MS plate.

Module 3 - Carpentry

6P

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Typical jobs that may be made in this practice module:

- i. To make wooden joints and/or a pattern or like.

Module 4 - Welding shop (Arc welding 3P + gas welding 3P) 3P

Typical jobs that may be made in this practice module:

- i. ARC WELDING (3P): To join two thick (approx 5mm) MS plates by manual metal arc welding.
- ii. GAS WELDING (3P): To join two thin mild steel plates or sheets by gas welding.

Module 5 - Electrical & Electronics 3P

House wiring, soft Soldering

Module 6 – Smithy 3P

Typical jobs that may be made in this practice module:

- i. A simple job of making a square rod from a round bar or similar.

For further study (Optional)

Module 7 - Casting 3P

Typical jobs that may be made in this practice module:

- i. One/ two green sand moulds to prepare, and a casting be demonstrated.

Module 8 - Plastic moulding & Glass Cutting 3P

Typical jobs that may be made in this practice module:

- i. For plastic moulding, making at least one simple plastic component should be made.
- ii. At least one sample shape on glass should be made using laser cutting machine.

Examinations could involve the actual fabrication of simple components, utilizing one or more of the techniques covered above.

Text Books:

1. Hajra Choudhury S.K., Hajra Choudhury A.K. and Nirjhar Roy S.K., “Elements of Workshop Technology”, Vol. I 2008 and Vol. II 2010, Media promoters and publishers private limited, Mumbai.
2. Rao P.N., “Manufacturing Technology”, Vol. I and Vol. II, Tata McGrawHill House, 2017.

Reference Books:

1. Gowri P., Hariharan and A. Suresh Babu, Manufacturing Technology – I, Pearson Education, 2008.
2. Roy A. Lindberg, “Processes and Materials of Manufacture”, 4th edition, Prentice Hall India, 1998.

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3. Kalpakjian S. and Steven S. Schmid, Manufacturing Engineering and Technology, 4th edition, Pearson Education India Edition, 2002.
4. Manufacturing Science by A. Ghosh and A.K. Mallick, Wiley Eastern.
5. Principles of Metal Cutting/Principles of Machine Tools by G.C. Sen and A. Bhattacharya, New Central Book Agency, Kolkata.

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	-	2	3	2	-	2	2	2	3
CO2	2	2	3	2	2	2	2	-	3	2	2	3
CO3	3	2	2	2	2	2	2	2	2	2	2	3
CO4	2	2	3	2	3	3	2	-	3	3	3	3

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1ST Year 2nd Semester: 2nd Semester								
Sl. No	Category	Course Code	Course Title	Hours per week				Credits
	THEORY			L	T	P	Total	
1	Basic Science courses	CH 201	Chemistry-I	3	0	0	3	3
2	Basic Science courses	M 201	Mathematics –II	4	0	0	4	4
3	Engineering Science Courses	EE 201	Basic Electrical Engineering	3	0	0	3	3
4	Engineering Science Courses	CS 201	Programming for Problem Solving	3	0	0	3	3
	PRACTICAL							
5	Basic Science course	CH 291	Chemistry-I Lab	0	0	3	3	1.5
6	Humanities and Social Sciences including Management courses	HSMC 291	Professional Communication LAB	0	0	2	2	1.0
7	Engineering Science Courses	EE 291	Basic Electrical Engineering Lab	0	0	3	3	1.5
8	Engineering Science Courses	ME 292	Engineering Graphics & Design Lab	0	0	3	3	1.5
9	Engineering Science Courses	CS 291	Programming for Problem Solving Lab	0	0	3	3	1.5
10	PROJECT	PR291	Theme based Project II	0	0	1	1	0.5
11	PROJECT	PR292	Skill Development II: Life Skill	1	0	0	1	0.5
	MANDATORY ACTIVITIES / COURSES							
12	Mandatory Course	MC281	NSS/Physical Activities / Meditation & Yoga / Photography	0	0	3	3	3 Units
	TOTAL CREDIT							21

Course Name: Chemistry

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Code: CH 201

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Pre requisites: A basic knowledge in 10+2 science with chemistry

Course Outcomes (COs):

After completion of this course students will be able to

CO1: Describe the fundamental properties of atoms & molecules, atomic structure and the periodicity of elements in the periodic table

CO2: Apply fundamental concepts of thermodynamics in different engineering applications.

CO3: Apply the knowledge of water quality parameters, corrosion control & polymers to different industries.

CO4: Determine the structure of organic molecules using different spectroscopic techniques.

CO5: Evaluate theoretical and practical aspects relating to the transfer of the production of chemical products from laboratories to the industrial scale, in accordance with environmental considerations.

Course Content

Module- I: Inorganic Chemistry 9L

(i) Atomic structure **5L**

Bohr's theory to hydrogen-like atoms and ions; spectrum of hydrogen atom. Quantum numbers, Introduction to the concept of atomic orbitals, diagrams of s, p and d orbitals, Pauli's exclusion principle, Hund's rule, exchange energy, Aufbau principle and its limitation, introduction to Schrodinger equation.

(ii) Periodic properties 4L

Modern Periodic table, group trends and periodic trends in physical properties: electron affinity, electronegativity, polarizability, oxidation states, effective nuclear charges, penetration of orbitals, variations of s, p and d orbital energies of atoms.

Module II: Physical Chemistry **8L**

(i) Use of free energy in chemical equilibria 6L

Thermodynamic functions: internal energy, enthalpy, entropy and free energy. 2nd Law of Thermodynamics, Estimations of entropy and free energies, Free energy and emf, Cell potentials, the Nernst equation and applications.

(ii) Real Gases **2L**

Reason for deviation of real gases from ideal behavior, Equations of state of real

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gases, Vander Waals' equation, pressure & volume correction, validity, critical state of gas.

Module III: Organic Chemistry **8L**

(i) *Stereochemistry* 4L

Representations of 3 dimensional structures, Chirality, optical activity, isomerism, structural isomerism, stereoisomers, enantiomers, diastereomers, configurations (D,L& cis trans), racemisation.

(ii) *Organic reactions* 4L

Concepts of inductive effect, resonance, hyperconjugation, introduction to reactions involving substitution, addition, elimination, oxidation (Baeyer villiger oxidation), reduction (Clemmensen reduction, Wolff-Kishner reduction).

Module IV: Industrial Chemistry **8L**

(i) *Water* 2L

Hardness, alkalinity, numerical

(ii) *Corrosion.* 2L

Types of corrosion: wet & dry, preventive measures

(iii) *Polymers* 3L

Classification of polymers, conducting polymers, biodegradable polymers

(iv) *Synthesis of a commonly used drug molecule.* 1L

Paracetamol, Aspirin

Module V: Spectroscopic techniques in Chemistry **3L**

Electromagnetic radiation, Principles of spectroscopy, spectrophotometer, infrared spectroscopy, fingerprint region, functional group region, UV-VIS spectroscopy, ^1H Nuclear magnetic resonance spectroscopy, chemical shift.

Textbooks

1. A Text Book of Organic Chemistry, Arun Bahl & Arun Bahl
2. General & Inorganic Chemistry, P.K. Dutt
3. General & Inorganic Chemistry, Vol I, R.P. Sarkar
4. Physical Chemistry, P.C. Rakshit

Reference Books

1. Chemistry: Principles and Applications, by M. J. Sienko and R. A. Plane
2. Fundamentals of Molecular Spectroscopy, by C. N. Banwell

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3. Engineering Chemistry (NPTEL Web-book), by B. L. Tembe, Kamaluddin and M. S.Krishnan
4. Physical Chemistry, by P. W. Atkins
5. Organic Chemistry: Structure and Function by K. P. C. Vollhardt and N. E. Schore, 5th Edition <http://bcs.whfreeman.com/vollhardtschore5e/default.asp>

Project Domain

1. Application of Thermodynamics
2. Application of polymers in daily life
3. Nanomaterials and its applications
4. Determination of water quality parameters
5. Electronic storage devices
6. Managing E –wastes
7. Application of chemistry in core engineering
8. Application of spectroscopy in medical field
9. Applications of green chemistry
10. Merits of commercial organic products
11. Bioplastics
12. Any other related topics

CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO 1	3	2	2	2	-	-	-	-	2	2	2	2
CO 2	3	3	3	3	-	-	-	-	2	2	2	3
CO 3	3	3	2	2	-	2	2	-	2	-	3	3
CO 4	3	2	3	2	-	-	2	-	2	2	3	3
CO 5	3	3	3	3	2	2	2	-	2	-	2	3

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Course Name: Mathematics-II

Course Code: M 201

Contact: 3:1:0

Total Contact Hours: 48

Credit: 4

Prerequisite: The students to whom this course will be offered must have the concept of (10+2) calculus.

Course Outcomes (COs):

On successful completion of the learning sessions of the course, the learner will be able to

CO1: Recall the properties and formula related to ordinary differential equations, improper integral, Laplace transform and numerical techniques.

CO2: Determine the solutions of the problems related to ordinary differential equations, improper integral, Laplace transform and numerical techniques.

CO3: Apply appropriate mathematical tools of ordinary differential equations, improper integral, Laplace transform and numerical techniques for the solutions of the problems.

CO4: Analyze engineering problems by using differential equation, Laplace Transform and Numerical Methods.

Course Content:

Module I: First Order Ordinary Differential Equations (ODE):

10L

Solution of first order and first degree ODE: Exact ODE, Rules for finding Integrating factors, Linear ODE, Bernoulli's equation, Solution of first order and higher degree ODE: solvable for p , solvable for y solvable for x and Clairaut's equation.

Module II: Second Order Ordinary Differential Equations (ODE):

10L

Solution of second order ODE with constant coefficients: C.F. & P.I., Method of variation of parameters, Cauchy-Euler equations, Solution of simultaneous linear ODEs.

Module III: Laplace Transform (LT):

14L

Improper integrals; Beta and Gamma functions and their properties.

Definition and existence of LT, LT of elementary functions, First and second shifting properties, Change of scale property, LT of $t f(t)$, LT of $\frac{f(t)}{t}$, LT of derivatives of $f(t)$, LT of integral of $f(t)$, Evaluation of improper integrals using LT, LT of periodic and step functions,

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Inverse LT: Definition and its properties, Convolution theorem (statement only) and its application to the evaluation of inverse LT, Solution of linear ODE with constant coefficients (initial value problem) using LT.

Module IV: Numerical Methods

14L

Introduction to error analysis, Calculus of finite difference. **Interpolation:** Newton forward and backward interpolation, Lagrange's interpolation, Newton's divided difference interpolation formula. Numerical integration: Trapezoidal rule, Simpson's 1/3 rule, Weddle's rule. Numerical solution of ordinary differential equation: Euler method, Modified Euler method, Fourth order Runge-Kutta method.

Project Domains:

1. Mathematical modeling using ODE.
2. Application of ODE.
3. Application of Laplace Transform in different engineering branches.
4. Application of Numerical Methods in different engineering branches.

Text Books:

1. Kreyszig, E., Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. Ramana, B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11th Reprint, 2010.
3. Veerarajan, T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2008.
4. Grewal, B.S., Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.
5. Bali, N.P. and Goyal, M., A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2008.
6. Samanta Guruprasad, A text book of Engineering Mathematics-II, New age International Publishers
7. Mollah, S. A, Numerical Analysis and Computational Procedures, Books and Allied (P) Ltd.

Reference Books:

1. Thomas, G.B. and Finney, R.L., Calculus and Analytic geometry, 9th Edition, Pearson, Reprint, 2002.
2. Boyce, W. E. and DiPrima, R. C., Elementary Differential Equations and Boundary Value Problems, 9th Edn., Wiley India, 2009.
3. Ross, S. L., Differential Equations, 3rd Ed., Wiley India, 1984.
4. Piskunov, N., Differential and Integral Calculus, Vol. I & Vol. II, Mir Publishers, 1969.
5. Coddington, E. A., An Introduction to Ordinary Differential Equations, Prentice Hall, India, 1995.

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6. Dey, Sukhendu, Gupta Sisir, Numerical Methods, MsGraw Hill Education(India) Private Limited.
7. Jain, M. K., Iyengar, S. R. K., Jain, R. K., Numerical Methods, New age International Publishers

CO-PO Mapping:

PO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	P12
CO1	3	2	-	-	-	-	-	-	-	-	-	2
CO2	3	2	-	-	-	-	-	-	-	-	-	2
CO3	3	2	2	-	-	-	-	-	-	-	-	2
CO4	2	3	2	2	-	-	-	-	-	-	-	2

Curriculum for B.Tech Under Autonomy
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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Basic Electrical Engineering

Course Code: Ee201

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Pre-requisite: Basic 12th standard Physics and Mathematics, Concept of components of electric circuit.

Course Outcomes (COs):

After attending the course students would be able to -

CO1: understand and analyze basic electric circuits

CO2: study the working principles of electrical machines.

CO3: introduce the components of low voltage electrical installations

CO4: study the fundamentals of electrical Power systems and Control Systems

Course Content

Module- I: DC Circuits

8L

Definition of electric circuit, linear circuit, non-linear circuit, bilateral circuit, unilateral circuit, Dependent source, node, branch, active and passive elements, Kirchhoff 's laws, Source equivalence and conversion, Network Theorems - Superposition Theorem, Thevenin's Theorem, Norton Theorem, Maximum Power Transfer Theorem, Star-Delta Conversions.

Module- II: AC Fundamentals

8L

Representation of sinusoidal waveforms, peak and rms values, phasor representation, real power, reactive power, apparent power, power factor. Analysis of single-phase ac circuits consisting of R, L, C, RL, RC, RLC combinations (series and parallel), resonance. Three phase balanced circuits, voltage and current relations in star and delta connections.

Module- III: Electrical Machines

10L

Transformer: Magnetic materials, BH characteristics, ideal and practical transformer, equivalent circuit, losses in transformers, regulation, and efficiency. Auto-transformer and three-phase transformer connections.

Rotating Machines - DC Machines: Brief idea on constructional features, classifications, working principle of both motor and generator. Simple problems on Voltage equation. Three-Phase Induction Motor: Basic concept of three phase circuit and production of rotating magnetic field. Working principle of three-phase induction motor and torque-speed characteristics (concept only).

Module- IV: Electrical Installations

3L

Module- V: Fundamentals of Power Systems **5L**

Module- VI: Introduction to Control Systems	2L
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Text books:

- ### Reference books:

- ### CO-PO Mapping:

[illegible]

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Course Name: Programming for Problem Solving

Course Code: CS 201

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: Number system, Boolean Algebra

Objective of the course:

The objective of the course is to make the students able to –

1. Understand the fundamental concept of Computer and mathematical knowledge for problem solving.
2. Understand the basic the principles of designing structured programs like C programming and use of data types/operators/input/output function for developing and implementing program.
3. Use conditional branching, iteration, recursion and formulate algorithms and programs.
4. Understand the concept of arrays, pointers, file and dynamic memory allocation and also able to create new data types using structure, union and enum.
5. Understand how to decompose a problem into functions and assemble into a complete program by means of modular programming.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the fundamental concept of Computer and mathematical knowledge and apply them in designing and analyzing solution to engineering problem.
CO2	Understand the basic concept of C programming and use of data types/operators/input/output function for developing and implementing complete program leading to solution of mathematical and engineering problem.
CO3	Use conditional branching, iteration, recursion and formulate algorithms and programs in solving mathematical/scientific/engineering problem and also analyze the same leading to lifelong learning.
CO4	Understand the concept of arrays, pointers, file and dynamic memory allocation and apply it for problem solving and also create new data types using structure, union and enum.
CO5	Understand how to decompose a problem into functions and assemble into a complete program by means of modular programming possibly as a team.

Course Content:

Module-I: Fundamentals of Computer

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History of Computer, Generation of Computer, Classification of Computers, Basic structure of Computer System, Primary & Secondary Memory, Processing Unit, Input & Output devices.

Number System: basic of Binary, Octal, Decimal and Hexadecimal number systems; Representation and interchanging of number in different number systems. Introduction to complements system, Representation of signed and unsigned numbers in signed magnitude signed 1's complement system and signed 2's complement system.

Arithmetic – Addition and Subtraction (using 1's complement and 2's complement).

Representation of Characters-ASCII Code

Basics of Compiler, Interpreter and Assembler

Problem solving – Basic concept of Algorithm. Representation of algorithm using flow chart and pseudo code. Some basic examples.

Module-II: Introduction to C Programming

5

Overview of Procedural vs Structural language; History of C Programming Language.

Variable and Data Types: The C character set identifiers and keywords, data type & sizes, variable names, declaration, statements.

Operators & Expressions: Arithmetic operators, relational operators, logical operators, increment and decrement operators, bitwise operators, assignment operators, conditional operators, special operators - type conversion, C expressions, precedence and associativity.

Input and Output: Standard input and output, formatted output - printf, formatted input scanf

Module-III: Branch and Loop

5

Branching: Concept of Statement and Blocks in C, Simple if, if - else, nested if-else and if-else ladder.

Switch Case: break and continue; switch-case, concept of goto and labels

Loops - while, for, do while

Module-IV: Program Structures

4

Function: Basics of Functions, function types, function prototypes, formal and actual parameter, function calling, functions returning values, functions not returning values. Recursion and Recursive Function.

Storage Class in C: Storage Class-auto, external, static and register storage class, scope rules and life time of variables

C preprocessor: Preprocessing directive and macro, parameterized macro

Module-V: Array and Pointer

7

Arrays: One dimensional arrays, Two-dimensional arrays, Passing an array to a function

Pointers: Pointers, Pointer and Array, Pointer and functions.

Strings: Character array and string, array of strings, Passing a string to a function, String related

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functions, Pointer and String.

Dynamic memory allocation: Malloc, calloc, realloc and free with example.

Module-VI: Structures, Unions and Enum

5

Basic of structures, arrays of structures, structures and pointers, bit fields. Basics of union and enum, difference between structure and union.

Module-VII: File in C

5

Files handling- opening and closing a file in different mode, formatted and unformatted files, Command line arguments, fopen, fclose, fgetc, fputc, fprintf, fscanf function.

Text book:

1. Byron Gottfried, Schaum's Outline of Programming with C, McGraw-Hill
2. Kanetkar Y. - Let us C, BPB Publication, 15th Edition

Reference Books:

1. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice Hall of India
2. K R Venugopal & S R Prasad – MASTERING C, TMH, 2nd Edition

CO – PO Mapping:+

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
C01	3	3	3	3	2	-	-	-	-	-	-	-	3	3	3
C02	2	2	3	2	-	-	-	-	-	-	-	-	3	2	3
C03	2	3	2	2	-	-	-	-	-	-	-	3	3	3	1
C04	3	2	2	2	3	-	-	-	-	-	-	-	3	3	1
C05	2	2	2	2	-	-	-	-	3	2	-	-	3	2	1

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Course Name: Chemistry Lab

Course Code: CH 291

Contact: 0:0:3

Credits: 1.5

Pre-requisite: A basic knowledge in 10+2 science with chemistry.

Course Outcomes (COs):

After attending this course, students would be

CO1: able to operate different types of instruments for estimation of small quantities chemicals used in industries and scientific and technical fields.

CO2: able to analyze and determine the composition of liquid and solid samples working as an individual and also as a team member.

CO3: able to analyze different parameters of water considering environmental issues.

CO4: able to synthesize drug and polymer materials.

CO5: Capable to design innovative experiments applying the fundamentals of chemistry.

Course Content:

Choice of 10-12 experiments from the following:

1. Determination of surface tension and viscosity
2. Thin layer chromatography
3. Determination of hardness of water
4. Determination of chloride content of water
5. Determination of the rate constant of a reaction
6. Determination of cell constant and conductometric titration
7. pH metric titrations
8. Synthesis of a polymer/drug
9. Saponification/acid value of an oil

10. Chemical analysis of a salt
Chemical oscillations- Iodine clock reaction
11. Determination of the partition coefficient of a substance between two immiscible liquids
12. Adsorption of acetic acid by charcoal
13. Estimation of iron in Mohr's salt solution by permanganometry (Redox Titration)
14. Innovative experiments (any one)
 - Synthesis of silver nano-particles

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- Green synthesis

CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	3	2	-	2	3	-	-	-	-	2
CO2	2	2	2	2	-	2	-	-	-	2	-	2
CO3	-	-	-	-	-	-	-	-	3	3	2	2
CO4	2	2	2	2	-	-	2	-	-	-	-	2
CO5	3	3	3	3	2	2	2	2	-	-	2	2

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Course Name: Professional Communication Lab

Course Code: HSMC 291

Contact: 0:0:2

Credit: 1

Pre requisites: Basic knowledge of LSRW skills.

Course Outcomes (COs):

After attending the course students' would be

CO1: Able to explain advanced skills of Technical Communication in English through Language Laboratory.

CO2: Able to apply listening, speaking, reading and writing skills in societal and professional life.

CO3: Able to demonstrate the skills necessary to be a competent Interpersonal communicator.

CO4: Able to analyze communication behaviours.

CO5: Able to adapt to multifarious socio-economical and professional arenas with the help of effective communication and interpersonal skills.

Course Content:

Module- 1: Introduction to the Language Lab

- a. The Need for a Language Laboratory
- b. Tasks in the Lab
- c. Writing a Laboratory Note Book

Module- 2: Active Listening

- a. What is Active Listening?
- b. Listening Sub-Skills—Predicting, Clarifying, Inferencing, Evaluating, Note-taking
- c. Academic Listening vs Business Listening
- d. Listening in Business Telephony
- e. Study of Contextualized Examples based on Lab Recordings

Module- 3: Speaking

- a. Speaking—Accuracy and Fluency Parameters
- b. Pronunciation Guide—Basics of Sound Scripting, Stress and Intonation
- c. Fluency-focussed activities—JAM, Conversational Role Plays, Speaking using Picture/Audio Visual inputs
- d. Accuracy-focussed activities—Identifying Minimal Pairs, Sound Mazes, Open and Closed

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Pair Drilling, Student Recordings (using software)

e. Group Discussion: Principles and Practice

f. Business Meetings and Sales Talks

Module- 4: Lab Project Work

a. Making a brief Advertisement video (1-2 minutes)

b. Making a brief Business Documentary film (5-7 minutes)

c. Client interaction video (5-7 minutes)

d. Making a short video CV (1-2 minutes)

References:

1. IIT Mumbai, Preparatory Course in English syllabus
2. IIT Mumbai, Introduction to Linguistics syllabus
3. Sasikumar et al. A Course in Listening and Speaking. New Delhi: Foundation Books, 2005.
4. Tony Lynch, Study Listening. Cambridge: Cambridge UP, 2004.

CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1 0	PO1 1	PO1 2
CO1	2	-	-	-	-	-	2	-	-	3	-	2
CO2	2	3	2	-	-	2	2	2	-	3	-	3
CO3	2	3	-	-	-	3	3	3	-	3	-	3
CO4	-	-	-	-	-	3	3	3	-	3	-	3
CO5	-	-	-	-	-	-	3	3	-	3	-	3

Curriculum for B.Tech Under Autonomy
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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Basic Electrical Engineering Laboratory

Course Code: EE291

Contact: 0:0:3

Credits: 1.5

Prerequisite: Basic Physics and applied physics, Basic Mathematics, Basic concept of Electric Circuit.

Course Outcomes (COs):

After completion of this course students will be able to

CO1: Identify and use common electrical components.

CO2: To develop electrical networks by physical connection of various components and analyze the circuit behavior.

CO3: Apply and analyze the basic characteristics of transformers and electrical machines.

List of Experiments

1. Basic safety precautions – earthing, introduction to measuring instruments – Voltmeter, Ammeter, Multimeter, Wattmeter, Real life Resistor, Capacitor, Inductor.
2. Verification of Thevenin's and Norton's Theorem.
3. Verification of Superposition and Maximum Power Transfer Theorem.
4. Characteristics of Fluorescent, Tungsten and Carbon filament lamps.
5. Study of R-L-C series circuit.
6. Three-phase Power measurement with two wattmeter methods.
7. Demonstration of cut-out sections of machines: DC Machine (commutator-brush arrangement), Induction Machine (squirrel cage rotor).
8. Measurement of primary and secondary voltage and current of single-phase transformer – Open Circuit and Short Circuit Test.
9. Starting, Reversing and speed control of DC shunt motor.
10. Torque-Speed characteristics of DC Machine.
11. Torque-Speed characteristics of Three-phase Induction Motor.
12. Test on single-phase Energy Meter.
13. Innovative experiments

CO-PO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
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Curriculum for B.Tech Under Autonomy

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

CO 1	-	2	-	-	-	-	-	-	1	-	-	1
CO 2	-	2	2	2	-	-	-	-	2	-	-	1
CO 3	-	2	-	2	-	-	-	-	2	-	-	1

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Engineering Graphics & Design

Course Code: ME 292

Contact: 0:0:3

Credits: 1.5

Prerequisites: Basic knowledge of geometry

Course Outcomes (COs):

After completion of the course students will be able to

CO1: get introduced with Engineering Graphics and visual aspects of design.

CO2: know and use common drafting tools with the knowledge of drafting standards.

CO3: be able to apply computer aided drafting techniques to represent line, surface or solid models in different Engineering viewpoints.

CO4: be able to produce part models; carry out assembly operation and show working procedure of a designed project work using animation.

List of Drawing:

Traditional Engineering Graphics:

Principles of Engineering Graphics; Orthographic Projection; Descriptive Geometry; Drawing Principles; Isometric Projection; Surface Development; Perspective; Reading a Drawing; Sectional Views; Dimensioning & Tolerances; True Length, Angle; intersection, Shortest Distance.

Module 1: Introduction to Engineering Drawing

Principles of Engineering Graphics and their significance, Usage of Drawing instruments, lettering, Conic sections including Rectangular Hyperbola (General method only); Cycloid, Epicycloid and Involute; Scales – Plain, Diagonal and Vernier Scales.

Module 2: Orthographic & Isometric Projections

Principles of Orthographic Projections-Conventions - Projections of Points and lines inclined to both planes; Projections of planes on inclined Planes - Auxiliary Planes; Projection of Solids inclined to both the Planes- Auxiliary Views; Isometric Scale, Isometric Views of lines, Planes, Simple and compound Solids; Conversion of Isometric Views to Orthographic Views and Vice-versa.

Module 3: Sections and Sectional Views of Right Angular Solids

Drawing sectional views of solids for Prism, Cylinder, Pyramid, Cone and project the true shape of the sectioned surface, Auxiliary Views; Development of surfaces of Right Regular Solids -

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Prism, Pyramid, Cylinder and Cone; Draw sectional orthographic views of objects from industry and dwellings (foundation to slab only).

Computer Graphics:

Engineering Graphics Software; -Spatial Transformations; Orthographic Projections; Model Viewing; Co-ordinate Systems; Multi-view Projection; Exploded Assembly; Model Viewing; Animation; Spatial Manipulation; Surface Modeling; Solid Modeling.

Module 4: Overview of Computer Graphics

Demonstration of CAD software [The Menu System, Toolbars (Standard, Properties, Draw, Modify and Dimension), Drawing Area (Background, Crosshairs, Coordinate System), Dialog boxes and windows, Shortcut menus (Button Bars), Zooming methods, Select and erase objects].

Module 5: CAD Drawing, Customization, Annotations, layering

Set up of drawing page including scale settings, ISO and ANSI standards for dimensioning and tolerancing; Using various methods to draw straight lines, circles, applying dimensions and annotations to drawings; Setting up and use of Layers, changing line lengths (extend/lengthen); Drawing sectional views of solids; Drawing annotation, CAD modeling of parts and assemblies with animation, Parametric and nonparametric solid, surface and wireframe modeling, Part editing and printing documents.

Module 6: Demonstration of a simple team design project

Illustrating Geometry and topology of engineered components: creation of engineering models and their presentation in standard 2D blueprint form and as 3D wire-frame and shaded solids; Meshed topologies for engineering analysis and tool-path generation for component manufacture, use of solid-modeling software for creating associative models at the component and assembly levels.

Text Books:

1. Bhatt N.D., Panchal V.M. & Ingle P.R, (2014), Engineering Drawing, Charotar Publishing House

2. K. Venugopal, Engineering Drawing + AutoCAD, New Age International publishers

Reference Books:

1. Pradeep Jain, Ankita Maheswari, A.P. Gautam, Engineering Graphics & Design, Khanna Publishing House

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2. Agrawal B. & Agrawal C. M. (2012), Engineering Graphics, TMH Publication.
3. Shah, M.B. & Rana B.C. (2008), Engineering Drawing and Computer Graphics, Pearson Education
4. Narayana, K.L. & P Kannaiah (2008), Text book on Engineering Drawing, Scitech Publishers.

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	3	2	2	2	2	-	2	2	2	2
CO2	2	2	3	2	2	2	2	2	2	2	2	2
CO3	2	2	3	2	3	2	2	-	2	2	2	3
CO4	2	2	3	3	3	3	2	2	3	3	2	2

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Course Name: Programming for Problem Solving Lab

Course Code: CS 291

Contact: 3:0:0

Credits: 1.5

Prerequisites: Number system, Boolean Algebra

Objective of the course:

The objective of the course is to make the students able to –

6. Understand and propose appropriate command or function in running system or developing program.
7. Identify and propose appropriate data type, arithmetic operators, input/output functions and also conditional statements in designing effective programs.
8. Design and develop effective programs using iterative statements as well as recursive functions using modular programming approach.
9. Explain and organize data in arrays, strings and structures and manipulate them through programs and also define pointers of different types and use them in defining self-referential structures and also to construct and use files for reading and writing.
10. Prepare laboratory reports on interpretation of experimental results and analyze it for validating the same.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand, propose and apply appropriate command or function in running system or developing program for engineering and mathematical problems depending on the platform used even in changed environment leading to their lifelong learning.
CO2	Identify and apply appropriate data type, arithmetic operators, input/output functions and also conditional statements in designing effective programs to solve complex engineering problem using modern tools.
CO3	Design and develop effective programs after analyzing engineering and mathematical problems using iterative statements as well as recursive functions using modular programming approach possibly as a team maintaining proper ethics of collaboration.
CO4	Explain and organize data in arrays, strings and structures and manipulate them through programs and also define pointers of different types and use them in defining self-referential structures and also to construct and use files for reading and writing to and from leading to solution of engineering and mathematical problem.
CO5	Prepare laboratory reports on interpretation of experimental results and analyze it for validating the same maintaining proper ethics of collaboration.

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Course Content:

Module-I:

Familiarization with some basic commands of DOS and Linux. File handling and Directory structures, file permissions, creating and editing simple C program in different editor and IDE, compilation and execution of C program. Introduction to Codeblock.

Module-II:

Problem based on

- a) Basic data types
- b) Different arithmetic operators.
- c) printf() and scanf() functions.

Module-III:

Problem based on conditional statements using

- a) if-else statements
- b) different relational operators
- c) different logical operators

Module-IV:

Problem based on

- a) **for** loop
- b) **while** loop
- c) **do-while** loop

Module-V:

Problem based on

- a) How to write a menu driven program using **switch-case** statement
- b) How to write a function and passing values to a function
- c) How to write a **recursive function**.

Module-VI:

Problem based on

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- a) How to use **array** (both **1-D** and **2-D**).
- b) How to pass an **array** to a **function**.

Module-VII:

Problem based on manipulation of strings in different way.

Module-VIII:

Problem based on

- a) How to handle compound variables in C
- b) How to handle file in C
- c) How to use command line argument in C

Text book:

3. "Introduction to Algorithms" by Cormen, Leiserson, Rivest, Stein.
4. "The Design and Analysis of Computer Algorithms" by Aho, Hopcroft, Ullman. 3. "Algorithm Design" by Kleinberg and Tardos. 4. Design & Analysis of Algorithms, Gajendra Sharma, Khanna Publishing House, New Delhi

Reference Books:

1. "Design Analysis and Algorithms" by Hari Mohan Pandey.

CO – PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	3	2	-	-	-	-	-	-	-	3	3	3	-
CO2	3	3	2	3	3	-	-	-	-	-	-	-	3	3	3
CO3	2	3	3	3	-	-	-	3	3	-	-	-	-	-	3
CO4	2	2	2	2	-	-	-	-	-	-	-	-	3	-	-
CO5	-	-	-	-	-	-	-	3	-	3	-	-	3	-	3

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Course Name: Discrete Mathematics

Course Code: M301

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: Higher Secondary Level Mathematics

Course Objectives:

1. Comprehend the fundamental concepts of Set Theory, Mathematical Logic for inferencing and various Proof Techniques, Theory of Numbers, Algebraic Structures, Graph Theory.
2. Formulate problems based on Combinatorial Structures, Algebraic Structures, Graph Theory, Recurrence Relation etc.
3. Use the concepts of Discrete Mathematics to solve problems based on Combinatorial Structures, Algebraic Structures, Graph Theory, Recurrence Relation etc.
4. Provide proof of theorems using well known Proof Techniques and Mathematical Logic Frameworks to justify a claim.
5. Propose solutions to the complex problems of Discrete Mathematics and Analyze their effectiveness in solving the relevant problems

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the fundamental concepts of Set Theory to Explain or Illustrate and Identify problems where students can Apply the concept appropriately to Solve them.
CO2	Understand the fundamental concepts of Mathematical Logic and Proof Techniques so that they can Prove theorems using Proof Techniques and Mathematical Logic Frameworks to justify a claim.
CO3	Explain or Illustrate the fundamental Theory of Numbers and Identify problems where students can Use the concept appropriately to Solve them.
CO4	Explain or Illustrate the fundamental principles of Algebraic Structures and Identify problems where students can Apply the concept appropriately to Solve them.
CO5	Develop ideas to Propose solutions to the problems of Graph Theory and Identify problems where students can Apply the concept appropriately and Analyze the effectiveness as well as limitations of solutions making the students aware of its utilitarian importance for further explorations leading towards lifelong learning.

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Course Content:

MODULE-1: Set Theory [8L]

Set: Operations and Properties of set, Finite Set, Power Set, Cardinality of finite set, Cartesian Product, Relation: Types of Relations, Properties of Binary Relation, Equivalence Relation, Partial Ordering Relation and Poset, Lattice.

Combinatorics and Counting: Sum and product rule, Permutation and Combination Principle of Inclusion Exclusion. Pigeon Hole Principle.

Generating Functions and Recurrence Relations: Recursively defined relation and functions, Discrete Numeric Function, Growth of Functions, Problems on Recurrence Relations and their solutions using different methods.

MODULE-2: Mathematical Logic and Proof Techniques [8L]

Propositional Logic: Basics of Boolean Logic, Idea of Propositional Logic, well-formed formula, Truth tables, Tautology, Satisfiability, Contradiction, Algebra of proposition, Inference theory of Propositional Logic.

Predicate Logic: Idea of First Order Predicate Logic and Quantifiers, well-formed formula of predicate, Inference theory of Predicate Logic.

Proof Techniques: Some Terminology, Proof Methods and Strategies, Forward Proof, Proof by Contradiction, Proof by Contraposition, Proof of Necessity and Sufficiency.

MODULE-3: Theory of Numbers [4L]

Well-Ordering Principle, Divisibility theory and properties of Divisibility, Fundamental theorem of Arithmetic, Prime and Composite Numbers.

Greatest Common Divisor and Euclidean Algorithm, Congruence, Residue Classes.

MODULE-4: Algebraic Structures [8L]

Concepts of Groups, Subgroups and Order, Cyclic Groups, Cosets, Normal Subgroups, Permutation and Symmetric groups, Group Homomorphisms.

Elementary properties of Rings and related problems.

Elementary properties of Fields and related problems.

Elementary properties of Vector Space and related problems.

MODULE-5: Graph Theory [8L]

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Graph Terminologies and their properties: Degree, Connectivity, Path, Cycle, Sub-Graph, Isomorphism, Eulerian and Hamiltonian Walks, Matrix representation of graphs, Shortest Path in Graph.

Graph Colouring and Matching: Colouring Vertices and Chromatic Number, Colouring Edges and Total Colouring, Independence and Chromatic Partitioning, Cliques, Perfect Graphs, Bounds on Chromatic Numbers, Chromatic Polynomials, Matching.

Tree: Rooted Trees, Binary Search Tree and Tree Sorting, Spanning Tree, Weighted Trees and prefix codes.

Textbook:

1. Kenneth H. Rosen, Discrete Mathematics and its Applications, Tata McGraw – Hill.
2. Susanna S. Epp, Discrete Mathematics with Applications, 4th edition, Wadsworth Publishing Co. Inc.

Reference Books:

3. C L Liu and D P Mohapatra, Elements of Discrete Mathematics A Computer Oriented Approach, 3rd Edition by, Tata McGraw – Hill.
4. J.P. Tremblay and R. Manohar, Discrete Mathematical Structure and It's Application to Computer Science", TMG Edition, TataMcgraw-Hill
5. Seymour Lipschutz, Marc Lipson, Discrete Mathematics (Schaum's Outlines Series), Tata McGraw - Hill.

CO–PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	-	3	-	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	-	-	-	-	-	-	-	-	2	-	2
CO4	2	2	2	3	-	-	-	-	-	-	-	-	2	-	2
CO5	2	2	3	3	-	-	-	-	-	-	-	2	2	2	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Analog Electronics

Course Code: ESC301

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisite:

Concept of basic electronics devices, basic law of circuit analysis

Course Objective(s):

1. To provide a strong foundation on linear circuits
2. To familiarize students with applications of various IC's
3. Having a broad coverage in field that is relevant for engineers to design Linear circuits using OP-AMPS
4. Familiarize the conversion of data from Analog to Digital and Digital to Analog

Course Outcomes:

After completion of the course students will be able to:

CO1	Understand the characteristics of ICs, diodes, transistors, FET and apply them in designing solution to engineering problems
CO2	Understand the working principles of RC coupled amplifier, feedback amplifier, voltage gain and current gain in these circuits, handling frequencies and bandwidth related problems and demonstrate the solutions leading to mathematical and engineering problems.
CO3	Analyze the functioning of OP-AMP, feedback amplifiers, oscillator circuits and design different circuits leading to lifelong learning
CO4	Understand, implement and evaluate the performance of different types of analog amplifiers and multi-vibrator circuits
CO5	Understand different types of power amplifiers, power supply including ADC and DAC circuits and compare their performances and solve related problems possibly as a team

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Content:

Module1: [4L]

Small signal amplifiers: Introduction to Analog Integrated Circuits, BJT Modeling-hybrid model of transistors; Emitter follower circuits, High frequency model of transistors. FET Small signal analysis - Source follower

Module 2: [9L]

Transistor Amplifiers: RC coupled amplifier, functions of all components, equivalent circuit, derivation of voltage gain, current gain, input impedance and output impedance, frequency response characteristics, lower and upper half frequencies, bandwidth, and concept of wide band amplifier.

Feedback Amplifiers & Oscillators: Feedback concept, Voltage series-shunt, current series-shunt feedback Configurations, Barkhausen criterion, Colpitts, Hartley's, Phase shift, Wien bridge and crystal oscillators.

Module3: [14L]

Operational Amplifier: Introduction to Integrated Circuits, Differential Amplifier, Constant current source (current mirror etc.), level shifter, CMRR, Open & Closed loop circuits, importance of feedback loop (positive & negative), Block Diagram of OPAMP, Ideal OPAMP.

Applications of Operational Amplifiers: analog adder, subtractor, integrator, differentiator, comparator, Schmitt Trigger. Instrumentation Amplifier, Log & Anti-log amplifiers, Analog multiplier, Precision Rectifier, voltage to current and current to voltage converter, free running Multivibrator, zero crossing detector

Multivibrator – Monostable, Bistable, Astable multivibrators ; Monostable and astable operation using 555 timer.

Module 4: [9L]

Large signal Amplifiers: Introduction to power amplifiers (Class A, B, AB)

Power Supply: Analysis for DC voltage and ripple voltage with C, L-C and C-L-C filters in Rectifier Circuit - Regulated DC power supplies- Line regulation, output resistance and temperature coefficient, Series and Shunt Voltage Regulation – percentage regulation, Fixed output voltage IC regulator 78xx and 79xx series , Adjustable output voltage regulator, LM

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337 series power supply ICs, Concept of **Switched** Mode Power Supply

Text Books:

1. Millman Halkias – Integrated Electronics, McGraw Hill
2. Schilling & Belove—Electronic Circuit: Discrete & Integrated, 3/e, McGraw Hill
3. Ramakant A. Gayakwad —Op- Amps and linear Integrated Circuits, Pub: PHI
4. Boylested&Nashelsky- Electronic Devices and Circuit Theory- Pearson/PHI
5. “Operational Amplifiers and Linear Integrated Circuits” by Robert F. Coughlin, Frederick F. Driscoll

Reference Books:

1. Rashid-Microelectronic Circuits- Analysis and Design- Thomson (Cenege Learning)
2. Linear Integrated Circuits – D. Roy Choudhury & Shail B. Jain
3. Analog Integrated Circuits – J. B. Gupta

CO-PO & PSO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	-	-	-	-	-	-	-	2	3	2
CO2	3	3	3	3	3	-	-	-	-	-	-	-	2	3	2
CO3	3	3	3	3	3	-	-	-	-	-	-	3	2	3	2
CO4	3	3	3	3	3	-	-	-	-	-	-	-	2	3	2
CO5	3	3	3	3	3	-	-	-	3	-	2	-	2	3	2
CO	3	3	3	3	3	-	-	-	3	-	2	3	2	3	2

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Digital Electronics

Course Code: ESC302

Contact: 3:0:0

Credit: 3

Total Contact Hours: 36

Pre-requisite:

Basic concepts of Logic gates, Truth Tables, Concept of basic components of a digital computer

Course Objective(s):

Objective of Digital electronics is to acquire the basic knowledge of different logic gates and to prepare students to perform the analysis and design of various digital electronic circuits.

Course Outcomes:

After completion of the course students will be able to:

CO1	Understand boolean algebra, number system and execute and solve the problem related to number system conversion, minimization techniques and their application in digital design which leads to the solution of engineering problems
CO2	Understand the working principles of the logic gates, combinational circuits as well as different components of a digital computer and demonstrate the solutions leading to mathematical and engineering problems
CO3	Analyze and design different sequential circuits like flip flops, registers and counters leading to lifelong learning
CO4	Understand, implement and evaluate the performance of different types of digital circuits
CO5	Understand different logic families like TTL, ECL, MOS and CMOS, A/D and D/A conversion Programmable logic Array, programmable Array logic, Sequential Programmable Devices and compare their performances and solve related problems possibly as a team

Course Content:

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Module – 1: [8L]

Binary Number System [1L], BCD, ASCII, EBDIC, Gray codes and their conversions [1L], Introduction and laws of boolean algebra [1L], boolean functions, minterm and maxterm, prime implicants, representation in SOP and POS forms [2L], minimization of logic expressions by Karnaugh map and algebraic method [3L].

Module – 2: [8L]

Combinational circuits: Adder and Subtractor (half-full adder & subtractor) [2L], Serial & Parallel Adder, Carry look ahead adder and Parity Generator[2L], Encoder, Decoder, Multiplexer [2L], Demultiplexer, Comparator, Code Converters [2L]

Module – 3: [12L]

Sequential Circuits:

Flip-Flops, SR, JK, Master slave JK, D, T and, characteristic Tables, Excitation tables. [5L] Basic concept of Synchronous and Asynchronous counters, Up/Down Counters, Ring counter, Johnson counter, Design of Modulo-N Counter, Counter applications [5L] Registers (SISO, SIPO, PIPO, PISO) [2L]

Module – 4: [8L]

A/D and D/A conversion techniques – Basic concepts (D/A :R-2-R only [2L] , A/D: successive approximation [2L]), Logic families- TTL, ECL, MOS and CMOS - basic concepts [2L], Programmable logic Array, programmable Array logic, Sequential Programmable Devices [2L]

Text Book:

1. Salivahanan S - Digital Circuits and Design, Oxford
2. Morries Mano - Digital Logic Design- PHI

Reference Book:

1. R. P. Jain - Modern Digital Electronics, 2/e, Mc Graw Hill
2. Thomas L. Floyd - Digital Fundamentals: A Systems Approach, Pearson

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CO-PO & PSO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	-	-	-	-	-	-	-	2	3	2
CO2	3	3	3	3	3	-	-	-	-	-	-	-	2	3	2
CO3	3	3	3	3	3	-	-	-	-	-	-	3	2	3	2
CO4	3	3	3	3	3	-	-	-	-	-	-	-	2	3	2
CO5	3	3	3	3	3	-	-	-	3	-	2	-	2	3	2
CO	3	3	3	3	3	-	-	-	3	-	2	3	2	3	2

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name : Programming in PYTHON
Course Code : PCC-CS301
Contact : 3:0:0
Total Contact Hours : 35
Credits : 3

Prerequisite:

1. Basics of computer, Boolean Algebra

Course Objective(s)

- Describe the core syntax and semantics of Python programming language.
- Discover the need for working with the strings and functions.
- Illustrate the process of structuring the data using lists, dictionaries, tuples and sets.
- Indicate the use of regular expressions and built-in functions to navigate the file system.

Course Outcome(s)

On completion of the course students will be able to –

CO1	Interpret the fundamental Python syntax and semantics and be fluent in the use of Python control flow statements.
CO2	Develop proficiency in the handling of strings and functions
CO3	Identify the methods to design and modify Python programs by applying the data structures like lists, dictionaries, tuples and sets.
CO4	Articulate the concepts of Dictionaries while analyzing different case studies.
CO5	Identify the commonly used operations collaborating file systems and regular expressions.

Course Contents:

Module-I: Parts of Python Programming Language

7

Parts of Python Programming Language, Identifiers, Keywords, Statements and Expressions, Variables, Operators, Precedence and Associativity, Data Types, Indentation, Comments, Reading Input, Print Output, Type

Curriculum for B.Tech Under Autonomy

B.Tech. – Computer Science & Technology

L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Conversions, The type() Function and Is Operator, Dynamic and Strongly Typed Language, **Control Flow Statements**, The if Decision Control Flow Statement, The if...else Decision Control Flow Statement, The if...elif...else Decision Control Statement, Nested if Statement, The while Loop, The for Loop, The continue and break Statements, Catching Exceptions Using try and except Statement.

Module-II: Functions

3

Built-In Functions, Commonly Used Modules, Function Definition and Calling the Function, The return Statement and void Function, Scope and Lifetime of Variables, Default Parameters, Keyword Arguments, *args and **kwargs, Command Line Arguments.

Module-III: Strings

9

Creating and Storing Strings, Basic String Operations, Accessing Characters in String by Index Number, String Slicing and Joining, String Methods, Formatting Strings, **Lists**, Creating Lists, Basic List Operations, Indexing and Slicing in Lists, Built-In Functions Used on Lists, List Methods, The del Statement.

Module-IV: Self-Study

8

Dictionaries, Creating Dictionary, Accessing and Modifying key; value Pairs in Dictionaries, Built-In Functions Used on Dictionaries, Dictionary Methods, The del Statement, **Tuples and Sets**, Creating Tuples, Basic Tuple Operations, Indexing and Slicing in Tuples, Built-In Functions Used on Tuples, Relation between Tuples and Lists, Relation between Tuples and Dictionaries, Tuple Methods, Using zip() Function, Sets, Set Methods, Traversing of Sets, Frozenset.

Module-V: Files

8

Types of Files, Creating and Reading Text Data, File Methods to Read and Write Data, Reading and Writing Binary Files, The Pickle Module, Reading and Writing CSV Files, Python OS and os.path Modules, **Regular Expression Operations**, Using Special Characters, Regular Expression Methods, Named Groups in Python Regular Expressions, Regular Expression with glob Module.

Text books:

1. Gowrishankar S, Veena A, “Introduction to Python Programming”, 1st Edition, CRC Press/Taylor & Francis, 2018.

Recommended books:

1. Jake VanderPlas, “Python Data Science Handbook: Essential Tools for Working with Data”, 1st Edition, O'Reilly Media, 2016.
2. Aurelien Geron, “Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems”, 2nd Edition, O'Reilly Media, 2019.
3. Wesley J Chun, “Core Python Applications Programming”, 3rd Edition, Pearson Education India, 2015.
4. Miguel Grinberg, “Flask Web Development: Developing Web Applications with Python”, 2nd Edition, O'Reilly Media, 2018.

CO-PO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
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[illegible]

Curriculum for B.Tech Under Autonomy
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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Paper Name: Data Structures

Paper Code: PCC-CS302

Contact Hours/Week: 3:0:0

Credits: 3

Allotted Lectures: 36L

Prerequisite:

1. Familiarity with the fundamentals of C or other programming language
2. A solid background in mathematics, including probability, set theory.

Course Objectives:

The objective of the course is to make the students able to -

1. Understand the concept of data structure and create new data structure and also analyzing the efficiency of the same.
2. Identify and differentiate different types of data structures and implement the appropriate data structure and analyze the same.
3. Understand and implement stack, queue and dequeue by selecting appropriate methods.
4. Understand and implement different non-linear data structures by selecting appropriate methods.
5. Understand different factors of sorting and searching algorithm and select the appropriate algorithm and also implement and analyze the algorithm.

Course Outcome(s)

On completion of the course students will be able to

CO1	Understand the concept of data structure and create new data structure to propose efficient solution by writing appropriate algorithm and program for engineering and mathematical problem after analyzing the efficiency of the same.
CO2	Identify and differentiate different types of data structures and implement the appropriate data structure after analyzing complex engineering problem leading to their lifelong learning.

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CO3	Understand and implement stack, queue and dequeue by selecting appropriate methods and use it for solving real life and engineering problem choosing appropriate modern tools
CO4	Understand and implement different non-linear data structures by selecting appropriate methods and apply it for solving complex engineering problem and also argue and judge maintaining the professional ethics to validate the same.
CO5	Understand different factors of sorting and searching algorithm and select the appropriate algorithm for solving complex engineering problem and also implement and analyze the algorithm.

Course Content:

Module I: Introduction (4L):

Concepts of data and information; Concept of Abstract Data Type, Data Structure and Data Type. Classification of Data Structures- Primitive and Non-Primitive Data Structure, Linear and Non-Linear Data Structure. Need of Data Structures. (1L)

Concept of algorithms and programs, Different methods of representing algorithm; Algorithm analysis, time and space analysis of algorithms – Asymptotic notations like Big Oh (O), Big Omega(Ω) and Theta(Θ) notation(definition and significance). (3L)

Module II: Non-Restricted Linear Data Structure [9L]

List or Linear List: Definition and Example, List as ADT. Representation of Linear List- Sequential Representation and Linked Representation.

Array: Introduction to sequential representation, Linearization of multidimensional array. Application of array- representation of polynomial using array, Representation of Sparse matrix using array.

Linked List: Introduction to linked representation, Implementation of different types of linked list- Singly linked list, Doubly linked list, Circular linked list, Circular Doubly Linked List. Application of Linked list- Representation of polynomial.

Module III: Restricted Linear Data Structure [6L]

Stack: Definition of Stack, implementations of stack using array and linked list

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Applications of stack- infix to postfix conversion, Postfix Evaluation

Recursion: Principles of recursion - use of stack, tail recursion. Tower of Hanoi using recursion.

Queue: Definition of Queue; Implementation of queue using array- physical, linear and circular model; Implementation of queue using linked list.

Deque- Definition and different types of dequeue.

Module IV: Nonlinear Data structures [9L]

Trees and Binary Tree:

Basic terminologies; Definition of tree and binary tree. Difference between tree and binary tree, Representation of binary tree (using array and linked list)

Binary tree traversal (pre-, in-, post- order); Threaded binary tree- definition, insertion and deletion algorithm; Binary search tree- Definition, insertion, deletion, searching algorithm;

Height balanced binary tree: AVL tree- definition, insertion and deletion with examples only.

m –Way Search Tree: B Tree – Definition, insertion and deletion with examples only; B+ Tree – Definition, insertion and deletion with examples only.

Heap: Definition (min heap and max heap), creation, insertion and deletion algorithm. Application of heap (priority queue and sorting).

Graphs: Definition and representation (adjacency matrix, incidence matrix and adjacency list).

Graph traversal– Depth-first search (DFS), Breadth-first search (BFS) - concepts of edges used in DFS and BFS (tree-edge, back-edge, cross-edge, and forward-edge).

Module V: Sorting and Searching [8L]

Sorting Algorithms: Definition and need of sorting, different types of sorting algorithm(internal, external, stable, in-place, comparison based); Bubble sort, Insertion sort, Selection sort, Quick sort, Merge sort, Radix sort – algorithm with analysis (time complexity)

Searching: Sequential search – algorithm with analysis (time complexity); improvement using sentinel.

Binary search and Interpolation Search algorithm with analysis (time complexity)

Hashing: Introduction and purpose of Hashing and Hash functions (division, folding and mid-square), Collision resolution techniques.

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Text books:

1. Data Structures Through 'C' Language by Samiran Chattopadhyay, Debabrata Ghosh Dastidar, Matangini Chattopadhyay, Edition: 2001, BPB Publications
2. Fundamentals of Data Structures of C by Ellis Horowitz, Sartaj Sahni, Susan Anderson-freed 2nd Edition, Universities Press

Reference books:

1. Data Structures, Algorithms, and Software Principles in C by Thomas A. Standish, 1 Edition, Pearson.
2. Data Structures by S. Lipschutz, Special Indian Edition, Tata McGraw Hill Education (India) Private Limited
3. Data Structures and Program Design In C by Robert L. Kruse, Bruce P. Leung 2nd Edition, Pearson
4. Data Structures in C by Aaron M. Tenenbaum, 1st Edition, Pearson

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	1	-	-	-	-	-	-		3	3	3
CO2	3	3	3	3		-	-	-	-	-	-	3	3	3	3
CO3	3	3	3	3	3	-	-	-	-	-	-		3	3	1
CO4	3	3	3	3	1	-	-	2	-	-	-	-	3	3	3
CO5	3	3	3	3	1	-	-	-	-	-	-		3	1	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Universal Human Values2: Understanding Harmony

Course Code: HSMC 303

Contacts: 3:0:0

Total Contact Hours: 36

Credit: 3

Prerequisite: None

Course Outcome(s):

After completion of the course students will be able to –

CO1	Develop holistic perspective based on self-exploration about themselves (human being), family, society and nature/existence.
CO2	Cultivate the harmony in the human being, family, society and nature/ existence
CO3	Strengthen self-reflection
CO4	Build commitment and courage to act.
CO5	Understand different factors of Harmony of family, Society, nature and Implications on Professional Ethics.

Course Content:

Module1: Course Introduction-Need, Basic Guidelines, Content and Process for Value Education[8L]

Self-Exploration–what is it? -Its content and process; ‘Natural Acceptance’ and Experiential Validation-as the process for self-exploration. Continuous Happiness and Prosperity- A look at basic Human Aspirations. Right understanding, Relationship and Physical Facility-the basic requirements for fulfillment of aspirations of every human being with the incorrect priority. Understanding Happiness and Prosperity correctly-A critical appraisal of the current scenario. Method to fulfill the above human aspirations: understanding and living in harmony at various levels. Practice sessions to discuss natural acceptance in human being as the innate acceptance for living with responsibility (living in relationship, harmony and co-existence) rather than as

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arbitrariness in choice based on liking-disliking.

Module2: Understanding Harmony in the Human Being – Harmony in Myself! [6L]

Understanding human being as a co-existence of the sentient ‘I’ and the material ‘Body’. Understanding the needs of Self (‘I’) and ‘Body’ - happiness and physical facility. Understanding the Body as an instrument of ‘I’ (I being the doer, seer and enjoyer). Understanding the characteristics and activities of ‘I’ and harmony in ‘I’. Understanding the harmony of I with the Body: Sanyam and Health; correct appraisal of Physical needs, meaning of Prosperity in detail. Programs to ensure Sanyam and Health. Practice sessions to discuss the role others have played in making material goods available to me. Identifying from one’s own life. Differentiate between prosperity and accumulation. Ensuring health vs dealing with disease discussion.

Module 3: Understanding Harmony in the Family and Society- Harmony in Human-Human Relationship [7L]

Understanding values in human-human relationship; meaning of Justice (nine universal values in relationships) and program for its fulfilment to ensure mutual happiness; Trust and Respect as the foundational values of relationship. Understanding the meaning of Trust; Difference between intention and competence. Understanding the meaning of Respect, Difference between respect and differentiation; the other salient values in relationship. Understanding the harmony in the society (society being an extension of family): Resolution, Prosperity, fearlessness (trust) and co-existence as comprehensive Human Goals. Visualizing a universal harmonious order in society – Undivided Society, Universal Order-from family to world family. Practice sessions to reflect on relationships in family, hostel and institute as extended family, real life examples, teacher-student relationship, goal of education etc. Gratitude as a universal value in relationships. Elicit examples from students’ lives.

Module4: Understanding Harmony in the Nature and Existence-Whole existence as Coexistence[8L]

Understanding the harmony in the Nature. Interconnectedness and mutual fulfilment among the four orders of nature- recyclability and self- regulation in nature. Understanding Existence as Co-existence of mutually interacting units in all-pervasive space. Holistic perception of harmony at all levels of existence. Practice sessions to discuss human being as cause of imbalance in nature (film “Home” can be used), pollution, depletion of resources and role of Technology etc.

Module5: Implications of the above Holistic Understanding of Harmony on Professional Ethics [7L]

Natural acceptance of human values. Definitiveness of Ethical Human Conduct. Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order.

Competence in professional ethics:

- a. Ability to utilize the professional competence for augmenting universal human order

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- b. Ability to identify the scope and characteristics of people-friendly and eco-friendly production systems,
- c. Ability to identify and develop appropriate technologies and management patterns for above production systems.
- d. Case studies of typical holistic technologies, management models and production systems. Strategy for transition from the present state to Universal Human Order:
- e. At the level of individual: as socially and ecologically responsible engineers, technologists and managers
- f. At the level of society: as mutually enriching institutions and organizations.
- g. Practice Exercises and Case Studies in Practice (tutorial) Sessions to discuss the conduct as an engineer or scientist etc.

Text Books:

1. Human Values and Professional Ethics by RRGaur, RSangal, GPBagaria, Excel Books, New Delhi, 2010

Reference Books

1. Jeevan Vidya: Ek Parichaya, A Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.
2. Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3. The Story of Stuff (Book).
4. The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi
5. Small is Beautiful - E.F Schumacher.
6. Slow is Beautiful - Cecile Andrews
7. Economy of Permanence - JC Kumarappa
8. Bharat Mein Angreji Raj - Pandit Sunderlal
9. Rediscovering India - by Dharampal
10. Hind Swaraj or Indian Home Rule - by Mohandas K. Gandhi
11. India Wins Freedom - Maulana Abdul Kalam Azad
12. Vivekananda - Romain Rolland (English)
13. Gandhi - Romain Rolland (English)

CO-PO Mapping:

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	2	1	-	2	1	-	-	-	2	3	3	2	3	3
CO2	3	2	-	1	3	2	-	1	2	-	3	3	3	3	3
CO3	3	2	2	-	2	3	1	-	2	1	3	3	1	3	3
CO4	3	1	-	2	-	-	-	2	-	3	3	3	3	3	2

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

CO5	2	1	3	2	-	-	-	2	-	3	3	3	3	3	3
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Curriculum for B.Tech Under Autonomy
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Practical

Name of the Paper: Numerical Methods lab

Paper Code: M(CS)391

Contact :=1:0:3

Credit Point: 2.5

Prerequisite:

3. Basic programming knowledge

Course Objective(s)

The objective of the course is to make the students able to –

- Provide basic programming skills for solving the problems in numerical methods and statistics.

Course Outcome(s)

After completion of the course students will be able to

CO1	Describe and explain the theoretical workings of numerical techniques with the help of C/ MATLAB
CO2	Compute basic command and scripts in a mathematical programming language
CO3	Apply the programming skills to solve the problems using multiple numerical approaches.
CO4	Analyze if the results are reasonable, and then interpret and clearly communicate the results.
CO5	Apply the distinctive principles of numerical analysis and the associated error measures.

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List of Experiment:

1. Assignments on Newton forward /backward, Lagrange's interpolation.
 2. Assignments on numerical integration using Trapezoidal rule, Simpson's 1/3 rule
 3. Assignments on numerical solution of a system of linear equations using Gauss elimination, Gauss Jacobi and Gauss-Seidel iterations.
 4. Assignments on numerical solution of Algebraic Equation by Bisection method, Regula-Falsi method, Newton-Raphson method.
 5. Assignments on ordinary differential equation: Euler's method, Euler's modified method, Runge-Kutta methods.
 6. Simple problems as assignment on Measures of Central Tendency- mean, median, mode, Measures of Dispersion- variance, standard deviation. Problems related to engineering field.
 7. Innovative Experiments
- Implementation of numerical methods on computer through C/C++ and commercial Software Packages: Matlab / Scilab / Labview / Mathematica/NAG (Numerical Algorithms Group/Python).

CO-PO Mapping

CO #	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	2	1	1	-	2	-	1	2	2	2	2
CO2	3	3	3	2	2	1	1	-	2	-	1	2	2	2	2
CO3	3	3	3	2	2	1	1	-	2	-	1	2	2	2	2
CO4	3	3	3	2	2	1	1	-	2	-	1	2	2	2	2
CO5	3	3	3	2	2	1	1	-	2	-	1	2	2	2	2

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Course Name: Digital Logic Lab

Course Code: ESC391

Credit: 1.5

Pre-requisite:

Basic concepts of logic gates, truth tables, function realization – minimization of logic expressions by K-map, concept of basic components of a digital computer, binary arithmetic

Course Objective(s):

This subject will act as prerequisite for computer organization and architecture. The aims of this lab are to make students familiar with the principles of combinational and sequential digital logic design and optimization at a gate level and designing various circuits with ICs.

Course Outcomes:

After completion of the course students will be able to:

CO1	Comprehend the working principles of the logic gates and combinational circuits like half and full adder circuits and execute and solve problems related to digital circuit design which leads to the solution of engineering problems
CO2	Understand the working principle of half and full adder circuit, adder-subtractor composite unit and carry-look-ahead adder circuit and Demonstrate the solutions leading to engineering problems
CO3	Design and compare different circuits- adder, subtractor, multiplexer, De- multiplexer, decoder, encoder, comparator and flip flop
CO4	Analyze and implement sequential circuits like register, counter leading to lifelong learning
CO5	Understand different circuits and construct digital circuits to verify the operations of DAC and different types of logic families and PLDs and compare their performances and solve related problems possibly as a team

Course Content:

1. A) Realization of basic gates and universal gates.

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B) Realization of basic gates using universal gates.

2. Design a Half adder and Full Adder circuit using basic gates and verify its output.
3. Design a Half subtractor and Full Subtractor circuit using basic gates and verify its output
4. Design an Adder/Subtractor composite unit.
5. Design of a 'Carry-Look-Ahead' Adder circuit.
6. Realization of a) Encoder, b) Decoder c) Multiplexer , d) De-MUX , e) Comparator and their Truth Table verification.
7. Realization of RS / JK / D flip flops using logic gates.
8. Design of Shift Register using J-K / D Flip Flop.
9. Realization of Synchronous Up/Down counters.
10. Design of MOD- N Counter
11. Study of DAC
12. Study of logic families and PLDs

Text Book:

1. Salivahanan S - Digital Circuits and Design, Oxford
2. Morries Mano - Digital Logic Design- PHI

Reference Book:

1. R.P. Jain - Modern Digital Electronics, 2/e, Mc Graw Hill
2. Thomas L. Floyd - Digital Fundamentals: A Systems Approach , Pearson,

CO-PO & PSO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	-	-	-	-	-	-	-	2	3	2
CO2	3	3	3	3	3	-	-	-	-	-	-	-	2	3	2
CO3	3	3	3	3	3	-	-	-	-	-	-	3	2	3	2
CO4	3	3	3	3	3	-	-	-	-	-	-	-	2	3	2
CO5	3	3	3	3	3	-	-	-	3	-	2	-	2	3	2
CO	3	3	3	3	3	-	-	-	3	-	2	3	2	3	2

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Programming in PYTHON Lab

Course Code: PCC-CS301

Contact: 0:0:3

Credits: 1.5

Prerequisites: Number system, Boolean Algebra

Objective of the course:

The objective of the course is to make the students able to –

- Describe the core syntax and semantics of Python programming language.
- Discover the need for working with the strings and functions.
- Illustrate the process of structuring the data using lists, dictionaries, tuples and sets.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the fundamental Python syntax and semantics and be fluent in writing simple Python programs in the appropriate platform.
CO2	Develop proficiency in the handling of strings and be fluent in the use of Python control flow statements.
CO3	Solve different problems while applying the concepts of loops.
CO4	Develop different programming skills while correlating functions.
CO5	Articulate the Object-Oriented Programming concepts such as encapsulation, inheritance and polymorphism as used in Python.

Course Content:

Module-I: Fundamentals of Python

- Introduction to Python
- Running Python Programs
- Writing simple Python Code

Module-II: Working with Data

- Data Types and Variables
- Using Numeric Variables
- Using String Variables

Curriculum for B.Tech Under Autonomy
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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Module-III: Input and Output

- Printing with Parameters
- Getting Input from a User
- String Formatting

Module-IV: Making Decisions

- Logical Expressions
- The “if” Statement
- Logical Operators
- More Complex Expressions

Module-V: Lists and Loops

- Lists and Tuples
- List Functions
- “For” Loops
- “While” Loops

Module-VI: Functions

- Writing and Calling Functions
- Function Inputs and Outputs
- Local and Global Scope

Module-VII: Working with Strings

- Character Data
- String Functions
- Input Validation with “try / except”

Text books:

- 1 Gowrishankar S, Veena A, “Introduction to Python Programming”, 1st Edition, CRC Press/Taylor & Francis, 2018.

Reference books:

- CO – PO Mapping:**

[illegible]

Curriculum for B.Tech Under Autonomy
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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Code: Data Structures Lab

Course Code: PCC-CS392

Contact: 0:0:3

Credits: 1.5

Perquisites:

1. Computer Fundamentals and principal of computer programming Lab

Objective of the course:

The objective of the course is to make the students able to –

1. Identify and propose appropriate data structures and data types to implement list using array and linked list.
2. Design and develop effective programs using stack, queue and recursive functions after implementing stack and queue.
3. Implement different data structures like binary tree, heap and use them to explain and organize data and manipulate them.
4. Implement different sorting and searching algorithm selecting appropriate data structures and analyze the efficiency of the resulting program.
5. Prepare laboratory reports on interpretation of experimental results and analyze it for validating the same.

Course Outcomes:

On completion of the course students will be able to

CO1	Identify and propose appropriate data structures and data types to implement list using array and linked list and design effective programs to solve complex engineering problem using list and modern tools.
CO2	Design and develop effective programs for engineering and mathematical problems using stack, queue and recursive functions after implementing stack and queue using modular programming approach possibly as a team maintaining proper ethics of collaboration.
CO3	Implement different data structures like binary tree, heap and use them to explain and organize data and manipulate them through programs leading to solution of complex engineering problem.

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CO4	Implement different sorting and searching algorithm selecting appropriate data structures and analyze the efficiency of the resulting program using modern engineering tools and methods leading to lifelong learning.
CO5	Prepare laboratory reports on interpretation of experimental results and analyze it for validating the same maintaining proper ethics of collaboration.

Course Content:

Module I: Implementing Non-Restricted Linear Data Structure[2 Lab]

Problem based on Implementation of Non-Restricted Linear Data Structure like-

- a) Implementation of list as data structure using array.
- b) Implementation of listas data structure using linked list of different types.
- c) Implementation of polynomial as data structure using array and linked list.
- d) Implementation of sparse matrix as data stricture using array.

Module II: Implementing Restricted Linear Data Structure[3 Lab]

Problem based on Implementation of Restricted Linear Data Structure like-

- a) Implementation of stack as data structure using array.
- b) Implementation of stack as data structure using linked list.
- c) Implementation of queue as data structure using array (physical, linear and circular model).
- d) Implementation of queue as data structure using linked list.
- e) Converting infix to post-fix and evaluating post-fix expression using stack.
- f) Implementing Tower-of-Hanoi problem.

Module III: Implementing Non-Linear Data Structure[2 Lab]

Problem based on Implementation of Non-Linear Data Structure like

- a) Implementation of Binary Tree as data structure using array and linked list.
- b) Implementation of Binary Search Tree (BST) as data structure using linked list.
- c) Implementation of Heap as data structure using array.
- d) Implementation of Priority Queue as data structure using Heap.

Module IV: Implementing Sorting and Searching algorithm [5 Lab]

Problem based on Implementation of Sorting and Searching algorithm like

- a) Implementation of Bubble sort using appropriate data structure.
- b) Implementation of Selection sort using appropriate data structure.
- c) Implementation of Insertion sort using appropriate data structure.
- d) Implementation of Quick sort using appropriate data structure.
- e) Implementation of Merge sort using appropriate data structure.
- f) Implementation of Heap sort using appropriate data structure.
- g) Implementation of Radix sort using appropriate data structure.
- h) Implementation of Sequential Search using appropriate data structure.

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- i) Implementation of Binary Search using appropriate data structure.
- j) Implementation of hashing with collision resolution using linear and quadratic probing.

Text books:

- 3. Data Structures Through 'C' Language by Samiran Chattopadhyay, Debabrata Ghosh Dastidar, Matangini Chattopadhyay, Edition: 2001, BPB Publications
- 4. Fundamentals of Data Structures of C by Ellis Horowitz, Sartaj Sahni, Susan Anderson-freed 2nd Edition, Universities Press

Reference books:

- 5. Data Structures, Algorithms, and Software Principles in C by Thomas A. Standish, 1 Edition, Pearson.
- 6. Data Structures by S. Lipschutz, Special Indian Edition, Tata McGraw Hill Education (India) Private Limited
- 7. Data Structures and Program Design In C by Robert L. Kruse, Bruce P. Leung 2nd Edition, Pearson
- 8. Data Structures in C by Aaron M. Tenenbaum, 1st Edition, Pearson

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	-	-	-	-	-	-	-	3	3	3
CO2	3	3	3	3	-	-	-	3	3	-	-	-	2	2	3
CO3	3	3	3	3	-	-	-	-	-	-	-	-	3	2	3
CO4	3	3	3	3	3	-	-	-	-	-	-	3	3	2	3
CO5	-	-	-	-	-	-	-	3	-	3	-	-	3	2	3

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MANDATORY COURSE

Course Name: ENVIRONMENTAL SCIENCE

Course Code: MC 301

Credits: 0

Total Lectures: 36

Course Objective(s):

- Be able to understand the natural environment and its relationships with human activities.
- Be able to understand environmental laws and regulations to develop guidelines and procedures for health and safety issues.
- Be able to solve scientific problem-solving related to air, water, noise & land pollution.

Course Outcome(s):

CO1	To understand the natural environment and its relationships with human activities.
CO2	To apply the fundamental knowledge of science and engineering to assess environmental and health risk.
CO3	To develop guidelines and procedures for health and safety issues obeying the environmental laws and regulations.
CO4	Acquire skills for scientific problem-solving related to air, water, noise & land pollution.
CO5	apply the fundamental knowledge of science and engineering to assess environmental and health risk

Course Contents:

Module 1: General [11L]

Natural Resources: Forest Resource, water resource, mineral resource, energy resources: alternative source of energy

Population Growth: Exponential Growth, logistic growth, Maximum sustainable yield, demography

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Disaster Management: Types of disasters (Natural & Man-made), Floods, Earthquake, Tsunamis, Cyclones, landslides (cause, effect & control)

Ecology & Ecosystem: Elements of ecology, definition of ecosystem- components types and function, Food chain & Food web, Structure and function of the following ecosystem: Forest ecosystem, Grassland ecosystem, Desert ecosystem, Aquatic ecosystems

Environmental Management: Environmental impact assessment, Environmental laws and protection act of India(The Environment protection Act, Air pollution Act, Water Act, Wildlife Protection Act) , Hazardous waste(management and Handling) Rules.

Module 2: Air pollution and control [10L]

Sources of Pollutants: point sources, nonpoint sources and manmade sources primary & secondary pollutant

Types of air pollutants: primary & secondary pollutant ; Suspended particulate matter, oxides of carbon, oxides of nitrogen, oxides of sulphur, particulate, PAN, Smog (Photochemical smog and London smog),

Effects on human health & climate: Greenhouse effect, Global Warming, Acid rain, Ozone Layer Depletion

Air pollution and meteorology: Ambient Lapse Rate, Adiabatic Lapse Rate, Atmospheric stability & Temperature inversion control of air pollution (ESP, cyclone separator, bag house, catalytic converter, scrubber (ventury)).

Module 3: Water Pollution [9L]

Classification of water (Ground & surface water)

Pollutants of water, their origin and effects: Oxygen demanding wastes, pathogens, nutrients, Salts, heavy metals, pesticides, volatile organic compounds.

Surface water quality parameters: pH, DO, 5 day BOD test, BOD reaction rate constants, COD. Numerical related to BOD

Lake: Eutrophication [Definition, source and effect].

Ground water: Aquifers, hydraulic gradient, ground water flow (Definition only),ground water pollution (Arsenic & Fluoride; sources, effects, control)

Quality of Boiler fed water: DO, hardness, alkalinity, TDS and Chloride

Layout of waste water treatment plant (scheme only).

Module 4: Land Pollution [3L]

Types of Solid Waste: Municipal, industrial, commercial, agricultural, domestic, hazardous solid wastes (bio-medical), E-waste

Solid waste disposal method: Open dumping, Land filling, incineration, composting, recycling (Advantages and disadvantages).

Waste management: waste classification, waste segregation, treatment & disposal

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Module 5: Noise Pollution [3L]

Definition of noise, effect of noise pollution on human health, Average Noise level of some common noise sources

Definition of noise frequency, noise pressure, noise intensity, noise threshold limit value, equivalent noise level, L_{10} (18 hr Index) . Noise pollution control.

Textbook:

1. A Textbook of Environmental Studies, Shashi Chawla. Tata McGraw Hill Education Private Limited

References Books:

1. Environmental Studies, Dr. J P Sharma, University Science Press
2. Environmental Engineering, J K Das Mohapatra, Vikas Publication

CO–PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO2	3	3	3	3	-	-	-	-	-	-	-	3	3	2	3
CO3	3	3	3	3	-	-	-	-	-	-	-	-	3	3	1
CO4	3	3	3	3	-	-	-	-	3	-	-	-	3	2	3
CO5	3	3	3	3	-	-	-	-	3	2	-	-	3	2	3
CO	3	3	3	3	-	-	-	-	3	2	-	3	3	3	3

2nd Year 2 nd Semester: 4 th Semester								
Sl. No.	Category	Course Code	Course Title	Hours per week				Credits
				L	T	P	Total	
A. THEORY								
1	Program Core Course	PCC-CS401	Computer Organization and Architecture	3	0	0	3	3
2	Program Core Course	PCC-CS402	Design and Analysis of Algorithms	3	0	0	3	3
3	Program Core Course	PCC-CS403	Operating Systems	3	0	0	3	3
4	Program Core Course	PCC-CS404	Formal Language and Automata Theory	3	0	0	3	3
5	Humanities and Social Sciences including Management courses	HSMC 402	Gender Culture and Development	2	0	0	2	2
6	Basic Science course	M401	Probability & Statistics	3	0	0	3	3
B. PRACTICAL								
6	Program Core Course	PCC-CS491	Computer Organization and Architecture Lab	0	0	3	3	1.5
7	Program Core Course	PCC-CS492	Design and Analysis of Algorithms Lab	0	0	3	3	1.5
8	Program Core Course	PCC-CS493	Operating Systems Lab	0	0	3	3	1.5
9	Engineering Science Courses	ESC491	IT Workshop Lab (C++/python)	0	0	3	3	1.5
10	PROJECT	PR 491	Theme based Project IV	0	0	1	1	0.5
11	PROJECT	PR492	Skill Development IV: Soft Skill & Aptitude-I	1	0	0	1	0.5
C. MANDATORY ACTIVITIES / COURSES								
15	MC	MC 481	Learning an Art Form [vocal or instrumental, dance, painting, clay modeling, etc.] OR Environmental Protection Initiatives	0	0	0	3	3Units
	TOTAL CREDIT WITHOUT MOOCS COURSES							24

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D.MOOCs COURSES								
16	MOOCs COURSES	HM401	MOOCs COURSE-II	3	1	0	4	4
TOTAL CREDIT WITH MOOCs COURSES								28

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Course Name: Computer Organization and Architecture

Course Code: PCC-CS401

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Perquisites: Digital Electronics

Course Objective:

The objective of the course is to make the students able to

1. Illustrate the basic concept of computer structure & computer arithmetic and apply this knowledge to design real life problem.
2. Summarize the basic concept of Basic Computer Organization and Central processing unit for designing and implementing problem.
3. Illustrate the concept of register transfer, micro operation and micro programmed control to design engineering problem.
4. Illustrate the concept of register transfer, micro operation and micro programmed control.
5. Understand the concept of computer Input-output organization and design an optimized model.

Course Outcomes:

After completion of the course students will be able to

CO1	Illustrate the basic concept of computer structure & computer arithmetic and apply this knowledge in designing solutions for real life engineering problems.
CO2	Summarize the basic concept of Basic Computer Organization and Central processing unit for designing and implementing mathematical and engineering problems leading to lifelong learning.
CO3	Illustrate the concept of register transfer, micro operation and micro programmed control to design and solve engineering problems.
CO4	Compare computer memories and apply this knowledge for developing an approach by means of existing and new methods as a team work.

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CO5	Understand the concept of computer Input-output organization and design an optimized model for building a new solution as a professional engineering practice as a team.
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Course Content:**Module – I:**

6

Structure of Computers

Computer types[1L], Functional units, Basic operational concepts[1L], Von Neumann Architecture, Bus Structures, Software, Performance[1L].

Computer Arithmetic

Fixed-point multiplication - Booth's algorithm. [1L], Fixed-point division - Restoring and non-restoring algorithms. [1L] Floating-point number representation- IEEE 754 format and Floating-point arithmetic operation [1L]

Module – II:

9

Basic Computer Organization and Design

Instruction codes, Computer Registers [1L], Computer Instructions and Instruction cycle[1L]. Timing and Control [1L], Memory-Reference Instructions, Input-Output and interrupt [1L]

Central Processing Unit

Stack organization[1L], Instruction Formats[1L], Addressing Modes [1L], Data Transfer and Manipulation[1L], Complex Instruction Set Computer (CISC) Reduced Instruction Set Computer (RISC), CISC vs RISC[1L]

Module – III:

8

Register Transfer and Micro-Operations

Register Transfer Language, Register Transfer[1L], Bus and Memory Transfers[2L], Arithmetic Micro-Operations, Logic Micro-Operations, Shift Micro-Operations, Arithmetic logic shift unit[1L]

Micro-Programmed Control

Control Memory[1L], Address Sequencing[1L], Micro-Program example[1L], Design of Control Unit[1L]

Module – IV:

7

Computer Memory

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Memory Hierarchy, Semiconductor Memories, RAM (Random Access Memory), Read Only Memory (ROM), Types of ROM[1L], Main Memory[1L], Cache Memory [1L], Cache coherence and synchronization mechanisms[1L], Mapping Technique in cache memory: Direct, Full Associative and Set Associative [2L], Cache Performance and optimization techniques [1L]

Module – V:**6****Input-Output Organization**

Introduction to I/O operations [1L], Synchronous and asynchronous transfer [1L], Modes of transfer [1L], DMA[1L], Bus Arbitration [1L], Input-output processor [1L]

Text Books:

1. David A. Patterson and John L. Hennessy- Computer Organization and Design: The Hardware/Software Interface
2. Morris Mano- Computer System Architecture, 3rd Edition, Pearson

Reference Books:

1. Hayes J. P., “Computer Architecture & Organisation”, McGraw Hill,
2. William Stallings, Computer Organization and Architecture: Designing for Performance

CO – PO Mapping:

COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO2	3	3	3	3	-	-	-	-	-	-	-	3	3	2	3
CO3	3	3	3	3	-	-	-	-	-	-	-	-	3	3	1
CO4	3	3	3	3	-	-	-	-	3	-	-	-	3	2	3
CO5	3	3	3	3	-	-	-	-	3	2	-	-	3	2	3

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Design & Analysis of Algorithm

Course Code: PCC-CS402

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: To know data-structure and basic programming ability

Objective of the course:

11. The aim is to learn how to develop efficient algorithms for simple computational tasks and reasoning about the correctness of them
12. Through the complexity measures, different range of behaviors of algorithms and the notion of tractable and intractable problems will be understood.

Course Outcomes:

After completion of the course students will be able to

CO1	To understand and illustrate the concepts of time and space complexity, worst case, average case and best-case complexities and the asymptotic notation.
CO2	To analyze and apply the design principles and concepts to various basic algorithm design viz. dynamic programming, greedy method etc.
CO3	To understand and analyze various string matching and graph algorithms.
CO4	To understand, illustrate and analyze the different complexity classes
CO5	To discuss, implement and analyze, verify the efficiency of the randomized and approximation algorithms.

Course Content:

Module-I

4

Algorithm Development & Complexity Analysis: [4L]

Stages of algorithm development for solving a problem: Describing the problem, identifying a suitable technique, Design of an algorithm, Proof of Correctness of the algorithm. Time and Space Complexity, Different Asymptotic notations – their mathematical significance. Solving Recurrences: Substitution Method, Recurrence Tree Method, Master Theorem (Statement Only).

Module-II

14

Algorithm Design Techniques [L]

Brute force techniques – Travelling Sales man Problem, Divide and Conquer - Matrix multiplication: Strassen's algorithm, Greedy techniques - Fractional Knapsack problem, Job

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Sequencing with Deadline, Graph Coloring, Finding Minimum Cost Spanning Tree, Dynamic programming - 0/1 Knapsack problem, Matrix chain multiplication, Travelling Salesman Problem, Backtracking-N-Queens Problem, Knights Tour on Chess Board.

Module-III **3**

String matching problem: [3L]

Different techniques – Naive algorithm, string matching using finite automata, and Knuth, Morris, Pratt (KMP) algorithm with their complexities.

Module-IV **5**

Graph Algorithms [5L]

Single Source Shortest Path – Dijkstra's Algorithm, All pair shortest path – Floyd-Warshall Algorithm. Network Flows, Maximum Flows – Ford-Fulkerson Algorithm, Push Re-label Algorithm, Minimum Cost Flows – Cycle Cancelling Algorithm.

Module-V **5**

Complexity Classes: [5L]

The Class P, The Class NP, Reducibility and NP-completeness – SAT (without proof), 3-SAT, Vertex Cover, Independent Set, Maximum Clique.

Module-VI **5**

Approximation and Randomized Algorithms [3L]

Approximation Algorithms - The set-covering problem – Vertex cover, K-center clustering. Randomized Algorithms - The hiring problem, Finding the global Minimum.

Recent Trends [2L]

Text book:

5. "Introduction to Algorithms" by Cormen, Leiserson, Rivest, Stein.
6. "The Design and Analysis of Computer Algorithms" by Aho, Hopcroft, Ullman. 3. "Algorithm Design" by Kleinberg and Tardos. 4. Design & Analysis of Algorithms, Gajendra Sharma, Khanna Publishing House, New Delhi

Reference Books:

- CO – PO Mapping:**

[illegible]

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Operating Systems

Course Code: PCC-CS403

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites:

1. Computer organization
2. Computer Architecture
3. Data Structures
4. Algorithms & Programming Concept

Course Objectives:

The objective of the course is to make the students able to -

1. Students will learn how Operating System is Important for Computer System.
2. To make aware of different types of Operating System and their services.
3. To learn different process scheduling algorithms and synchronization techniques to achieve better performance of a computer system.
4. To know virtual memory concepts.
5. To learn secondary memory management.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the basic principles of operating systems and compare different types of operating systems.
CO2	Understand the main principles and techniques for the implementation of processes, threads as well as the different algorithms for process scheduling and inter process communication.
CO3	Solve the main problems related to concurrency and the different synchronization mechanisms.
CO4	Explain the device and I/O management functions in operating systems as part of a uniform device abstraction.
CO5	Formulate the rationale view for virtual memory abstractions and explain the disk organization and file system structure.

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Content:

Module I: Introduction 4L

Functionalities of Operating System, Evolution of Operating System. Types of Operating System: batch, multi-programmed, time-sharing, real-time, distributed, parallel, Structural overview, Protection & Security.

Module II: 10L

Processes: Concept of processes, process states, PCB, process scheduling, co-operating processes, independent process, suspended process, Interaction between processes and OS, Inter-process communication: Message passing 3L

Threads: overview, benefits of threads, user and kernel level threads, Thread models. 2L

CPU scheduling: Scheduling criteria, preemptive & non-preemptive scheduling, scheduling algorithms (FCFS, SJF, SRTF, RR, priority, multilevel queue, multilevel feedback queue scheduling). 5L

Module III: 10L

Process Synchronization: background, critical section problem, synchronization hardware, classical problems of synchronization (producer-consumer, readers-writer, dining philosophers, etc), semaphores, monitors. 5L

Deadlocks: deadlock characterization, methods for handling deadlocks, deadlock prevention, deadlock avoidance, deadlock detection, recovery from deadlock. 5L

Module IV: 6L

Memory Management: Background, logical vs. physical address space, swapping, contiguous memory allocation, paging, Segmentation, TLB. 3L

Virtual Memory: background, demand paging, page replacement algorithms (FCFS, LRU, Optimal), thrashing, Working set model. 3L

Module V: 6L

Disk structure, disk scheduling (FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK etc), disk reliability, disk formatting, boot block, bad blocks. 2L

File: File concept, access methods, directory structure, file system structure, UNIX file structure, allocation methods (contiguous, linked, indexed), free-space management (bit vector). 2L

I/O: I/O hardware, polling, interrupts, DMA, caching, buffering, blocking-non blocking I/O. 2L

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Text Book:

1. Abraham Silberschatz, Peter B. Galvin, Greg Gagne, Operating System Concepts.
2. Operating Systems& Systems Programming by P Balakrishna Prasad

Reference Book:

1. Dietel H. N., “An Introduction to Operating Systems”, Addison Wesley.
2. Andrew Tanenbaum, Modern Operating Systems, Prentice Hall.
3. William Stallings, Operating Systems, Prentice Hall.

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO2	3	3	3	3	-	-	-	-	-	-	-	-	3	2	3
CO3	3	3	3	3	-	-	-	-	-	-	-	3	3	3	1
CO4	3	3	3	3	-	-	-	-	-	-	-	-	3	3	1
CO5	3	3	3	3	-	-	-	-	3	2	-	-	3	2	1
CO	3	3	3	3	-	-	-	-	3	2	-	3	3	3	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Formal Language and Automata Theory

Course Code: PCC-CS404

Contacts: 3:0:0

Total Contact Hours: 36

Credits:3

Prerequisites: Digital Logic, Computer Fundamentals

Course Objectives:

The objectives of this course are to enable students to

1. construct finite state machines and the equivalent regular expressions.
2. prove the equivalence of languages described by finite state machines and regular expressions.
3. construct pushdown automata and the equivalent context free grammars.
4. prove the equivalence of languages described by pushdown automata and context free grammars.
5. construct Turing machines and to prove the equivalence of languages described by Turing machines

Course Outcome(s):

After completion of the course students will be able to

CO1	Understand the fundamental concepts of Finite State Automata to Explain or Illustrate and Identify problems where students can Apply the concept appropriately to Solve them.
CO2	Understand the fundamental concepts of Regular Expressions and its relation with DFA so that they can Develop regular expression for a specified language and Validate it.
CO3	Understand the fundamental concepts of Context Free Grammar so that they can Design grammar for a specified language and Validate it.
CO4	Explain or Illustrate the fundamental operating principles of Push Down Automata and Use it appropriately to Solve problems.

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

CO5	Understand the operating principles of Turing Machine and Design Turing Machines to Propose solutions to the related problems appropriately and validate the effectiveness as well as limitations of computations making the students aware of its utilitarian importance for further explorations leading towards lifelong learning.
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Course Contents:

Module-1:[9L]

Fundamentals: Basic definition of sequential circuit, block diagram, mathematical representation, concept of transition table and transition diagram [1L]

Introduction to Finite State Model (FSM), Design of sequence detector, Finite State Machine, Finite Automata, Deterministic Finite Automation (DFA) and Non-deterministic Finite Automation (NFA), Transition diagrams, Transition tables and Language recognizers. [3L]

NFA with empty transitions, Equivalence between NFA with and without empty transitions. NFA to DFA conversion. [2L]

Minimization of FSM: Minimization Algorithm for DFA, Introduction to Myhill-Nerode Theorem [2L]

Limitations of FSM, Application of Finite Automata [1L]

Module-2:[7L]

Finite Automata with output – Moore & Mealy machine. Representation of Moore & Mealy Machine, Processing of the String through Moore & Mealy Machine, Equivalence of Moore & Mealy Machine – Inter-conversion. [2L]

Equivalent states and Distinguishable States, Equivalence and k-equivalence, Minimization of Mealy Machine [1L]

Minimization of incompletely specified machine – Merger Graph, Merger Table, Compatibility Graph [2L]

Lossless and Lossy Machine – Testing Table, Testing Graph [2L]

Module-3:[5L]

Regular Languages, Regular Sets, Regular Expressions, Algebraic Rules for Regular Expressions, Arden's Theorem statement and proof [1L]

Constructing Finite Automata (FA) for given regular expressions, Regular string accepted by FA [2L]

Constructing Regular Expression for a given Finite Automata [1L]

Pumping Lemma of Regular Sets. Closure properties of regular sets [1L]

Module-4:[9L]

Grammar Formalism –

Context Free Grammars, Derivation trees, sentential forms. Rightmost and leftmost derivation of strings, Parse Tree, Ambiguity in context free grammars. [1L]

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Minimization of Context Free Grammars.[1L], Removal of null and unit production[1L]

Chomsky normal form and Greibach normal form.[1L]

Pumping Lemma for Context Free Languages.[1L]

Enumeration of properties of CFL, Closure property of CFL, Ogden's lemma & its applications[1L],

Regular grammars – right linear and left linear grammars[1L]

Pushdown Automata: Pushdown automata, definition. Introduction to DCFL, DPDA, NCFL, NPDA[1L]

Acceptance of CFL, Acceptance by final state and acceptance by empty state and its equivalence.[1L]
 Equivalence of CFL and PDA, inter-conversion.[1L]

Module-5:[5L]

Turing Machine: Turing Machine, definition, model[1L]

Design of TM, Computable functions [1L] Church's hypothesis, counter machine [1L]

Types of Turing machines[1L]

Universal Turing Machine, Halting problem[1L]

Textbook:

1. "Introduction to Automata Theory Language and Computation", Hopcroft H.E. and Ullman J.D., Pearson Education.
2. Peter Linz, "An Introduction to Formal Languages and Automata", Sixth Edition, Jones & Bartlett, 2016. ISBN: 978-9384323219

Reference Books:

1. "Formal Languages and Automata Theory", C.K. Nagpal, Oxford
2. "Switching and Finite Automata Theory", Zvi Kohavi, 2nd Edition., Tata McGraw Hill

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	-	3	-	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	-	-	-	-	-	-	-	-	2	-	2
CO4	2	2	2	3	-	-	-	-	-	-	-	-	2	-	2
CO5	2	2	3	3	-	-	-	-	-	-	-	2	2	2	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Gender, Culture and Development

Course Code: HSMC 402

Contacts: 2:0:0

Total Contact Hours: 24

Credit: 2

Prerequisite: None

Course Outcome(s):

On successful completion of the learning sessions of the course, the learner will be able to:

CO1	Provide an analysis of the location of women in the processes of economic development; to understand what economic development is, the scales or levels at which it occurs, and the centrality of gender at every level.
CO2	Examine theoretical and conceptual frameworks for that analysis.
CO3	Reflect upon linkages between the global economy and the gendered macro and micro process of development and transitions from 'government' to 'governance.'
CO4	Explain the usefulness of a rights based approach to gender justice.
CO5	Provide basis for research, practical action and policy formulation and or evaluating for evaluating directions and strategies for social change from a gender perspective.

Course Content:

Module1:

Introduction to Gender, Definition of Gender, Basic Gender Concepts and Terminology, Exploring Attitudes towards Gender, Social Construction of Gender .[4L]

Module 2:

Gender Roles and Relations, Types of Gender Roles, Gender Roles and Relationships Matrix,

Curriculum for B.Tech Under Autonomy
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Gender-based Division and Valuation of Labour .[6L]

Module 3:

Gender Development Issues , Identifying Gender Issues, Gender Sensitive Language, Gender, Governance and Sustainable Development, Gender and Human Rights, Gender and Mainstreaming. [5L]

Module 4:

Gender-based Violence, The concept of violence, Types of Gender-based violence, The relationship between gender, development and violence, Gender-based violence from a human rights perspective. [5L]

Module5:

Gender and Culture Gender and Film, Gender and Electronic Media, Gender and Advertisement, Gender and Popular Literature. [4L]

Text Books:

1. Beneria, Lourdes. (2004). Gender, Development, and Globalization: Economics as if All People Mattered. Routledge Press. (GDGE)
2. Molyneux and Razavi. (2002). Gender Justice, Development and Rights. Oxford University Press (GJDR or WGD)
3. Visvanathan, Duggan, Wiegiersma and Nisonoff. (2011). The Women, Gender and Development Reader. 2nd Edition. Zed Press (WGD)

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	3	2	-	-	-	-	-	-	-	-	3	-	-
CO2	2	3	2	3	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	-	-	-	-	-	-	-	-	2	-	2
CO4	2	2	2	3	-	-	-	-	-	-	-	-	2	-	2
CO5	2	2	3	3	-	-	-	-	-	-	-	2	2	2	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Probability and Statistics

Course Code: M401

Total Contact Hours: 36

Credit: 3

Prerequisite:

The students to whom this course will be offered must have the concept of (10+2) standard algebra and calculus.

Course Objective(s):

The purpose of this course is

1. Recall the distinctive principles of probability and statistics.
2. Understand the theoretical workings of theory of probability and tests of hypotheses.
3. Apply statistical methods to compute and explain point estimators and interval estimators for mean, variance and proportion.
4. Analyze statistical data from engineering experiments.

Course Outcome(s):

On successful completion of the learning sessions of the course, the learner will be able to:

CO1	Understand the basic concepts of Probability and Random Variables to Explain or Illustrate and Identify problems where students can Apply the concept appropriately to Solve them.
CO2	Understand the basic concepts of Two dimensional random variables to Explain or Illustrate and Identify problems where students can Apply the concept appropriately to Solve them.
CO3	Understand the basic concepts of Sampling Distribution to Explain or Illustrate and Identify problems where students can Apply the concept appropriately to Solve them.
CO4	Understand the basic concepts of Parameter Estimation to Explain or Illustrate and Identify problems where students can Apply the concept appropriately to Solve them.
CO5	Understand the basic concepts of Testing of Hypotheses to Explain or Illustrate and Identify problems where students can Apply the concept appropriately to Solve them.

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Course Content**Module 1: (Probability and Random Variables) (15L)**

The axioms of probability, Conditional probability, Baye's theorem, Discrete and continuous random variables, probability mass, probability density and cumulative distribution functions, mathematical expectation, Moments, Moment generating functions, Binomial, Poisson, Geometric, Uniform, Exponential and Normal distributions.

Module 2: (Two dimensional random variables) (5 L)

Joint distributions, Marginal and conditional distributions, Covariance, Correlation and linear regression, Transformation of random variables, Central limit theorem (for independent and identically distributed random variables).

Module 3: (Sampling Distribution) (3 L)

Distributions of the sample mean and the sample variance for a normal population, Chi-Square, t and F distributions, problems

Module 4: (Parameter Estimation) (4 L)

Unbiasedness, consistency, the method of moments and the method of maximum likelihood estimation, confidence intervals for parameters in one sample and two sample problems of normal populations, confidence intervals for proportions, problems

Module 5: (Testing of Hypotheses) (9 L)

Null and alternative hypotheses, the critical and acceptance regions, two types of error, power of the test, the most powerful test and Neyman-Pearson Fundamental Lemma, tests for one sample and two sample problems for normal populations, tests for proportions, Chi square goodness of fit test and its applications, problems.

Text Books:

1. Sheldon M. Ross, "Introduction to Probability and Statistics for Engineers and Scientists", Academic Press, (2009).
2. D. C. Montgomery and G.C. Runger, "Applied Statistics and Probability for Engineers", 5th edition, John Wiley & Sons, (2009)..

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3. Robert H. Shumway and David S. Stoffer, “Time Series Analysis and Its Applications with R Examples”, Third edition, Springer Texts in Statistics, (2006).

Reference Books:

1. N. G. Das: Statistical Methods, TMH.
2. Sancheti , D. S. & Kapoor , V.K. : Statistics Theory , Method & Application, Sultan chand& sons , New Delhi
3. N.K.Dutta (2004). Fundamentals of Biostatistics, Kanishka Publishers.

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	3	2	-	-	-	-	-	-	-	-	3	-	-
CO2	2	3	2	3	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	-	-	-	-	-	-	-	-	2	-	2
CO4	2	2	2	3	-	-	-	-	-	-	-	-	2	-	2
CO5	2	2	3	3	-	-	-	-	-	-	-	2	2	2	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Practical

Course Name: Computer Organization and Architecture Lab

Course Code: PCC-CS491

Contact: 0:0:3

Credits:1.5

Prerequisites: Digital Electronics

Course Objective:

The objective of the course is to make the students able to

1. Interpret and use proper method in an appropriate platform to design and to solve problems
2. Use the codes and number systems converting circuits and Compare different types of logic families which are the basic unit of different types of logic gates to design the problem using modern tools
3. Outline different types of digital electronic circuit using various mapping and logical tools and summarize the techniques to prepare the most simplified circuit using various mapping and mathematical methods.
4. Apply the knowledge of digital electronic circuits to design memory and ALU.
5. Interpret the result of the experiments, prepare laboratory reports based on observed output and analyse it.

Course Outcomes:

After completion of the course students will be able to

CO1	Interpret and use proper method in an appropriate platform to design and to solve problems related to Mathematics and Engineering field leading to lifelong learning.
CO2	Use the codes and number systems converting circuits and Compare different types of logic families which are the basic unit of different types of logic gates to design the problem using modern tools for solving complex engineering problems.

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B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

CO3	Outline different types of digital electronic circuit using various mapping and logical tools and summarize the techniques to prepare the most simplified circuit using various mapping and mathematical methods for solving the problem as a professional engineering practice as a team.
CO4	Apply the knowledge of digital electronic circuits to design memory and ALU and analyse the same to solve engineering-related computational problems as a team.
CO5	Interpret the result of the experiments, prepare laboratory reports based on observed output and analyse it to validate professional ethics and responsibilities and norms of the engineering practice.

Course Content:

Module-1: Study and verify the truth table of

- a) Logic gates
- b) Universal gates

Module-2: Study and realize the logic gates from universal gates

Module-3: Study and verify the truth table of

- a) Half Adder
- b) Full Adder

Module-4: Study and verify the truth table of

- a) Half Subtractor
- b) Full Subtractor

Module-5: Study and verify the truth table of

- a) Multiplexer
- b) De-Multiplexer

Module-6: Study and verify the truth table of

- a) Encoder
- b) Decoder

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Module-7: Study and verify the truth table of

- a) 7 segment Decoder
- b) Comparator (1 bit, 2 bits)
- c) Carry look ahead adder

Module-8: Study and verify the truth table of

- a) SR flip-flop
- b) D flip-flop
- c) JK flip-flop
- d) T flip-flop

Module-9: Study, design and verify the truth table of 4 bits Serial In-Parallel Out Shift Registers

Module-10: Study, design and verify the 4 bit Synchronous or Asynchronous Counter using JK flip-flop

Module-11: Design a composite ALU for multi bit arithmetic operation

Module-12: Design of RAM

Textbook:

1. M. Morris Mano, Digital Design, Prentice-Hall
2. Digital Circuits and Design, S. Salivahanan and S. Arivazhagan, McGraw-Hill Education

Reference book:

1. Hayes J. P., “Computer Architecture & Organisation” , McGraw Hill,
2. William Stallings, Computer Organization and Architecture: Designing for Performance

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	-	3	3	3	3

Curriculum for B.Tech Under Autonomy

B.Tech. – Computer Science & Technology

L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

CO2	3	3	3	3	3	-	-	-	-	-	-	-	3	3	3
CO3	3	3	3	3	-	-	-	3	3	-	-	-	3	3	3
CO4	3	3	3	3	-	-	-	3	3	-	-	-	3	3	3
CO5	-	-	-	-	-	-	-	3	-	3	-	-	3	2	3
CO	3	3	3	3	3	-	-	3	3	3	-	3	3	3	3

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B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Design & Analysis of Algorithm Lab

Course Code: PCC-CS492

Contact: 0:0:2

Credits: 1.5

Prerequisites: Programming Knowledge

Objective of the course:

13. Design and implement efficient algorithms for a specified application.
14. Strengthen the ability to identify and apply the suitable algorithm for the given real world problem.

Course Outcomes:

After completion of the course students will be able to

CO1	To identify and prove the correctness and analyze the running time of the basic algorithms for those classic problems in various domains.
CO2	To understand and illustrate methods for analyzing the efficiency and correctness of algorithms (such as exchange arguments, recurrence, induction, and average case analysis)
CO3	To analyze and design algorithms using the dynamic programming, greedy method, Backtracking, Branch and Bound strategy, and recite algorithms that employ this strategy.
CO4	To understand, compare, contrast, and choose appropriate implementation of the algorithmic design techniques to present an algorithm that solves a given problem.
CO5	To Identify and analyze criteria and specifications appropriate to new problems.

Course Content:

- A.** Implementation of various Divide & Conquer Methods; viz. Matrix Multiplication.
- B.** Implementations of various Dynamic Programming Methods, viz. Matrix Chain Multiplication Method, Travelling Salesman Problem etc.
- C.** Implementations of various Brunch & Bound Techniques, viz.
- D.** Implementations of various Backtracking Methods, viz. n-Queen Problem.
- E.** Implementations of Greedy Method, viz. Fractional Knapsack Problem, Job Sequencing Problem etc.
- F.** Implementations of String Matching Algorithm viz. Naïve Algorithm, String Matching with Finite Automata etc.
- G.** Implementations of Various Graph Algorithm, viz. Dijkstra's Algorithm, Floyd Algorithm etc.

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

H. Implementation of some Real Life Trendy Problems.

CO – PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
C01	3	3	3	3	-	-	-	-	-	-	2	-	2	2	3
C02	3	3	3	3	-	-	-	-	-	-	-	-	2	3	3
C03	3	3	3	3	-	-	-	-	-	-	2	-	3	3	3
C04	3	3	3	3	-	-	-	-	-	-	2	-	3	2	3
C05	3	3	3	3	-	-	-	-	-	-	2	-	3	3	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Operating Systems Lab

Course Code: PCC-CS493

Contact: 0:0:3

Total Contact Hours: 36

Credits: 1.5

Prerequisites: Operating Systems

Course Objectives:

The objective of the course is to make the students able to -

- 1 . Understand and execute basic commands of shell script
- 2 . Implement various CPU scheduling Algorithms
3. Implement process creation and inter process communication
- 4 Implement Deadlock Avoidance and detection Algorithms
- 5 Implement Page replacement , file organization and file allocation strategies

Course Outcomes:

After completion of the course students will be able to

CO1	Analyze and simulate CPU Scheduling Algorithms like FCFS, Round Robin, SJF, and Priority.
CO2	Understand the concepts of deadlock in operating systems and implement them in
CO3	Create process creation and implement inter process communication
CO4	Analyze the performance of the various page replacement schemes
CO5	Understand the concepts of file organization and file allocation strategies

Course Content:

Preliminaries of Operating System: 6P

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Basics of UNIX Commands, managing users, managing systems, file managements, useful commands, Shell scripting : shell syntax, executing shell scripts.

CPU Scheduling 12P

1. FCFS, SJF, SRTF, RR, PRIORITY algorithms. Display/Gantt chart and Compute and print the average waiting time and average turnaround time.

Process Communication : 3P

Inter-Process Communication (using shared memory, pipes or message queues).

Synchronization : 6P

Producer-Consumer problem using semaphores (using UNIX system calls)

Memory Management and File : 9P

Memory management schemes (paging and segmentation) page replacement algorithm, Memory allocation schemes (First fit, Best fit and Worst fit)

Text book:

1. Mukesh Singhal and Niranjana Shivaratri, Advanced Concepts in Operating Systems, McGraw-Hill.

Reference Books:

1. Nancy Lynch, Distributed Algorithms, Morgan Kaufmann.

CO – PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO2	3	3	3	3	-	-	-	-	-	-	-	-	3	2	3
CO3	3	3	3	3	-	-	-	-	-	-	-	3	2	3	2
CO4	3	3	3	3	-	-	-	-	-	-	-	-	3	2	2
CO5	3	3	3	3	-	-	-	-	2	3	-	-	2	2	2
CO	3	3	3	3	-	-	-	-	3	3	-	3	3	3	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: IT Workshop Lab(C++/PYTHON)

Course Code: ESC491

Contact: 3:0:0

Credits: 1.5

Prerequisites: Basic Object Oriented /C++/Python programming knowledge

Objective of the course:

The objective of the course is to make the students able to –

- Develop programs with reusability
- Develop programs for file handling
- Handle exceptions in programming
- Develop applications for a range of problems using object-oriented programming techniques

Course Outcomes:

After completion of the course students will be able to

CO1	Articulate the basic Object-Oriented Programming concepts such as class and objects as used in Python/C++.
CO2	Illustrate exceptions in Python along with creating their own based on the problems or analyze various problems based on the knowledge of classes and abstractions in C++.
CO3	Understand the concept of multithreading in Python or write various programs while applying the concept of inheritance in C++.
CO4	Use databases in Python or articulate the Object-Oriented Programming concepts such as polymorphism in C++.
CO5	Execute programs while applying the concepts of database connections, transactions etc. in Python or illustrate exceptions in C++ along with creating their own based on the problems.

Course Content:

Module-I: Python Object Oriented

- Overview of OOP
- Creating Classes and Objects
- Accessing attributes
- Built-In Class Attributes
- Destroying Objects

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Or Module-I: Introduction to C++

- Fundamental idea of OOP
- Basic terminologies in C++
- Simple programs on C++

Module-II: Python Exceptions Handling

- What is Exception?
- Handling an exception
- try....except...else
- try-finally clause
- Argument of an Exception
- Python Standard Exceptions
- Raising an exceptions
- User-Defined Exceptions

Or Module-II: Classes and abstractions in C++

- Class structure
- Class objects
- Class scope
- this pointer
- Friends to a class
- Static class members
- Constant member functions
- Constructors and Destructors
- Dynamic creation and destruction of objects
- Data abstraction
- ADT and information hiding.

Module-III: Python Multithreaded Programming

- What is multithreading?
- Starting a New Thread
- The Threading Module
- Synchronizing Threads
- Multithreaded Priority Queue

Or Module-III: Inheritance and Virtual Functions and Polymorphism in C++

- Defining a class hierarchy
- Different forms of inheritance
- Defining the Base and Derived classes
- Access to the base class members
- Base and Derived class construction
- Destructors
- Virtual base class
- Static and Dynamic binding

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- virtual functions
- Dynamic binding through virtual functions
- Virtual function call mechanism
- Pure virtual functions
- Abstract classes
- Implications of polymorphic use of classes
- Virtual destructors

Module-IV: Using Databases in Python

- Python MySQL Database Access
- Install the MySQLdb and other Packages
- Create Database Connection
- CREATE, INSERT, READ, UPDATE and DELETE Operation
- DML and DDL Operation with Databases
- Performing Transactions
- Handling Database Errors

Or Module-IV: Exception Handling in C++

- Benefits of exception handling
- Throwing an exception
- The try block
- Catching an exception
- Exception objects
- Exception specifications
- Stack unwinding
- Rethrowing an exception
- Catching all exceptions.

Text books:

1. Gowrishankar S, Veena A, “Introduction to Python Programming”, 1st Edition, CRC Press/Taylor & Francis, 2018.
2. E. Balagurusamy, “Object-Oriented Programming with C++”, 8th Edition, Mc Graw Hill, 2020.

Reference books:

3. Wesley J Chun, “Core Python Applications Programming”, 3rd Edition, Pearson Education India, 2015.
4. Miguel Grinberg, “Flask Web Development: Developing Web Applications with Python”, 2nd Edition, O'Reilly Media, 2018.

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MANDATORY COURSE

Course Name: Learning an Art Form [vocal or instrumental, dance, painting, clay modelling, etc.] OR Environmental Protection Initiatives

Course Code: MC 481

Credits: 3

Total Lectures: 36

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3 rd Year 1 st Semester: 5 th Semester								
Sl. No.	Category	Course Code	Course Title	Hours per week				Credits
				L	T	P	Total	
A. THEORY								
1	Humanities and Social Sciences including Management courses	HSMC 505	Principles of Management	2	0	0	2	2
2	Program Core Course	PCC-CS501	Compiler Design	3	0	0	3	3
3	Program Core Course	PCC-CS502	Database Management Systems	3	0	0	3	3
4	Program Core Course	PCC-CS503	Object Oriented Programming using Java	3	0	0	3	3
5	Professional Elective courses	PEC-CS-T-501	Microprocessors and Microcontrollers	3	0		3	3
		PEC-CS-S-501	Advanced Computer Architecture					
		PEC-CS-D-501	Artificial Intelligence					
		PEC-CS-A-501	Computer Graphics					
B. PRACTICAL								
7	Program Core Course	PCC-CS591	Compiler Design Lab	0	0	3	3	1.5
8	Program Core Course	PCC-CS592	Database Management Systems Lab	0	0	3	3	1

								5
9	Program Core Course	PCC-CS593	Object Oriented Programming using Java Lab	0	0	3	3	1 · 5
1 0	Professional Elective courses	PEC-CS-T-591	Microprocessors and Microcontrollers Lab	0	0	3	3	1 · 5
		PEC-CS-S-591	Advanced Computer Architecture Lab					
		PEC-CS-D-591	Artificial Intelligence Lab					
		PEC-CS-A-591	Computer Graphics Lab					
1 1	PROJECT	PR 591	Minor Project I	0	0	3	2	1
1 2	PROJECT	PR 592	Skill Development V: Soft Skill & Aptitude-II	1	0	0	1	0 · 5
C. MANDATORY ACTIVITIES / COURSES								
1 4	MC	MC 501	Constitution of India	3	0	0	3	3Units
	TOTAL CREDIT WITHOUT MOOCS COURSES							21.5
D. MOOCS COURSES**								
1 5	MOOCS COURSES	HM501	MOOCS COURSE-III	3	1	0	4	4
TOTAL CREDIT WITH MOOCS COURSES								25.5

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Paper name: Principles of Management

Paper Code: HSMC 501

Credits: 2

No. of lectures: 24

Course Outcome(s):

After completion of the course students will be able to

CO1	To recall and identify the relevance of management concepts
CO2	To apply management techniques for meeting current and future management challenges faced by the organization
CO3	To compare the management theories and models critically to solve real-life problems in an organization.
CO4	To apply principles of management in order to execute the role as a manager in an organization
CO5	Analysis Quality control and Understand the Functions of Marketing .

Course Content:

Module-1: Management Concepts: Definition, roles, functions and importance of Management, Evolution of Management thought-contribution made by Taylor, Fayol, Gilbreth, Elton Mayo, McGregor, Maslow **(4L)**

Module - 2: Planning and Control: Planning: Nature and importance of planning, -types of planning, Levels of planning - The Planning Process. –MBO, SWOT analysis, McKinsey's 7S Approach.

Organizing for decision making: Nature of organizing, span of control, Organizational structure – line and staff authority.

Basic control process -control as a feedback system – Feed Forward Control –Requirements for effective control – control **(4L)**

Module - 3: Group dynamics: Types of groups, characteristics, objectives of Group Dynamics. Leadership: Definition, styles & functions of leadership, qualities for good leadership, Theories of leadership **(4L)**

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Module – 4: Work Study and work measurement: Definition of work study, Method Study Steps, Tools and Techniques used in the Method Study and Work Measurement Time Study: Aim & Objectives, Use of stopwatch procedure in making Time Study. Performance rating, allowances and its types. Calculation of Standard Time. Work sampling (4L)

Module - 5: Marketing Management: Functions of Marketing, Product Planning and development, Promotional Strategy (2L)

Module - 6: Quality management: Quality definition, Statistical quality control, acceptance sampling, Control Charts –Mean chart, range chart, c chart, pchart, np chart, Zero Defects, Quality circles, Kaizen & Six Sigma, ISO -9000 Implementation steps, Total quality management (6L)

Text Books:

1. Essentials of Management, by Harold Koontz & Heinz Weihrich Tata McGraw
2. Production and Operations Management-K.Aswathapa, K. Shridhara Bhat, Himalayan Publishing House

References:

1. Organizational Behavior, by Stephen Robbins Pearson Education, New Delhi
2. New era Management, Daft, 11th Edition, Cengage Learning
3. Principles of Marketing, Kotlar Philip and Armstrong Gary, Pearson publication

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	-	-	2	-	-	-	-	3	-	3	2	3	3	2
CO2	-	-	-	-	-	2	-	3	3	-	3	3	3	-	3
CO3	2	-	-	2	-	-	-	-	2	-	3	3	3	2	-
CO4	3	-	-	-	-	2	-	2	3	-	3	-	2	3	-
CO5	2	2	-	2	-	-	-	-	2	-	3		-	3	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name : Compiler Design
Course Code : PCC-CS501
Contact : 3:0:0
Total Contact Hours : 36
Credits : 3

Prerequisites:

- Mathematics
- Concept of programming languages
- Data structures
- Computer architecture
- Formal languages and automata theory
- Some advanced math might be required if you adventure in code optimization

Objective of the course:

The objective of the course is to make the students able to –

1. Make the student understand the process involved in a compiler
2. Create an overall view of various types of translators, linkers, loaders, and phases of a compiler
3. Understand the concepts of syntax analysis, various types of parsers especially the top down approach
4. Create awareness among students the various types of bottom up parsers,
5. Understand the syntax analysis and, intermediate code generation, type checking, the role of symbol table and its organization, Code generation, machine independent code optimization and instruction scheduling.

Course Outcomes:

After completion of the course students will be able to

CO1	Illustrate the basic concept of compilers and discuss on the components as well as the strengths and weaknesses of various phases of designing a compiler.
CO2	Design and analyze algorithms for syntactic or parsing techniques and semantic analysis of the process of designing compilers.
CO3	Develop the parsers and experiment the knowledge of activation tree, activation record and dynamic storage allocation techniques

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CO4	Construct the intermediate code representations and generation.
CO5	Apply for various optimization techniques for dataflow analysis.

Course Content:

Module I

[7L]

Compilers, Cousins of the Compiler, Analysis-synthesis model, The phases of the compiler. The role of the lexical analyzer, Tokens, Patterns, Lexemes, Input buffering, Specifications of a token, Recognition of tokens, Finite automata, From a regular expression to an NFA, From a regular expression to DFA, Design of a lexical analyzer generator (Lex).

Module II

[10L]

The role of a parser, Context free grammars, Writing a grammar, Top down Parsing, Non recursive Predictive parsing (LL), Bottom up parsing, Handles, Viable prefixes, Operator precedence parsing, LR parsers (SLR, LALR, Canonical LR), Parser generators (YACC), Error Recovery strategies for different parsing techniques. Syntax directed translation: Syntax directed definitions, Construction of syntax trees, Bottom-up evaluation of S-attributed definitions, L-attributed definitions, Bottom-up evaluation of inherited attributes.

Module III

[7L]

Type systems, Specification of a simple type checker, Equivalence of type expressions, Type conversions Source language issues (Activation trees, Control stack, scope of declaration, Binding of names), Symbol tables, dynamic storage allocation techniques.

Module IV

[4L]

Intermediate languages, Graphical representation, Three-address code, Implementation of three address statements (Quadruples, Triples, Indirect triples).

Module V

[8L]

Consideration for Optimization, scope of optimization, local optimization, loop optimization, folding, DAG representation, Flow Graph, Data flow equation, global optimization, redundant sub expression elimination, induction variable elimination, copy propagation, basic blocks & flow graphs, transformation of basic blocks, DAG representation of basic blocks, peephole optimization Object code forms, machine dependent code optimization, register allocation and assignment, generic code generation algorithms, DAG for register allocation.

1. Alfred Aho, V. Ravi Sethi, D. Jeffery Ullman, "Compilers Principles, Techniques and Tools", Addison Wesley, 2nd edition
2. Holub Allen. Compiler Design in C, PHI, 1993.

1. Chattopadhyay, Santanu. Compiler Design. PHI Learning Pvt. Ltd., 2005
2. Tremblay and Sorenson Compiler Writing-McgrawHill International

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B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Paper Name: Database Management System

Paper Code: PCC-CS502

Contact (Periods/Week): 3:0:0

Total Credit Hours: 36

Credit: 3

Prerequisite:

1. Logic of programming language
2. Basic concepts of data structure and algorithms

Course Objectives(s):

1. To Understand and Describe the basic concepts and utility of Database management system, different data models of Database management system.
2. To Design an Entity Relationship (E-R) Diagram and relational model for any kind of real-life application and able to Apply relational algebra operations, SQL, Neo4j for solving query.
3. To Analyze and Create the relational database for any real-life applications based on normalization.
4. To Apply the query optimization techniques, different file organization techniques and determine whether the transaction satisfies the ACID properties.

Course Outcome(s):

On completion of the course students will be able to:

CO1	To Understand and Describe the basic concepts and utility of Database management system, different data models of Database management system.
CO2	To Design an Entity Relationship (E-R) Diagram and relational model for any kind of real-life application and able to Apply relational algebra operations, SQL, Neo4j for solving query.
CO3	To Analyze and Create the relational database for any real-life applications based on normalization.
CO4	To Apply the query optimization techniques, different file organization techniques and determine whether the transaction satisfies the ACID properties.

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CO5	Explore DBMS based ideas through developing software programs with adequate documentation in collaborative environment for successfully carrying out projects on DBMS Problems and investigate their effectiveness by analyzing the performances using proper techniques and tools and Assess the limitations of solutions underscoring utilitarian importance for further explorations leading towards lifelong learning.
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Course Content:

Module 1:

Introduction [3L]

Concept & Overview of DBMS, Data Models, Database Languages, Database Administrator, Database Users, Three Schema architecture of DBMS.

Module 2:

Entity-Relationship and Relational Database Model [11L]

Basic concepts, Design Issues, Mapping Constraints, Keys, Entity-Relationship Diagram, Weak Entity Sets, Extended E-R features, case study on E-R Model. Structure of relational Databases, Relational Algebra, Relational Calculus, Extended Relational Algebra Operations, Views, Modifications of the Database.

Module 3:

Graph Based Model [4L]

Concept of graph-based model, difference between relational model and graph-based model, application, overview of Neo4j CQL.

Module 4:

SQL and Integrity Constraints [6L]

Concept of DDL, DML, DCL. Basic Structure, Set operations, Aggregate Functions, Null Values, Domain Constraints, Referential Integrity Constraints, assertions, views, Nested Subqueries, Database security application development using SQL, Stored procedures and triggers.

Module 5:

Relational Database Design [8L]

Functional Dependency, Different anomalies in designing a Database., Normalization using functional dependencies, Decomposition, Boyce-Codd Normal Form, 3NF, Normalization using multi-valued dependencies, 4NF, 5NF, Case Study

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Module 6:**Internals of RDBMS [8L]**

Physical data structures, Query optimization: join algorithm, statistics and cost bas optimization. Transaction processing, Concurrency control and Recovery Management: transaction model properties, state serializability, lock base protocols; two phase locking, Dead Lock handling.

Module 7:**File Organization & Index Structures [3L]**

File & Record Concept, Fixed and Variable sized Records, Types of Single-Level Index (primary, secondary, clustering), Multilevel Indexes

Text Books:

1. Henry F. Korth and Silberschatz Abraham, “Database System Concepts”, Mc.Graw Hill.
2. Elmasri Ramez and Navathe Shamkant, “Fundamentals of Database Systems”, Benjamin Cummings Publishing. Company.

Reference:

1. “Fundamentals of Database Systems”, Ramez Elmasri, Shamkant B. Navathe, Addison Wesley Publishing.
2. Ramakrishnan: Database Management System, McGraw-Hill

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	-	3	-	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	-	-	-	-	-	-	-	-	2	-	2
CO4	2	2	2	3	-	-	-	-	-	-	-	-	2	-	2
CO5	2	2	3	3	-	-	-	-	-	-	-	2	2	2	3

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name : Object Oriented Programming using Java
Course Code : PCC-CS503
Contact : 3:0:0
Total Contact Hours : 36
Credits : 3

Prerequisites:

- Computer Fundamentals
- Basic understanding of Computer Programming and related Programming Paradigms
- Problem Solving Techniques with proper logic Implementation.

Objective of the course:

The objective of the course is to make the students able to –

- It allows to map with real world Object (Object orientation) rather than action(Procedure) that comes to produce software as separated code modules which rise up decoupling and increases code re-usability.
- It demonstrates that how can you change the implementation of an object without affecting any other code by increasing data security and protecting unwanted data access. (Encapsulation).
- It allows you to have many different functions, all with the same name, all doing the same job, but depending upon different data. (Polymorphism).
- It guides you to write generic code: which will work with a range of data, so you don't have to write basic stuff over, and over again. (Generics).
- It lets you write a set of functions, then expand them in different direction without changing or copying them in any way. (Inheritance).

Course Outcomes:

After completion of the course students will be able to

CO1	Design the process of interaction between Objects, classes & methods w.r.t. Object Oriented Programming.
CO2	Acquire a basic knowledge of Object Orientation with different properties as well as different features of Java.
CO3	Analyze various activities of different string handling functions with various I/O operations.

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CO4	Discuss basic code reusability feature w.r.t. Inheritance, Package and Interface.
CO5	Implement Exception handling, Multithreading and Applet (Web program in java) programming concept in Java.

Course Content:

Module 1: Introduction

[2L]

Object Oriented Analysis (OOA) & Object Oriented Design (OOD) - Concepts of object oriented programming language, Relationships among objects and classes-Generalization, Specialization, Aggregation, Association, Composition, links, Meta-class. [1L]; Object Oriented Programming concepts - Difference between Java and C++; Different features of Java [1L];

Module 2: Java Basics

[10L]

Basic concepts of java programming - Advantages of java, Byte-code & JVM, Data types, Different types of Variables.[1L] ;Java Operators & Control statements [1L]; Java loops. [1L]; Array.[1L] ;Creation of class, object, method. [1L]; Constructor- Definition, Usage of Constructor, Different types of Constructor.[1L]; finalize method and garbage collection, Method & Constructor overloading. [1L]; this keyword, use of objects as parameter & methods returning objects.[1L]; Call by value & call by reference. [1L]; Static variables & methods.Nested & inner classes.[1L].

Module 3: Basic String handling & I/O

[5L]

Basic string handling concepts- Concept of mutable and immutable string, Methods of String class- charAt(), compareTo(), equals(), equalsIgnoreCase(), indexOf(), length() , substring(). [1L]; toCharArray(), toLowerCase(), toString(), toUpperCase() , trim() , valueOf() methods, Methods of String buffer class- append(), capacity(), charAt(), delete(), deleteCharAt(). [1L];ensureCapacity(), getChars(), indexOf(), insert(), length(), setCharAt(), setLength(), substring(), toString(). [1L] ;Command line arguments, basics of I/O operations – keyboard input using BufferedReader [1L] ; Scanner class in Java I/O operation [1L];

Module 4: Inheritance and Java Packages

[8L]

Inheritance - Definition, Advantages, Different types of inheritance and their implementation. [1L] ;Super and final keywords, super() method. [1L]; Method overriding, Dynamic method dispatch.[1L]; Abstract classes & methods.[1L]; Interface - Definition, Use of Interface.[1L]; Multiple inheritance by using Interface.[1L] ;Java Packages -Definition, Creation of packages. [1L]; Java Access Modifiers - public, private, default and protected,Importing packages, member access for packages. [1L]

Module 5: Exception handling, Multithreading and Applet Programming

[11L]

Exception handling - Basics, different types of exception classes.Difference between Checked & Unchecked Exception.[1L]; Try & catch related case studies.[1L]; Throw, throws & finally.

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[1L]; Creation of user defined exception. [1L]; Multithreading - Basics, main thread [1L]; Thread life cycle.[1L]; Creation of multiple threads-yield(), suspend(), sleep(n), resume(), wait(), notify(), join(), isAlive().[1L] ;Thread priorities, thread synchronization.[1L];Interthread communication, deadlocks for threads[1L]; Applet Programming - Basics, applet life cycle, difference between application & applet programming[1L]; Parameter passing in applets. [1L]

Textbooks:

1. Herbert Schildt – "Java: The Complete Reference " – 9th Ed. – TMH
2. E. Balagurusamy – " Programming With Java: A Primer " – 3rd Ed. – TMH.

Reference Books:

1. R.K Das – " Core Java for Beginners " – VIKAS PUBLISHING.
2. Rambaugh, James Michael, Blaha – " Object Oriented Modelling and Design " – Prentice Hall, India.

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	3	2	-	-	-	-	-	-	-	-	-	2	3
CO2	3	2	3	1	-	-	-	-	-	-	-	-	3	2	-
CO3	3	3	2	3	-	-	2	-	1	-	-	-	2	-	3
CO4	2	-	2	2	-	-	-	-	-	-	-	-	3	2	-
CO5	2	-	3	1	2	-	-	-	2	-	2	-	-	2	3

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B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Microprocessors & Microcontrollers

Course Code: PEC-CS-T-501

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Perquisites: Digital Electronics

Course Objective:

The objective of the course is to make the students able to

1. Understand the architecture of 8085 microprocessors and its address bus, data bus, interrupt, I/O device interfacing and apply this knowledge in designing solution to solve real life problems.
2. Summarize the working principle of 8085 microprocessors, addressing modes, counter and time delays, stack and subroutine, timing diagram of the instructions.
3. Identify and understand the 8086 microprocessors and its architecture, pin details, addressing modes, interrupts, instruction set, examples of simple assembly language, memory interfacing with 8086.
4. Understand the working principles of 8051 microcontrollers and its architecture, pin details, addressing modes, instruction set, examples of simple assembly language.
5. Understand the concept of AVR Family architecture, register File, ALU, memory access and instruction execution. I/O memory, EEPROM. I/O ports, timers. UART, and Interrupt structure.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the architecture of 8085 microprocessors and its address bus, data bus, interrupt, I/O device interfacing and apply this knowledge in designing solutions for real life engineering problems.
CO2	Summarize the working principle of 8085 microprocessors, addressing modes, counter and time delays, stack and subroutine, timing diagram of the instructions and use this knowledge for designing and implementing mathematical and engineering problems leading to lifelong learning.

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CO3	Identify and understand the concept of 8086 microprocessors and its architecture, pin details, addressing modes, interrupts, instruction set, examples of simple assembly language, memory interfacing with 8086 to solve engineering problems.
CO4	Understand the working principles of 8051 microcontrollers and its architecture, pin details, addressing modes, instruction set, examples of simple assembly language and apply this knowledge for developing an approach by means of existing and new methods as a team work.
CO5	Understand the concept of AVR Family architecture, register File, ALU, memory access and instruction execution. I/O memory, EEPROM. I/O ports, timers. UART, Interrupt Structure and design an optimized model for building a new solution as a professional engineering practice as a team.

Course Content:

Module -I:

9

Introduction to Microcomputer based system. [1L] History of evolution of Microprocessors and Microcontrollers and their advantages and disadvantages. [1L] Architecture of 8085 Microprocessor, Pin description of 8085. [2L] Address/data bus De- multiplexing, Status Signals and the control signals. [1L] Interrupts of 8085 processors (software and hardware) [2L] I/O Device Interfacing - I/O Mapped I/O and Memory Mapped I/O, Memory interfacing with 8085 [2L]

Module -II:

9

Instruction set of 8085 microprocessors, Addressing modes. [3L] Assembly language programming with examples, Counter and Time Delays, Stack and Subroutine. [4L] Timing diagram of the instructions (a few examples) [2L]

Module III:

7

The 8086 microprocessor- Architecture, Pin Details, addressing modes, Interrupts [3L] Instruction set, Examples of Simple Assembly Language [2L], Memory interfacing with 8086 [2L]

Module -IV:

6

Introduction to 8051 Microcontroller – Architecture, Pin Details [3L], Addressing modes, Instruction set, Examples of Simple Assembly Language. [3L]

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Module -V:

5

Introduction, AVR Family architecture[1L], Register File, The ALU[1L], Memory access and Instruction execution. I/O memory, EEPROM. I/O ports [2], Timers.UART. Interrupt Structure[1L]

Text Books:

1. MICROPROCESSOR architecture, programming and Application with 8085 - R. Gaonkar (Penram international Publishing LTD.) [For Module 1 and2]
2. Fundamentals of Microprocessor and Microcontrollers - B. Ram (Paperback) [For Module3]
3. 8051 Microcontroller – K. Ayala (Cengage learning) [For Module4]

Reference Books:

1. B.Ram , Fundamental of Microprocessor and Microcontrollers, DhanpatRai Publications.
2. Intel Corp: The 8085 / 8085A. Microprocessor Book – Intel marketing communication, Wiley interscience publications, 1980.
3. Walter A. Tribel – The 8088 and 8086 Microprocessors, Pearson Education

CO – PO Mapping:

C Os	P O1	PO2	PO3	PO4	PO 5	PO 6	PO 7	PO8	PO9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO3
C O 1	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
C O 2	3	3	3	3	-	-	-	-	-	-	-	3	3	2	3
C O 3	3	3	3	3	-	-	-	-	-	-	-	-	3	3	1
C O 4	3	3	3	3	-	-	-	-	3	-	-	-	3	2	3

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C O 5	3	3	3	3	-	-	-	-	3	2	-	-	3	2	3
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Curriculum for B.Tech Under Autonomy
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Course Name: Advanced Computer Architecture

Course Code: PEC-CS-S-501

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: Computer Organization and Architecture

Course Objective:

The objective of the course is to make the students able to

1. Illustrate the basic concept of computer architecture and its performance measurement, parallel processing, Flynn's classification and Amdahl's law to design real life engineering problem.
2. Summarize the basic concept of pipeline, instruction pipeline, arithmetic pipeline, hazards detection and prevention to design and implement mathematical and engineering problem.
3. Identify the concept of Instruction-Level Parallelism to solve engineering problem.
4. Illustrate and Compare concept of Multiprocessor architecture and parallel architecture.
5. Understand the concept of message passing architecture and interconnection network and design an optimized model

Course Outcomes:

After completion of the course students will be able to

CO1	Illustrate the basic concept of computer architecture and its performance measurement, parallel processing, Flynn's classification and Amdahl's law and apply this knowledge in designing solution for real life engineering problem.
CO2	Summarize the basic concept of pipeline, instruction pipeline, arithmetic pipeline, hazards detection and prevention and use this knowledge for designing and implementing mathematical and engineering problem leading to lifelong learning.
CO3	Identify the concept of Instruction-Level Parallelism to solve engineering problem.

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CO4	Illustrate and Compare concept of Multiprocessor architecture and parallel architecture and apply this knowledge for developing an approach by means of existing and new methods as a team work.
CO5	Understand the concept of message passing architecture and interconnection network and design an optimized model for building a new solution as a professional engineering practice as a team.

Course Content:

Module – 1: [7]

Introduction [3]

Review of basic computer architecture [1], Measuring, reporting and summarizing performance [1], quantitative principles of Computer design [1]

Parallel processing [4]

Parallel processing - concept, architecture [1], Flynn's classification-SISD, SIMD, MISD, MIMD [1], SIMD architecture, MIMD architecture, loosely coupled and tightly coupled computers [1], Amdahl's law [1].

Module-2[8]

Pipelining

Introduction of pipelining, classification of pipeline [1], Instruction pipeline [1], Arithmetic pipeline [1], hazard detection and resolution in pipeline [3], dynamic pipeline [1], reservation table and case study [2].

Module-3[6]

Instruction-Level Parallelism

Instruction-Level Parallelism: Basic Concepts [2L], Techniques for Increasing ILP, Superscalar, Super Pipelined and VLIW Processor Architectures [2L], Array Processor [1], Vector Processors [1L]

Module -4[8]

Multiprocessor architecture

Shared memory architecture - classification, bus based symmetric multiprocessors [1], cache coherence methods, snooping protocols, directory based protocols [1], Centralized and Shared-memory architecture: synchronization[1L]

Introduction to Parallel Architecture-Different Classification scheme, Performance of Parallel Computers, PRAM model (EREW, CREW, CRCW) [3L], Program flow mechanisms –Control flow, data-driven and demand driven [2]

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Module -5[7]

Message passing architecture [4]

Basic concept, Routing in message passing network [1], switching mechanisms in message passing [1], Examples of message passing architecture [1], Message passing vs Shared memory architectures [1]

Interconnection Network [3]

Interconnection Network- taxonomy, bus based dynamic interconnection networks [1], switch-based interconnection networks [1], static interconnection networks [1], performance analysis [1L]

Text Books:

1. J. L. Hennessey and D. A. Patterson: Computer Architecture: A Quantitative Approach, 5th edition, Morgan Kaufmann, 2012.
2. K. Hwang and F. A. Briggs: Computer Architecture and Parallel Processing, Tata McGraw Hill, New Delhi.

Reference Books:

1. Tse-yun Feng, A Survey of Interconnection Networks, IEEE, 1981.
2. Selim G. Akl, The Design and Analysis of Parallel Algorithms, Prentice-Hall, 1989.

CO–PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO2	3	3	3	3	-	-	-	-	-	-	-	3	3	2	3
CO3	3	3	3	3	-	-	-	-	-	-	-	-	3	3	1
CO4	3	3	3	3	-	-	-	-	3	-	-	-	3	2	3
CO5	3	3	3	3	-	-	-	-	3	2	-	-	3	2	3
CO	3	3	3	3	-	-	-	-	3	2	-	3	3	3	3

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Artificial Intelligence

Course Code: PEC-CS-D-501

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: Data Structure, Design and Analysis of Algorithms, Statistics

Course Objectives:

The objective of the course is to enable students to

1. Comprehend the fundamental concepts of Knowledge Representation and Inferencing in Artificial Intelligence and its utilitarian importance in current technological context
2. Formulate a problem as State-Space Exploration Framework or an Inferencing Framework of Artificial Intelligence.
3. Use the strategies of AI-Heuristics to find acceptable solutions avoiding brute-force techniques.
4. Design AI-Frameworks for Inferencing based on knowledge base.
5. Analyze the effectiveness of AI-Inferencing Model in offering solutions to the respective problem.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand and explain the fundamental concepts of Knowledge Representation and Inferencing in Artificial Intelligence and its utilitarian importance in current technological context for further exploration leading towards lifelong learning.
CO2	Identify and formulate an engineering problem primarily to fit a State-Space Exploration Framework or an Inferencing Model/Agent Design Framework within the scope of Artificial Intelligence paradigm.
CO3	Explore relevant literature and apply the concept of Heuristic Techniques of Artificial Intelligence to solve problems.
CO4	Develop Inferencing Models for proposing solutions to the problems of Artificial Intelligence.
CO5	Implement Inferencing Models of Artificial Intelligence through developing feasible algorithms and investigate their effectiveness by analyzing their performances in solving the relevant problems.

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Course Content:

MODULE-1: Introduction to Artificial Intelligence [1L]

Basic Concepts, History of Artificial Intelligence, Architecture of an Artificial Intelligent Agent, Applications of Artificial Intelligence

MODULE-2: Artificial Intelligence Problem Formulation as State-Space Exploration Problem for Goal Searching [5L]

Basic Concepts, State-Space Exploration Formulation for Water Jug Problem, Missionaries and Cannibals Problems, Farmer-Wolf-Goat-Cabbage Problem, 8-Puzzle Problem, Constraint Satisfaction Problem and Production System for Goal Searching.

Blind Search Techniques for Goal Searching: Breadth First Search, Depth First Search, Depth Limited Search, Iterative Deepening Search, Uniform Cost Search, Bi-directional Search.

MODULE-3: Heuristic Techniques for Goal Searching [8L]

Basic Concepts of Heuristic Techniques and Properties of Heuristic Functions, Hill Climbing Search, Best First Search, A* Search, Memory-bounded heuristic search: Iterative-deepening A* Search, Recursive Best First Search, Simplified Memory Bounded A* Search.

Simulated Annealing Based Stochastic Search, Genetic Algorithm Based Evolutionary Search, Ant Colony Optimization, Particle Swarm Optimization.

MODULE-4: Adversarial Search for Game Playing [2L]

Basic Concepts, Minimax Search, Alpha-Beta Pruning.

MODULE-5: Knowledge Representation and Inference using Propositional Logic and Predicate Logic [5L]

Propositional Logic: Knowledge Representation and Inference using Propositional Logic

Predicate Logic: Knowledge Representation, Inference and Answer Extraction using First Order Predicate Logic

MODULE-6: Slot-and-Filler Structure for Knowledge Representation [2L]

Weak Slot-and-Filler Structure for Knowledge Representation: Semantic Nets and Frames.

Strong Slot-and-Filler Structure for Knowledge Representation: Conceptual Dependency and Script.

MODULE-7: Reasoning under Uncertainty [5L]

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Bayesian Inferencing and Bayesian Belief Network, Dempster-Shafer Theory, Overview of Fuzzy Logic and Inferencing, Overview of Hidden Markov Model.

MODULE-7: Planning [5L]

Basic Concepts, Problem of Blocks World, Components of a Planning System, Algorithms for Planning: Goal Stack, Nonlinear Planning Using Constraint Posting, Hierarchical Planning, Algorithms for Planning as State-Space Search, Heuristics for planning, Planning Graphs and GRAPHPLAN Algorithm.

MODULE-7: Introduction to Natural Language Processing [1L]

Basic Concepts, Steps of Natural Language Processing, Morphological, Syntactic and Semantic Analysis, Discourse Integration and Pragmatic Analysis, Applications of Natural Language Processing.

MODULE-8: Introduction to Machine Learning [2L]

Basic concepts of Machine Learning Model, Supervised Learning, Unsupervised Learning, and Reinforced Learning, Overview of Artificial Neural Network

Textbook:

3. Russell, S. and Norvig, P. 2015. Artificial Intelligence - A Modern Approach, 3rd edition, Prentice Hall.
4. Rich, E., Knight, K and Shankar, B. 2009. Artificial Intelligence, 3rd edition, Tata McGrawHill.

Reference Books:

6. Padhy, N.P. 2009. Artificial Intelligence and Intelligent Systems, Oxford University Press.
7. Deepak Khemani, “A First Course in Artificial Intelligence”, McGraw Hill.

CO–PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	3	3	-	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	-	-	-	-	-	-	-	-	2	-	2
CO4	2	2	2	3	-	-	-	-	-	-	-	2	2	-	2
CO5	2	2	3	3	2				-	-	-	2	2	2	3

Curriculum for B.Tech Under Autonomy
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Course Name: Computer Graphics

Course Code: PEC-CS-A-501

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: Mathematics, Computer Fundamentals & Principle of Computer Programming

Course Objectives:

1. Use of the components of a graphics system and become familiar with building approach of graphics system components and algorithms related with them.
2. Understand the basic principles of 2D and 3D computer graphics.
3. Understand of how to scan convert the basic geometrical primitives, how to transform the shapes to fit them as per the picture definition.
4. Understand the mapping from a world coordinates to device coordinates, clipping, and projections.
5. Discuss the application of computer graphics concepts in the development of computer games, information visualization, and business applications.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the fundamental concept of Computer graphics and mathematical knowledge and explain the foundations of computer graphics and different display technology and devices.
CO2	Demonstrate different scan conversion algorithms, drawing algorithms, polygon filling algorithms, curves and surface drawing algorithms, clipping algorithms, surface removal algorithms using graphics tools.
CO3	Understand the basic concept of graphics programming and implement clipping with the comprehension of windows, view-ports in relation to images display on screen.
CO4	Analyze and compare different drawing algorithms, polygon filling algorithms, curves and surface drawing algorithms hidden surface illumination methods

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CO5	Develop the concept of geometric models, mathematical and algorithmic approach necessary for programming computer graphics leading to lifelong learning.
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Course Content:

Module-I: Introduction

4

Introduction: Objective, applications, GKS/PHIGS, normalized co-ordinate system, aspect ratio.

Module-II: Computer Graphics System

4

Graphics System: Vector and raster graphics, various graphics display devices, graphics interactive devices, segmented graphics, attribute table.

Module-III: Computer Graphics System

4

Raster Scan Graphics: Line drawing algorithms, circle/ellipse drawing algorithms, polygon filling algorithms.

Module-IV: Geometric Transformation

4

Geometric Transformation: Homogeneous co-ordinate system, 2D and 3D transformations, projection—orthographic and perspective.

Module-V: Curves and Surfaces

4

Curves and Surfaces: Curve approximation and interpolation, Lagrange, Hermite, Bezier and BSpline curves/surfaces and their properties, curves and surface drawing algorithms.

Module-VI: Curves and Surfaces

4

Geometric modelling: 3D object representation and its criteria, edge/vertex list, constructive solid geometry, wire-frame model, generalized cylinder, finite element methods.

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Module-VII: Viewing and Clipping**4**

Clipping :Window and viewport, 2D and 3D clipping algorithms.

Module-VIII: Hidden Surfaces**4**

Hidden Lines and Hidden Surfaces: Concept of object- and image-space methods, lines and surface removal algorithms.

Module-IX: Illumination and Color models**4**

Intensify, Coloring and Rendering: RGB, YIQ, HLS and HSV models and their conversions, gamma correction, halftoning. Illumination models, polygon mesh shading, transparency, shadow, texture.

Text Books

1. D. Hearn and P. M. Baker: Computer Graphics, 2nd ed. Prentice Hall of India, New Delhi, 1997.
2. W. M. Newman and R. F. Sproull: Principles of Interactive Computer Graphics, McGraw Hill, New Delhi, 1979.

Reference Books

1. F. S. Hill: Computer Graphics, McMillan, New York, 1990.
2. D. P. Mukherjee: Fundamentals of Computer Graphics and Multimedia, Prentice Hall of India, New Delhi, 1999.
3. J. D. Foley et al.: Computer Graphics, 2nd ed., Addison-Wesley, Reading, Mass., 1993.
4. W. K. Giloi: Interactive Computer Graphics: Data Structure, Algorithms, Languages, Prentice Hall, Englewood Cliffs, 1978.

CO – PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO2	3	3	3	3	3	-	-	-	-	-	-	-	3	2	3
CO3	3	3	3	3	-	-	-	-	-	-	-	-	2	3	2
CO4	2	3	2	3	-	-	-	-	-	-	-	-	3	2	3
CO5	3	3	3	3	-	-	-	-	3	2	-	3	3	2	2

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Practical

Course Name : Compiler Design Lab
Course Code : PCC-CS591
Contact : 0:0:3
Total Contact Hours : 36
Credits : 1.5

Objective of the course:

The objective of the course is to make the students able to –

- To provide an Understanding of the language translation peculiarities by designing complete translator for mini language.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the practical approaches of how a compiler works.
CO2	Develop and analyze the role of syntax and semantics of Programming languages in compiler construction.
CO3	Apply the techniques and algorithms used in Compiler Construction in compiler component design.
CO4	Create intermediate code representations and generation.
CO5	Construct different tools in construction of the phases of a compiler for the mini language.

Course Contents:

Module 1

[13]

- Programming to Scan and Count the number of characters, words, and lines in a file.
- Design a lexical analyzer for the given language. The lexical analyzer should ignore redundant spaces, tabs and new lines, comments etc.
- Implement the lexical analyzer using JLex, flex or other lexical analyzer generating tools.

- Module 2 [13]

- Module 3 [3]

- Module 4 [4]

- Module 5 [3]

- Text Books:**

- ### Reference Books:

- [illegible]

Curriculum for B.Tech Under Autonomy
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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Paper Name: Database Management System Lab

Paper Code: PCC-CS592

Contact: 3P/Week

Total Credit Hours:

Credits: 1.5

Prerequisite:

1. Logic of programming language
2. Basic concepts of data structure and algorithms

Course Outcome(s):

On completion of the course students will be able to:

CO1	Demonstrate and Explain the database management system and different database languages.
CO2	Understand and Apply the SQL queries related to management of data and transaction processing for solving real life problems.
CO3	Explain and Analyze about query processing techniques involved in query optimization.
CO4	Demonstrate and Apply the PL/SQL programming, the concept of Cursor Management, Error Handling, Package and Triggers for solving real life complex problems.
CO5	Design and Assess the commercial database systems.

Course Content:

Module 1

Conceptual Designing using ER Diagrams (Identifying entities, attributes, keys and relationships between entities, cardinalities, generalization, specialization etc.)

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Module 2

Converting ER Model to Relational Model (Represent entities and relationships in Tabular form, Represent attributes as columns, identifying keys) and apply the normalization techniques.

Module 3

Creation of Tables using SQL- Overview of using SQL tool, Data types in SQL, Creating Tables (along with Primary and Foreign keys), Altering Tables and Dropping Tables

Module 4

Practicing DML commands- Insert, Select, Update, Delete

Module 5

Practicing Queries using ANY, ALL, IN, EXISTS, NOT EXISTS, UNION, INTERSECT, CONSTRAINTS etc., Practicing Sub queries (Nested, Correlated) and Joins (Inner, Outer and Equi).

Module 6

Practice Queries using COUNT, SUM, AVG, MAX, MIN, GROUP BY, HAVING, VIEWS Creation and Dropping, Practicing on Triggers - creation of trigger, Insertion using trigger, Deletion using trigger, Updating using trigger

Module 7

Procedures- Creation of Stored Procedures, Execution of Procedure, and Modification of Procedure, PL/SQL, Cursors- Declaring Cursor, Opening Cursor, Fetching the data, closing the cursor.

CO-PO Mapping:

CO#	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	2	2	2	-	-	-	-	-	-	3	3	2
CO2	3	3	3	3	3	1	1	-	-	-	1	1	3	3	3
CO3	3	3	3	3	3	2	1	-	-	1	1	1	2	3	2
CO4	3	3	3	3	3	2	1	-	-	1	1	-	3	3	3
CO5	3	3	3	3	3	2	1	1	-		2	1	3	3	3

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Course Name : Object Oriented Programming using Java Lab
Course Code : PCC-CS593
Contact : 0:0:3
Total Contact Hours : 36
Credits : 1.5

Objective of the course:

The objective of the course is to make the students able to –

- It demonstrates that how can you change the implementation of an object without affecting any other code by increasing data security and protecting unwanted data access. (Encapsulation).
- It allows you to have many different functions, all with the same name, all doing the same job, but depending upon different data. (Polymorphism).
- It guides you to write generic code: which will work with a range of data, so you don't have to write basic stuff over, and over again. (Generics).
- It lets you write a set of functions, then expand them in different direction without changing or copying them in any way. (Inheritance)

Course Outcomes:

After completion of the course students will be able to

CO1	Create the procedure of communication between Objects, classes & methods.
CO2	Understand the elementary facts of Object Orientation with various characteristics as well as several aspects of Java.
CO3	Analyze distinct features of different string handling functions with various I/O operations.
CO4	Discuss simple Code Reusability notion w.r.t. Inheritance, Package and Interface.
CO5	Apply Exception handling, Multithreading and Applet (Web program in java) programming concept in Java.

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Course Contents:**Module 1: Java Basics****[10]**

1. Simple Java programming using operators, control statements & loops, array.
2. Programming on class, object, and method, access specifier.
3. Programming on constructor, method/constructor overloading.
4. Programming on this keyword, call by value, static variables & methods, inner classes.

Module 2: Basic String handling & I/O**[12]**

1. Programming to show the use of String class methods - charAt(), compareTo(), equals(), equalsIgnoreCase(), indexOf(), length(), substring(), toCharArray(), toLowerCase(), toString(), toUpperCase(), trim(), valueOf() methods.
2. Programming to show the use of StringBuffer class methods - append(), capacity(), charAt(), delete(), deleteCharAt(), ensureCapacity(), getChars(), indexOf(), insert(), length(), setCharAt(), setLength(), substring(), toString() methods.
3. Programming on Command line arguments.
4. Programming using keyboard input by implementing BufferedReader & Scanner classes.

Module 3: Inheritance, Interface and Java Packages**[6]**

1. Programming on Simple Inheritance, super and final keywords, super() method.
2. Programming on method overriding, dynamic method dispatch, abstract classes & methods, multiple inheritance by using interface.
3. Programming on importing system package, creating user-defined package, importing user-defined package, using protected access specifier, subclassing an imported class of a package, using same names for classes of different packages, adding multiple public classes to a package.

Module 4: Exception handling, Multithreading and Applet Programming**[8]**

1. Programming on exception handling using try-catch block, implementing throw and throws keywords, using finally block, creating user-defined exception.
2. Programming on creating child threads i) by extending thread class ii) by implementing runnable interface, creating child threads by assigning thread priorities.
3. Programming on creating simple applet to display some message, creating applet to add 2 integers, creating applet to do GUI based programming.

Textbooks:

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

1. Herbert Schildt – "Java: The Complete Reference " – 9th Ed. – TMH
2. E. Balagurusamy – " Programming With Java: A Primer " – 3rd Ed. – TMH.

Reference Books:

1. R.K Das – " Core Java for Beginners " – VIKAS PUBLISHING.
- 2 Rambaugh, James Michael, Blaha – " Object Oriented Modelling and Design " – Prentice Hall, India

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	-	-	-	1	-	-	1	-	3	3
CO2	3	2	2	-	1	-	-	-	1	-	-	2	3	2	3
CO3	2	3	2	3	-	-	-	-	2	-	-	-	-	3	3
CO4	1	-	-	-	-	-	-	-	1	2	-	2	3	3	-
CO5	2	1	1	-	1	-	-	-	2	-	-	2	-	2	3

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Microprocessor & Microcontroller Lab

Course Code: PEC-CS-T-691

Contact: 0:0:3

Credits: 1.5

Prerequisites: Digital Electronics

Course Objective:

The objective of the course is to make the students able to

1. Illustrate and use proper code in appropriate platform using suitable syntax for developing program to solve problems.
2. Apply the knowledge of architecture in the computational area to efficient programming codes to design programs, interface with peripherals using modern tools.
3. Outline different types of basic instruction set (data transfer, load/store, arithmetic, logical), subroutine calls and IN/OUT instructions and trainer kits to design and solve the problems.
4. Analyse the problem by applying the knowledge of microprocessor and microcontroller and implement a new approach for solving engineering-related computational problems.
5. Interpret the result of the experiments, prepare laboratory reports based on observed output and analyse it.

Course Outcomes:

After completion of the course students will be able to

CO1	Illustrate and use proper code in appropriate platform using suitable syntax for developing program to solve problems related to Mathematics and Engineering field leading to lifelong learning.
CO2	Apply the knowledge of architecture in the computational area to efficient programming codes to design programs, interface with peripherals using modern tools like trainer kits for solving complex engineering problems.
CO3	Outline different types of basic instruction set (data transfer, load/store, arithmetic, logical), subroutine calls and IN/OUT instructions and trainer kits to design and solve the problem as a professional engineering practice as a team.
CO4	Analyse the problem by applying the knowledge of microprocessor and microcontroller and implement a new approach for solving engineering-related

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	computational problems as a team.
CO5	Interpret the result of the experiments, prepare laboratory reports based on observed output and analyse it to validate professional ethics and responsibilities and norms of the engineering practice.

Course Content:

Module-1: Familiarization with 8085 register level architecture and trainer kit components including the memory map. Familiarization with process of storing and viewing the contents of memory as well as registers.

Module-2: Familiarization with prewritten program on 8085 trainer kit using the basic instruction set (data transfer, load/store, arithmetic, logical)

Module 3: Implement the following programs using 8085 trainer kit

- a. Table lookup
- b. Copying a block of memory
- c. Shifting a block of memory
- d. Packing and unpacking of BCD numbers
- e. BCD to Binary Conversion and vice-versa
- f. HCF of two numbers
- g. Addition of numbers using subroutine
- h. Clearing the flag register
- i. Addition and subtraction
- j. Copying and shifting a block of memory
- k. Packing and unpacking of BCD numbers
- l. Addition of BCD numbers
- m. Binary to ASCII conversions
- n. String matching
- o. Multiplication of two numbers
- p. Sorting of array of numbers

Module 4: Implement the following programs using subroutine calls and IN/OUT instructions using 8255 PPI on the trainer kit.

- a. Reading switch state & glowing LEDs accordingly
- b. finding out the frequency of a pulse train
- c. subroutine for delay.

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Module 5: Familiarization with 8051 Micro controller.

Module 6: Implement the following programs using 8051 Micro Controller trainer kit

- a. Addition and subtraction
- b. Copying and shifting a block of memory
- c. Packing and unpacking of BCD numbers
- d. Addition of BCD numbers
- e. Binary to ASCII conversions
- f. String matching
- g. Multiplication of two numbers
- h. Sorting of array of numbers

Text Books:

1. MICROPROCESSOR architecture, programming and Application with 8085 - R. Gaonkar (Penram international Publishing LTD.)
2. Fundamentals of Microprocessor and Microcontrollers - B. Ram (Paperback)

Reference books:

1. The 8051 microcontroller and Embedded systems - Mazidi, Mazidi and McKinley (PEARSON)
2. 8051 Microcontroller – K. Ayala (Cengage learning)

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	-	3	3	3	3
CO2	3	3	3	3	3	-	-	-	-	-	-	-	3	3	3
CO3	3	3	3	3	-	-	-	3	3	-	-	-	3	3	3
CO4	3	3	3	3	-	-	-	3	3	-	-	-	3	3	3
CO5	-	-	-	-	-	-	-	3	-	3	-	-	3	2	3

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Course Name: Advanced Computer Architecture Lab

Course Code: PEC-CS-S-591

Contact: 0:0:3

Credits: 1.5

Perquisites: Computer Organization and Architecture

Course Objective:

The objective of the course is to make the students able to

1. Illustrate and use proper syntax in appropriate platform using suitable syntax for developing program to solve problems.
2. Apply the knowledge of algorithms in the computational area to efficient programming codes to design the problem using modern tools.
3. Outline different types of digital electronic circuits such as adder, subtractor, encoder, decoder, multiplexer, de-multiplexer, flip-flops, register, counter using various mapping and modern tools to prepare the most simplified circuit.
4. Apply the knowledge of digital electronic circuits to design memory and ALU and analyse the same to solve engineering-related computational problems.
5. Interpret the result of the experiments, prepare laboratory reports based on observed output and analyse it.

Course Outcomes:

After completion of the course students will be able to

CO1	Illustrate and use proper syntax in appropriate platform for developing program to solve problems related to Mathematics and Engineering field leading to lifelong learning.
CO2	Apply the knowledge of algorithms in the computational area to efficient programming codes to design the problem using modern tools for solving complex engineering problems.
CO3	Outline different types of digital electronic circuits such as adder, subtractor, encoder, decoder, multiplexer, de-multiplexer, flip-flops, register, counter using various mapping and modern tools to prepare the most simplified circuit and optimize using various mapping and mathematical methods for solving the

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	problem as a professional engineering practice as a team.
CO4	Apply the knowledge of digital electronic circuits to design memory and ALU and analyse the same to solve engineering-related computational problems as a team.
CO5	Interpret the result of the experiments, prepare laboratory reports based on observed output and analyse it to validate professional ethics and responsibilities and norms of the engineering practice.

Course Content:

Module-1: Familiarization with Xilinx environment setup, syntax, variables, commands, data types, operators, decisions, loops, vectors, matrix, arrays, functions, and advanced part, creating and editing, compilation and execution of VHDL programs.

Module-2: Study and simulate for truth table verification in Xilinx environment of

- a. Logic gates
- b. Universal gates

Module-3: Study and simulate for truth table verification in Xilinx environment of

- a. Half Adder
- b. Full Adder

Module-4: Study and simulate for truth table verification in Xilinx environment of

- a. Half Subtractor
- b. Full Subtractor

Module-5: Study and simulate for truth table verification in Xilinx environment of

- a. Multiplexer
- b. De-Multiplexer
- c. Encoder
- d. Decoder

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Module-6: Study and simulate for truth table verification in Xilinx environment of

- a. 7 Segment Decoder
- b. Comparator (1 bit, 2 bits)
- c. Carry look ahead adder

Module-7: Study and simulate for truth table verification in Xilinx environment of

- a. SR flip-flop
- b. D flip-flop
- c. JK flip-flop
- d. T flip-flop

Module-8: Study and simulate for truth table verification in Xilinx environment of

- a. SISO
- b. SIPO
- c. PISO
- d. PIPO

Module-9: Study, design and simulate in Xilinx environment of

- a. ALU
- b. RAM

Textbook:

- 3. M. Morris Mano, Digital Design, Prentice-Hall
- 4. Digital Circuits and Design, S. Salivahanan and S. Arivazhagan, McGraw-Hill Education

Reference book:

- 1. Hayes J. P., "Computer Architecture & Organisation", McGraw Hill,
- 2. William Stallings, Computer Organization and Architecture: Designing for Performance
- 3. P. J. Ashenden, "The Designer's Guide to VHDL"
- 4. Doulos: "VHDL Golden Reference Guide"

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CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	-	3	3	3	3
CO2	3	3	3	3	3	-	-	-	-	-	-	-	3	3	3
CO3	3	3	3	3	-	-	-	3	3	-	-	-	3	3	3
CO4	3	3	3	3	-	-	-	3	3	-	-	-	3	3	3
CO5	-	-	-	-	-	-	-	3	-	3	-	-	3	2	3
CO	3	3	3	3	3	-	-	3	3	3	-	3	3	3	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Artificial Intelligence Lab

Course Code: PEC-CS-D-591

Contact: 3:0:0

Credits: 1.5

Prerequisites: Data Structure, Design and Analysis of Algorithms, Statistics

Course Objectives:

The objective of the course is to enable students to

1. Gain foundational knowledge of PROLOG to implement an Artificial Intelligent Agent as an executable computer program for Knowledge Representation and Inferencing
2. Formulate a problem by analyzing its characteristics to fit a State-Space Exploration Framework or an Inferencing Framework of Artificial Intelligence.
3. Apply the concepts of Artificial Intelligence to solve a problem by implementing well-known Artificial Intelligence strategies using proper techniques and tools of PROLOG.
4. Build expert systems offering solutions to the challenging problems of Artificial Intelligence.
5. Implement Artificial Intelligence based ideas as executable PROLOG programs through developing intelligent heuristic strategies.

Course Outcomes:

After completion of the course students will be able to

CO1	Acquire foundational knowledge of PROLOG to implement an Artificial Intelligent Agent as an executable computer program for Knowledge Representation and Inferencing and understand the working principle of the agent and assess its utilitarian importance in current technological context leading
CO2	Identify and formulate an engineering problem by analyzing its characteristics to fit a State-Space Exploration Framework or an Inferencing Agent Formulation Framework of Artificial Intelligence.
CO3	Explore relevant literature and apply the concepts of Artificial Intelligence to solve a problem by implementing well-known Artificial Intelligence strategies using proper techniques and tools of PROLOG.
CO4	Develop ideas and propose expert systems offering solutions to the challenging problems of Artificial Intelligence.

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CO5	Plan and Implement Artificial Intelligence based ideas as executable PROLOG programs through developing intelligent heuristic strategies or expert systems with adequate documentation in collaborative environment for successfully carrying out projects on Artificial Intelligence Problems and investigate their effectiveness by analyzing the performances using proper techniques and tools.
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Course Content:

WEEK-1: Introduction to PROLOG Programming along with the IDE and its Basic Components

Assignments for understanding the Basic Components of Knowledge Representation and Inferencing in Artificial Intelligence using PROLOG Programming and its working strategy.

WEEK -2: Arithmetic, Boolean Expression, Decision Making Strategies

Assignments for understanding implementation of Arithmetic Expression, Boolean Expression, and Decision-Making Strategies.

WEEK -3: Recursion and Looping through Recursion

Assignments for understanding implementation of Recursion and Looping through Recursion.

WEEK -4: List of Data Items in PROLOG

Assignments for understanding the utility of List in solving various problems.

WEEK -5: Blind Search Techniques – BFS, DFS

Implementation of BFS and DFS Algorithms for Goal Searching to solve Puzzles (8-Puzzle, Water Jug Puzzle)

WEEK -6: Heuristic Search Techniques – A* Search

Implementation of A* Search Algorithm for Goal Searching to solve Puzzles (8-Puzzle, Route Finding Puzzle)

WEEK-7: Constraint Satisfaction Problem Solving

Implementation of Backtracking Strategies to solve Constraint Satisfaction Problems (Graph Coloring Problem, 8-Queens Problem)

WEEK -8: Game Playing

WEEK -9: Discussion on Project Problems and Allocation (Problem Description Report Submission)

WEEK -11: Project Implementation, Verification and Documentation

WEEK -12: Project Demonstration and Project Report Review

5. Ivan Bratko, Prolog Programming for Artificial Intelligence, 4th Edition, Addison-Wesley
6. Russell, S. and Norvig, P. 2015. Artificial Intelligence - A Modern Approach, 3rd edition, Prentice Hall.
7. Rich, E., Knight, K and Shankar, B. 2009. Artificial Intelligence, 3rd edition, Tata McGrawHill.

8. Padhy, N.P. 2009. Artificial Intelligence and Intelligent Systems, Oxford University Press.
9. Deepak Khemani, “A First Course in Artificial Intelligence”, McGraw Hill.

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Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Computer Graphics Lab

Course Code: PEC-CS-A-591

Contact: 3:0:0

Total Contact Hours: 36

Credits: 1.5

Prerequisites: Mathematics – I,III,Computer Fundamentals & Principle of Computer Programming, Programming with C/C++

Course Objectives:

To understand the need of developing graphics application

To learn algorithmic development of graphics primitives like: line, circle, polygon etc.

To learn the representation and transformation of graphical images and pictures.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the fundamental concept of Computer graphics and mathematical knowledge and Implement different scan conversion algorithms and geometric transformations techniques.
CO2	Demonstrate different scan conversion algorithms, drawing algorithms, polygon filling algorithms, curves and surface drawing algorithms, clipping algorithms, surface removal algorithms using graphics tools and Implement boundary fill, flood fill and scan line polygon fill algorithms.
CO3	Understand the basic concept of graphics programming and implement window view port transformations and clipping algorithms
CO4	Analyze and Apply computer graphics principles and concepts to animation and game design
CO5	Develop the concept of geometric models, mathematical and algorithmic approach necessary for programming computer graphics leading to lifelong learning.

Course Content:

Module-I: Introduction of Computer Graphics System

9

Study of basic graphics functions defined in “graphics.h”. Interactive Graphics Systems), CGM (Computer Graphics Metafile), CGI (Computer Graphics Interface).

12

6

6

3

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

C02	3	3	2	2	-	-	-	-	-	-	-	-	3	2	3
C03	3	3	3	3	-	-	-	-	-	-	-	-	3	3	2
C04	3	3	3	3	-	-	-	-	3	-	-	-	3	3	3
C05	3	2	3	2	3	-	-	-	3	2	3	3	3	2	2

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
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Mandatory Courses

Course Name: Constitution of India

Course Code: MC 501

Contacts: 2:0:0

Total Contact Hours: 24

Credit: 0

Prerequisite: None

Total Contact Hours: 32

Course Outcome(s): Student will be able to:

CO1	Develop human values, create a wareness about lawratification and significance of Constitution
CO2	Comprehend the Fundamental Rights and Fundamental Duties of the Indian Citizentoim plant morality, social values and their social responsibilities.
CO3	Create understand ing of their Surroundings, Society, Social problems and their suitable solutions.
CO4	Familiarize with distribution of powers and functions of Local Self Government.
CO5	Realize the National Emergency Financial Emergency and their impact on Economy Of the country

Course Content:

Module 1: Introduction:

4L

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“Constitution”- Historical Background of the Constituent Assembly, Indian Constitution and its Salient Features, the Preamble of the Constitution.

**Module 2: Fundamental Rights, Fundamental Duties,
 Directive Principles of State Policy:**

8L

The Right to Equality
 The Right to Freedom: I (Article 19)
 The Right to Freedom: II (Articles 20, 21 and 22)
 The Right against Exploitation
 The Right to freedom of Religion
 Cultural and Educational rights
 The Right to Property
 The Right to Constitutional Remedies
 The Directive Principles
 Fundamental Duties

Module 3: Union Government and its Administration

6L

Structure of the Indian Union, Parliamentary System, Federal System, Centre-State Relations, Amendment of the Constitutional Powers and Procedure, The historical perspectives of the constitutional amendments in India, Emergency Provisions: National Emergency, President Rule, Financial Emergency, and Local Self Government – Constitutional Scheme in India.

Module 4: The Machinery of Government in the State

6L

Powers of Indian Parliament Functions of Rajya Sabha, Functions of Lok Sabha, Powers and Functions of the President, Comparison of powers of Indian President with the United States, Powers and Functions of the Prime Minister, Judiciary – The Independence of the Supreme Court, Appointment of Judges
 State Executives – Powers and Functions of the Governor, Powers and Functions of the Chief Minister, Functions of State Cabinet, Functions of State Legislature, Functions of High Court and Subordinate Courts.

Text / Reference Books:

- 1) Indian Constitution by D.D.Basu, The Publisher, LexisNexis
- 2) Constitution of India by Subhas C Kasyap, Vitasta Publishing
- 3) The Constitution of India, P.M Bakshi, Universal Law Publishing Co.Ltd, New Delhi, 2003.
- 4) Indian Constitution Text Book - Avasthi, Avasthi, Publisher: LAKSHMI NARAIN AGARWAL

CO-PO Mapping:

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO2	3	-	2	2	2	-	-	-	-	-	-	-	3	2	3
CO3	3	2	3	-	-	2	-	-	-	-	-	-	2	3	2
CO4	3	3	3	3	-	-	-	-	3	-	-	-	3	3	3
CO5	3	2	3	2	3	-	-	-	3	-	3	3	3	2	2

Curriculum for B.Tech Under Autonomy
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3 rd Year2 nd Semester: 6 th Semester								
Sl. No.	Category	Course Code	Course Title	Hours per week				Credits
				L	T	P	Total	
A. THEORY								
1	Humanities and Social Sciences including Management courses	HSMC 604	Economics for Engineers	2	0	0	2	2
2	Program Core Course	PCC-CS601	Computer Networks	3	0	0	3	3
3	Program Core Course	PCC-CS602	Software Engineering	3	0	0	3	3
4	Professional Elective courses	PEC-CS-T-601	Advanced Algorithms	3	0	0	3	3
		PEC-CS-S-601	Advanced Operating Systems					
		PEC-CS-D-601	Machine Learning					
		PEC-CS-A-601	Web and Internet Technology					
5	Professional Elective courses	PEC-CS-T-602	Parallel and Distributed Algorithms	3		0	3	3
		PEC-CS-S-602	Embedded Systems					
		PEC-CS-D-602	Soft Computing					
		PEC-CS-A-602	Human Computer Interaction					
6	Open Elective courses	OEC-CS-601A	Introduction to Internet of Things			0	3	3
		OEC-CS-601B	Bio-informatics					

		OEC-CS-601C	Robotics					
B. PRACTICAL								
7	Program Core Course	PCC-CS691	Computer Networks Lab	0	0	3	3	1.5
9	Program Core Course	PCC-CS692	Software Engineering Lab	0	0	3	3	1.5
10	Professional Elective courses	PEC-CS-T-691	Advanced Algorithms Lab	0	0	3	3	1.5
		PEC-CS-S-691	Advanced Operating Systems Lab					
		PEC-CS-D-691	Machine Learning Lab					
		PEC-CS-A-691	Web and Internet Technology Lab					
11	PROJECT	PR 691	Minor Project II	0	0	3	2	1
12	PROJECT	PR 692	Skill Development VI: Soft Skill & Aptitude-III	1	0	0	1	0.5
C. MANDATORY ACTIVITIES / COURSES								
13	MC	MC 601	Intellectual Property Right	3	0	0	3	3Units
	TOTAL CREDIT WITHOUT MOOCS COURSES							23.0
D.MOOCS COURSES**								

1 4	MOOCS COURSES	HM601	MOOCS COURSE-IV	3	1	0	4	4
TOTAL CREDIT WITH MOOCS COURSES								27

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Economics for Engineers

Course Code: HSMC 604

Contact: 2:0:0

Total Contact Hours: 24

Credits:2

Pre-requisites:

MATH – College Algebra, Pre-Calculus Algebra and Trigonometry.

Course Outcome(s):

On completion of the course students will be able to

CO1	Apply the appropriate engineering economics analysis method(s) for problem solving: present worth, annual cost, rate-of-return, payback, break-even, benefit-cost ratio.
CO2	Evaluate the cost effectiveness of individual engineering projects using the methods learned and draw inferences for the investment decisions.
CO3	Compare the life cycle cost of multiple projects using the methods learned, and make a quantitative decision between alternate facilities and/or systems.
CO4	Evaluate the profit of a firm, carry out the break even analysis and employ this tool to make production decision.
CO5	Discuss and solve advanced economic engineering analysis problems including taxation and inflation.

Course Contents:

MODULE I Introduction[3L]

Managerial Economics-Relationship with other disciplines-Firms: Types, Objectives and goals-Managerial Decisions-Decision Analysis.

MODULE II Demand and Supply Analysis[5 L]

Demand-Types of demand-determinants of demand-Demand function-Demand Elasticity-Demand forecasting-Supply-Determinants of supply-Supply function-Supply Elasticity.

MODULE III Cost Analysis[5 L]

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Element of costs, Marginal cost, Marginal Revenue, Sunk cost, Opportunity cost, Break-even analysis– PV ratio.

MODULE IV Elementary economic Analysis [4 L]

Inflation-Meaning of inflation, types, causes, measures to control inflation. National Income- Definition, Concepts of national income, Method of measuring national income.

MODULE V: Financial Accounting [5 L]

Concepts and Definition of Accounting, Journal, Ledger, Trial Balance. Trading A/C, Profit & Loss A/C and Balance Sheet.

MODULE VI: Investment Decision[2L]

Time value of money- Interest - Simple and compound, nominal and effective rate of interest, Cash flow diagrams, Principles of economic equivalence. Evaluation of engineering projects- Present worth method, Future worth method, Annual worth method, Internal rate of return method, Cost benefit analysis for public projects.

Textbooks:

1. Riggs, Bedworth and Randhwa, “Engineering Economics”, McGraw Hill Education India
2. Principles of Economics, Deviga Vengedasalam; Karunakaran Madhavan, Oxford University Press.

Reference Books:

1. Engineering Economy by William G. Sullivan, Elin M. Wicks, C. Patric Koelling, Pearson
2. R. Paneer Seelvan, “ Engineering Economics”, PHI
3. Ahuja, H.L., “Principles of Micro Economics” , S.Chand & Company Ltd
4. Jhingan, M.L., “Macro Economic Theory”
5. Macro Economics by S.P. Gupta, TMH
6. Haniff and Mukherjee, Modern Accounting, Vol-1, TMG
7. Modern Economic Theory – K.K. Dewett (S.Chand)

Reference Books:

1. Engineering Economy by William G. Sullivan, Elin M. Wicks, C. Patric Koelling, Pearson
2. R. Paneer Seelvan, “ Engineering Economics”, PHI
3. Ahuja, H.L., “Principles of Micro Economics” , S.Chand & Company Ltd
4. Jhingan, M.L., “Macro Economic Theory”
5. Macro Economics by S.P. Gupta, TMH
6. Haniff and Mukherjee, Modern Accounting, Vol-1, TMG
7. Modern Economic Theory – K.K. Dewett (S.Chand)

CO-PO MAPPING:

[illegible]

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Computer Networks

Course Code: PCC-CS601

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisite:

1. Familiarity and knowledge of Operating Systems and Computer Architecture
2. Also require little bit programming languages concepts like C, Java.

Course Objective(s):

- To be familiar with the basics of data communication
- To be familiar with various types of computer networks
- To have experience in designing communication protocols
- To be exposed to the TCP/IP protocol suite

Course Outcome(s):

After completion of the course the student able to do

CO1	Understand OSI and TCP/IP models.
CO2	Analyze MAC layer protocols and LAN technologies.
CO3	Design applications using internet protocols.
CO4	Implement routing and congestion control algorithms.
CO5	Develop application layer protocols and understand socket programming

Course Contents:

Module 1: Introduction [6L]

Introduction (3L):

Introduction: Computer Network, data communication, topology, OSI & TCP/IP Reference Models, layers and characteristics, Wireless Network, comparison to wired and wireless network.

Physical Layer: [3L]

Overview of data (analog & digital), signal (analog & digital), transmission (analog & digital) & transmission media (guided & unguided); Circuit switching: time division & space division switch, TDM bus; Telephone Network;

Module 2: Data Link Layer [10L]

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Framing, Error Control, Error Detection and Correction, Flow Control, Data Link Protocols, Simple Stop-and-Wait Protocol, ARQ mechanism, Sliding Window Protocols, One-Bit Sliding Window Protocol, Go-Back-N and Selective Repeat, HDLC, PPP Medium Access Control Sub-layer, The Channel Allocation. [5L]

Multiple Access Protocols: ALOHA, Carrier Sense Multiple Access Protocols, IEEE 802.x Ethernet, Switched Ethernet, Fast Ethernet, Gigabit Ethernet, 10 Gigabit Ethernet, Wireless LANs - IEEE 802.xx, Bluetooth, RFID, Bridges, Virtual LANs, Switching. [5L]

Module 3: Network Layer [10L]

IP Addressing, IPv4 and IPv6. Difference IPv4 and IPv6, Conversion of IPv4 and IPv6, Subnetting, Supernetting, Design Issues, Store-and-Forward Packet Switching, Virtual-Circuit and Datagram Networks, ARP, IP, ICMP, IPV6, BOOTP and DHCP–Delivery protocols Other Protocols such as mobile IP in wireless Network. [5L] Routing: Shortest Path Algorithms, Flooding, Distance Vector Routing, Link State Routing, Hierarchical Routing, Broadcast Routing, Multicast Routing, Anycast Routing: RIP, OSPF, BGP; Routing for Mobile Hosts. [5L]

Module 4: Transport layer: [6L]

Process to Process delivery; UDP; TCP, SCTP, TCP RENO, TCP/IP in Wireless environment, Congestion control in TCP: Congestion Control: Open Loop, Closed Loop choke packets; Quality of service: techniques to improve QoS: Leaky bucket algorithm, Token bucket algorithm. [5L] Advanced topic such as Remote Procedure Call, Delay Tolerant Networks. [1L]

Module 5: Application Layer [3L]

Introduction to DNS, SMTP, SNMP, FTP, HTTP & WWW: Cryptography (Public, Private Key based), Digital Signature, Firewalls

Module 6: Socket Programming [1L]

Introduction to Socket Programming, UDP socket and TCP Socket

Text books:

1. B. A. Forouzan – “Data Communications and Networking (3rd Ed.)” – TMH
2. S. Tanenbaum – “Computer Networks (4th Ed.)” – Pearson Education/PHI
3. W. Stallings – “Data and Computer Communications (5th Ed.)” – PHI/ Pearson Education
4. Zheng & Akhtar, Network for Computer Scientists & Engineers, OUP

Reference books:

1. Kurose and Rose – “Computer Networking -A top-down approach featuring the internet” – Pearson Education
2. Leon, Garica, Widjaja – “Communication Networks” – TMH
3. Walrand – “Communication Networks” – TMH.
4. Comer – “Internetworking with TCP/IP, vol. 1, 2, 3(4th Ed.)” – Pearson Education/PHI

CO-PO Mapping:

[illegible]

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Software Engineering

Course Code: PCC-CS 602

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: Programming for Problem Solving

Course Objective(s)

- 1.To understand the working environment in industry and aware of cultural diversity, who conduct themselves ethically and professionally.
2. Graduates use effective communication skills and technical skills to assure production of quality software, on time and within budget.
3. Graduates build upon and adapt knowledge of science, mathematics, and engineering to take on more expansive tasks that require an increased level of self-reliance, technical expertise, and leadership.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the basic concept of Software Engineering and mathematical knowledge and apply them in designing solution to engineering problem including he specification, design , implementation, and testing of software systems that meet specification, performance, maintenance and quality requirements
CO2	Analyze , elicit and specify software requirements through a productive working relationship with various stakeholders of the project
CO3	Design applicable solutions in one or more application domains using software engineering approaches that integrates ethical, social, legal and economic concerns.
CO4	Develop the code from the design and effectively apply relevant standards and perform testing, and quality management and practice team work.
CO5	Identify and Use modern engineering tools necessary for software project management time management and software reuse, and an ability to engage in life-long learning.

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Content:

Module-I: Introduction

6

Software Engineering, Characteristics, Components, Application, Definitions. Software Project Planning-Feasibility Analysis, Technical Feasibility, Cost-Benefit Analysis, Basics of estimation : COCOMO (Basic, intermediate, Complete) model.

Module-II: Software life cycle models

6

Evolution and impact of Software engineering, software life cycle models Waterfall, prototyping, Evolutionary, and Spiral models. Feasibility study, Functional and Non-functional requirements, Requirements gathering, Requirements analysis and specification.

Module-III: Software design

8

Basic issues in software design, modularity, cohesion, coupling and layering, function-oriented software design: DFD and Structure chart, object modeling using UML, Object-oriented software development, user interface design. Coding standards and Code review techniques.

Module-IV: Software Testing

7

Fundamentals of testing, White-box, and black-box testing, Test coverage analysis and test case design techniques, mutation testing, Static and dynamic analysis, Software reliability metrics, reliability growth modeling.

Module-V: Software project management

9

Software project management, Project planning and control, cost estimation, project scheduling using PERT and GANTT charts, cost-time relations: Rayleigh-Norden results, quality management, ISO and SEI CMMI, PSP and Six Sigma. Computer aided software engineering, software maintenance, software reuse, Component-based software development.

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Text Books:

1. Fundamentals of Software Engineering by Rajib Mall, –PHI-3rd Edition, 2009.
2. Software Engineering- Pankaj Jalote (Wiley-India)

Reference Books:

1. Software Engineering –Agarwal and Agarwal (PHI)
2. Software Engineering, by Ian Sommerville, Pearson Education Inc., New Delhi, (2009).
3. Software Engineering: A Practitioner’s Approach”, by Roger S. Pressman, McGraw-Hill.(2005)

CO – PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO2	3	3	2	2	-	-	-	-	-	-	-	-	3	2	3
CO3	3	3	3	3	-	-	-	3	-	-	-	3	3	3	2
CO4	3	3	3	3	-	-	-	-	3	-	-	-	3	3	3
CO5	3	2	3	2	3	-	-	-	3	2	3	3	3	2	2

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Advance Algorithm

Course Code: PEC-CS-T-601

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: Design & Analysis of Algorithm (PCC-CS402)

Objective of the course:

15. The aim is to learn how to develop efficient algorithms for simple computational tasks and reasoning about the correctness of them
16. Through the complexity measures, different range of behaviors of algorithms and the notion of tractable and intractable problems will be understood.

Course Outcomes:

After completion of the course students will be able to

CO1	Analyze the complexity/performance of different algorithms.
CO2	Determine the appropriate algorithm for solving a particular set of problems.
CO3	Categorize the different problems in various classes according to their complexity.
CO4	Achieve an insight of recent activities in the field of the advanced algorithm.
CO5	Design and build solutions for a real-world problem by applying relevant distributions

Course Content:

Module-I

4

Sorting:

Review of various sorting algorithms, topological sorting

Graph:

Definitions and Elementary Algorithms: Shortest path by BFS, shortest path in edge-weighted case (Dijkasra's), DFS and computation of strongly connected components, emphasis on correctness proof of the algorithm and time/space analysis, example of amortized analysis.

Module-II

6

Matroids:

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Introduction to greedy paradigm, algorithm to compute a maximum weight maximal independent set. Application to MST.

Graph Matching:

Algorithm to compute maximum matching. Characterization of maximum matching by augmenting paths, Edmond's Blossom algorithm to compute augmenting path.

Module-III

16

Flow-Networks:

Maxflow-Mincut Theorem, Ford Fulkerson Method to compute Maximum Flow, Edmond-Karp maximum-flow algorithm.

Matrix Computations:

Strassen's algorithm and introduction to divide and conquer paradigm, inverse of a triangular matrix, relation between the time complexities of basic matrix operations, LUP-decomposition

Shortest Path in Graphs:

Floyd-Warshall algorithm and introduction to dynamic programming paradigm. More examples of dynamic programming.

Modulo Representation of integers/ polynomials:

Chinese Remainder Theorem, Conversion between base-representation and modulo-representation. Extension to polynomials. Application: Interpolation problem.

Discrete Fourier Transform (DFT):

In complex field, DFT in modulo ring. Fast Fourier Transform algorithm. Schonhage-Strassen Integer Multiplication algorithm

Amortized Analysis:

Aggregate, Accounting, and Potential Method

Module-IV

10

Linear Programming:

Geometry of the feasibility region and Simplex algorithm

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

NP-completeness:

Examples, proof of NP-hardness and NP-completeness. One or more of the following topics based on time and interest Approximation algorithms, Randomized Algorithms, Interior Point Method, Advanced Number Theoretic Algorithm

Problem Solving Application

Recent Trends in problem solving paradigms using recent searching and sorting techniques by applying recently proposed data structures.

Text book:

1. "Introduction to Algorithms" by Cormen, Leiserson, Rivest, Stein.
2. "The Design and Analysis of Computer Algorithms" by Aho, Hopcroft, Ullman. 3. "Algorithm Design" by Kleinberg and Tardos. 4. Design & Analysis of Algorithms, Gajendra Sharma, Khanna Publishing House, New Delhi

Reference Books:

1. "Algorithm Design" by Kleinberg and Tardos.
2. Design & Analysis of Algorithms, Gajendra Sharma, Khanna Publishing House, New Delhi

CO – PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	-	2	2	2	3
CO2	3	3	3	3	-	-	-	-	-	-	-	2	2	3	3
CO3	3	3	3	3	-	-	-	-	-	-	-	3	3	3	3
CO4	3	3	3	3	-	-	-	-	-	-	-	3-	3	2	3
CO5	3	3	3	3	-	-	-	-	-	-	-	3	3	3	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Advanced Operating Systems

Course Code: PEC-CS-S-601

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: Operating Systems

Course Objectives:

The objective of the course is to make the students able to -

1. Understand design issues of Advanced Operating systems.
2. Understand basic concepts and need of Distributed operating systems
3. Understand the architecture of Distributed operating system.
4. Understand concepts and working of different advanced Operating systems like Multiprocessor Operating System, Real-time Operating System, Mobile Operating System.
5. Understand some real time Applications of Operating System

Course Outcomes:

After completion of the course students will be able to -

CO1	Demonstrate understanding of design issues of advanced operating systems and compare different types of operating systems.
CO2	Analyze the design aspects and issues of distributed operating systems.
CO3	Demonstrate understanding of different architectures used in Distributed Operating System.
CO4	Demonstrate understanding of different architectures used in Multiprocessor Operating System.
CO5	Formulate the solutions to schedule the real time applications.

Course Content:

Module I: Architectures of Distributed Systems: 7L

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

System Architecture Types, Distributed Operating Systems, Issues in Distributed Operating Systems, Communication Primitives. Theoretical Foundations: Inherent Limitations of a Distributed System, Lamports Logical Clocks, Vector Clocks, Causal Ordering of Messages, Termination Detection.

Module II: Distributed Mutual Exclusion: 7L

The classification of Mutual Exclusion Algorithms, Non-Token-Based Algorithms: Lamports Algorithm, The Ricart-Agarwala Algorithm, Maekawas Algorithm, Token-Based Algorithms: Suzuki-Kasamis Broadcast Algorithm, Singhals Heuristics Algorithm, Raymonds Heuristic Algorithm.

Module III: Distributed Deadlock Detection: 7L

Preliminaries, Deadlock Handling Strategies in Distributed Systems, Issues in Deadlock Detection and Resolution, Control Organizations for Distributed Deadlock Detection, Centralized- Deadlock – Detection Algorithms, Distributed Deadlock Detection Algorithms, Hierarchical Deadlock Detection Algorithms.

Module IV: Multiprocessor System Architectures 8L

: Introduction, Motivation for multiprocessor Systems, Basic Multiprocessor System Architectures, Multi Processor Operating Systems: Introduction, Structures of Multiprocessor Operating Systems, Operating Design Issues, Threads, Process Synchronization, Processor Scheduling, Distributed File Systems: Architecture, Mechanisms for Building Distributed File Systems, Design Issues

Module V: Distributed Scheduling : 7L

Issues in Load Distributing, Components of a load Distributed Algorithm, Stability, Load Distributing Algorithm, Requirements for Load Distributing, Task Migration, Issues in task migration. Distributed Shared Memory: Architecture and Motivation, Algorithms for Implementing DSM, Memory Coherence, Coherence Protocols, Design Issues

Text book:

1. Mukesh Singhal and Niranjana Shivaratri, Advanced Concepts in Operating Systems, McGraw-Hill.
2. Andrew S. Tanenbaum, Distributed Operating Systems, ACM Press.

Reference Books:

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

2. Nancy Lynch, Distributed Algorithms, Morgan Kaufmann.
3. Jie Wu, Distributed Systems, CRC Press.
4. Hagit Attiya, Jennifer Welch, Distributed Computing: Fundamentals, Simulations and Advanced Topics, McGraw-Hill.
5. Sape Mullender (ed.), Distributed Systems, Addison-Wesley

Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO2	3	3	3	3	-	-	-	-	-	-	-	-	3	2	3
CO3	3	3	3	3	-	-	-	-	-	-	-	3	3	3	1
CO4	3	3	3	3	-	-	-	-	-	-	-	-	3	3	1
CO5	3	3	3	3	-	-	-	-	3	2	-	-	3	2	1
CO	3	3	3	3	-	-	-	-	3	2	-	3	3	3	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Machine Learning

Course Code: PEC-CS-D-601

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: Data Structure, Design and Analysis of Algorithms, Statistics, Artificial Intelligence

Course Objectives:

6. Comprehend the fundamental concepts of the evolving technologies in machine learning such as Supervised and Unsupervised Learning
7. Formulate an engineering problem within the scope of machine learning paradigm.
8. Apply the concepts of machine learning to solve problems of making automated decisions dealing with large scale data.
9. Develop and Implement ideas for proposing solutions to the challenging problems of machine learning
10. Analyze the effectiveness of various machine learning Frameworks.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the basic concepts of machine learning to Explain or Illustrate and Identify problems where students can Apply the concept appropriately to Solve them.
CO2	Understand the fundamental concepts of regression analysis so that they can propose models for predicting values based on exemplary data and Analyze their performances.
CO3	Explain or Illustrate the fundamental strategies of unsupervised machine learning paradigm to solve clustering problems and Analyze their performances.
CO4	Explain or Illustrate the concepts of Mining Frequent Patterns, Associations and Data Streams and Apply them to solve the relevant problems and Analyze their performances.
CO5	Develop ideas to Propose solutions to the problems of supervised learning and Identify problems where students can Apply the concept appropriately and Analyze the effectiveness as well as limitations of solutions making the students aware of its utilitarian importance for further explorations leading towards lifelong learning.

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
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CourseContent:

MODULE-1: Introduction to Machine Learning [4L]

Basic Concepts, Various types of Machine Learning Techniques and related applications, Issues in Machine Learning Strategies, Data Exploration for Machine Learning: Data Types, Data Attributes, Statistical Description of Data, Data Visualization, Data Similarity Measures; Data Preprocessing: Data Cleaning, Data Integration, Data Reduction, Data Transformation & Discretization.

MODULE-2: Classification and Regression [14L]

Basic Concepts, assessing and visualizing performance of classification, k-Nearest-Neighbor Classifier, Decision Tree Classifier, Naïve Bayes Classifier; Ensemble Classification, Random Forest Strategy, Linear and Nonlinear Regression Methods and their performance analysis.

MODULE-3: Clustering, Association and Outlier Analysis [10L]

Basic Concepts, Partitioning Methods: k-Means and k-Medoids, Hierarchical Methods: Agglomerative and Divisive Hierarchical Clustering, Density-Based Methods: DBSCAN: Density-Based Clustering Based on Connected Regions with High Density; Outlier Analysis.

MODULE-4: Mining Frequent Patterns, Associations and Data Streams [3L]

Basic Concepts, Association analysis and Frequent Itemset Mining Methods: The Apriori Algorithm, Mining Time Series Data.

MODULE-5: Advanced Concepts [5L]

Introduction to advanced concepts of machine learning like Support Vector Machines and Artificial Neural Network and their applications in solving machine learning problems.

Textbook:

8. Christopher Bishop, Pattern Recognition and Machine Learning, Springer, 2007
9. Dr. Rajiv Chopra, Machine Learning, Khanna Publishing House, 2018
10. Machine Learning by SaikatDutt, Subramanian Chandramouli, Amit Kumar Das, Pearson.

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ReferenceBooks:

10. Machine Learning using Python, Manaranjan Pradhan and U Dinesh Kumar, Wiley
11. Hands-on Machine Learning with Scikit-Learn, Keras, and TensorFlow, AurélienGéron, O'Reilly
12. Han J &Kamber M, “Data Mining: Concepts and Techniques”, Morgan Kaufmann Publishers, Third Edition.

CO–PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	-	3	-	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	-	-	-	-	-	-	-	-	2	-	2
CO4	2	2	2	3	-	-	-	-	-	-	-	-	2	-	2
CO5	2	2	3	3	-	-	-	-	-	-	-	2	2	2	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Name of the Paper: Web and Internet Technology

Paper Code: PEC-CS-A-601

Contact (Periods/Week): 3L/Week

Credit Point: 3

No. of Lectures: 36

Course Objective(s):

- To impart the design, development, and implementation of Static and Dynamic Web Pages.
- To develop programs for Web using Scripting Languages and .net framework.
- To give an overview of Server Side Programming in Web.

Course Outcomes:

After completion of the course students will be able to:

CO1	Understand networks, IP, DNS, routing and other related technologies used in internet and execute and solve problems related to them leading to engineering problems solutions
CO2	Understand different web based technologies like HTML, DHTML, CSS, XML and demonstrate their use in design of web based solutions leading to engineering problems
CO3	Comprehend and analyze different client and server side technologies like JavaScript, Servlet, CGI and design appropriate engineering solutions leading to life long learning
CO4	Understand and implement different types of technologies like JSP, JavaBean, JDBC and ODBC and evaluate their performances
CO5	Understand different web based applications and network security techniques and apply them to protect the network against different attacks and solve related problems preferably as a team

Course Contents:

Module 1: [6L]

Introduction (1L): Overview, Network of Networks, Intranet, Extranet, and Internet. World Wide Web (1L): Domain and Sub domain, Address Resolution, DNS, Telnet, FTP, HTTP. Review of TCP/IP (1L): Features, Segment, Three-Way Handshaking, Flow Control, Error Control, Congestion control, IP Datagram, IPv4 and IPv6. IP Subnetting and addressing (1L): Classful

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and Classless Addressing, Subnetting. NAT, IP masquerading, IP tables. Internet Routing Protocol (1L): Routing -Intra and Inter Domain Routing, Unicast and Multicast Routing, Broadcast. Electronic Mail (1L): POP3, SMTP, Clients - Servers Communication.

Module -2: [9L]

HTML, DHTML & CSS: Introduction, Elements, Attributes, Heading, Paragraph. Formatting[1L]; Link, Table, List, Block, Layout, Html Forms, and input [1L]; Iframe, Colors[1L], Image Maps and attributes of image area [1L]; Introduction to CSS, basic syntax and structure of CSS, different types internal, external and inline CSS [1L]; Basic Introduction of DHTML, Difference between HTML and DHTML, Documentary Object Model (DOM) [1L]. Extended Markup Language (XML) : Introduction, Difference between HTML & XML, XML-Tree [1L]; Syntax, Elements, Attributes, Validation and parsing, DTD [2L].

Module 3: [15L]

Java Scripts: Basic Introduction, Statements, comments, variable, operators, data types [1L]; condition, switch, loop, break [1L]; Java script functions, objects, and events[1L]. CGI Scripts: Introduction, Environment Variable, GET and POST Methods [1L]. Java Servlet: Servlet environment and role, Servlet life cycle [1L]; Servlet methods- Request, Response, Get and post [1L]; Cookies and Session [1L].

Java Server Page (JSP):

JSP Architecture [1L]; JSP Servers, JSP Life Cycle [1L]; Understanding the layout of JSP, JSP Scriptlet Tag [1L]; JSP implicit object (request and response) [1L]; Variable declaration, methods in JSP [1L]; JSP directive (Taglib and Include), JavaBean- inserting JavaBean in JSP [1L]; JSP Action tags (Forward & Include) [1L]; Creating ODBC data source name, Introduction to JDBC, prepared statement and callable statement [1L].

Module-4: [6L]

Threats[1L]: Malicious code-viruses, Trojan horses, worms; eavesdropping, spoofing, modification, denial of service attacks. Network security techniques: Password and Authentication; VPN, IP Security [1L], security in electronic transaction, Secure Socket Layer (SSL), Secure Shell (SSH)[1L]. Firewall (1L): Introduction, Packet filtering, Stateful, Application layer, Proxy. Search Engine and Web Crawler: Definition, Meta data, Web Crawler [1L], Indexing, Page rank, overview of SEO[1L].

Textbooks:

1. “Web Technology: A Developer's Perspective”, N.P. Gopalan and J. Akilandeswari, PHI Learning, Delhi, 2013. (Topics covered: html, CSS, imagemap, xml)
2. “Learning PHP, MySQL & JavaScript”, Robin Nixon, O'Reilly Publication. (Topics covered: Java Script)
3. “Head First Servlet's & JSP”, Bryan Basham, Kathy Sterra, Bert Bates, O'Reilly Publication. (Topics covered: Servlet, JSP)

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4.Cryptography and Network Security by William Stallings Publisher: Pearson Education India(Topics covered: Threats, Security techniques, Firewall)

Recommended books:

1. "Programming the World Wide Web", Robert. W. Sebesta, Fourth Edition, Pearson Education, 2007.
2. “Core Web Programming”- Second Edition-Volume I and II, Marty Hall and Larry Brown, Pearson Education, 2001

CO-PO & PSO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	-	-	-	-	-	-	-	2	3	3
CO2	3	3	3	3	2	-	-	-	-	-	-	-	3	3	3
CO3	3	3	3	3	2	-	-	-	-	-	-	3	3	3	3
CO4	3	3	3	3	2	-	-	-	-	-	-	-	3	3	3
CO5	3	3	3	3	2	-	-	-	3	-	2	-	3	3	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: PARALLEL AND DISTRIBUTED SYSTEMS

Course Code: PEC-CS-T-602

Course: B.Tech.

Stream: Computer Science and Engineering

Total Contact Hours: 36

Contact: 3:0:0

Credits: 3

Prerequisite:

1. The student have the basic knowledge of Computer Organization Architecture, Distributed System and Operating System.

Course Objective(s)

1. To introduce the concepts of Modern Processors.
2. To introduce Optimization techniques for serial code.
3. To introduce Parallel Programming using OpenMP and MPI.

Course Outcome(s)

After completion of the course the student able to do

CO1	Understand of the parallel architecture and Programming
CO2	Design parallel programs develop CUDA programs for GPU.
CO3	Analyze and apply various parallel algorithms.
CO4	Capable to optimize algorithms for better performance
CO5	Understanding the distributed system

Detail Syllabus

PARALLEL AND DISTRIBUTED SYSTEM

Module 1:

Introduction of Parallel Programming: [7L]

Introduction:[2L] Scope, issues, applications and challenges of Parallel and Distributed Computing **Parallel Programming Platforms:[2L]** Implicit Parallelism: Trends in Microprocessor Architectures, Dichotomy of Parallel Computing Platforms, Physical

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Organization, Communication Costs in Parallel Machines, Routing Mechanisms for Interconnection Networks, GPU, coprocessing.

Principles of Parallel Algorithm Design:[3L] Decomposition Techniques, Characteristics of Tasks and Interactions, Mapping Techniques for Load Balancing.

Module 2:

CUDA programming model: [7L]

CUDA programming model: Overview of CUDA, Isolating data to be used by parallelized code, API function to allocate memory on parallel computing device, to transfer data, Concepts of Threads, Blocks, Grids, Developing a kernel function to be executed by individual threads, Execution of kernel function by parallel threads, transferring data back to host processor with API function.

Module 3:

Analytical Modeling of Parallel Programs: [7L]

Analytical Modeling of Parallel Programs: Sources of Overhead in Parallel Programs, Performance Metrics for Parallel Systems, The Effect of Granularity on Performance, Scalability of Parallel Systems, Minimum Execution Time and Minimum Cost-Optimal Execution Time.

Module 4:

Algorithms: [11L]

Dense Matrix Algorithms:[4L] Matrix-Vector Multiplication, Matrix-Matrix Multiplication, Issues in Sorting on Parallel Computers, Bubble Sort and Variants, Quick Sort, Other Sorting Algorithms

Graph Algorithms:[4L] Minimum Spanning Tree: Prim's Algorithm, Single-Source Shortest Paths: Dijkstra's Algorithm, All-Pairs Shortest Paths, Transitive Closure, Connected Components, Algorithms for Sparse Graph.

Search Algorithms for Discrete Optimization Problems:[3L] Sequential Search Algorithms, Parallel Depth-First Search, Parallel Best-First Search, Speedup Anomalies in Parallel Search Algorithms.

Module 5:

Models of distributed computation [4L]

Models of distributed computation; Design issues; Operating systems for distributed computing; Distributed algorithms and applications, Clock synchronization algorithms;

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Distributed memory systems; Message passing; Middleware; Point to-point communication; Fault Tolerance; Fault tolerant routing.

Text books:

1. A Grama, AGupra, G Karypis, V Kumar. Introduction to Parallel Computing (2nd ed.). Addison Wesley, 2003. 2. C Lin, L Snyder.
2. Principles of Parallel Programming. USA: Addison-Wesley Publishing Company, 2008.
3. J Jeffers, J Reinders. Intel Xeon Phi Coprocessor High-Performance Programming.

Reference Books

1. Morgan Kaufmann Publishing and Elsevier, 2013. 4. T Mattson, B Sanders, B Massingill. Patterns for Parallel Programming. Addison-Wesley Professional, 2004.

CO-PO & PSO Mapping:

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
CO 1	3	2	2	2	1	2	-	-	-	2	2	2	3	-	-
CO 2	2	3	3	2	2	1	-	-	-	1	2	3	-	1	3
CO 3	2	2	3	1	3	2	2	-	-	1	3	2	-	-	3
CO 4	1	3	1	3	2	3	2	-	-	3	1	1		1	3
CO 5	2	1	2	3	2	2	3	-	-	1	2	3	3	-	-

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Embedded System

Course Code: PEC-CS-S-602

Contact: 3:0:0

Credits: 3

Total Contact Hours: 36

Prerequisite: Knowledge of microprocessor and microcontroller.

Course Objective(s):

1. An ability to design a system, component, or process to meet desired needs within realistic constraints.
2. Ability to understand microcontroller, microcomputer, embedded system.
3. Understand different components of a micro-controller and their interactions.
4. To become familiar with the programming environment used to develop embedded systems.
5. Understand key concepts of embedded systems like IO, timers, interrupts, interaction with peripheral devices
6. Learn debugging techniques for an embedded system

Course Outcome(s):

After completion of the course the student able to do –

CO1	Understand the architecture and classifications of different embedded systems and the related programming knowledge..
CO2	Identify and understand the concepts of embedded systems like I/O, timers, interrupts, interaction with peripheral devices
CO3	Choose case-specific debugging technique for an embedded system
CO4	Design various real time systems using embedded systems.
CO5	Understand the working principles of microcontrollers and apply this knowledge for developing an approach by means of existing and new methods as a team work.

Course Content:

Module 1:

[5L]

Introduction to the Embedded System: Embedded system Vs General computing systems, Purpose of Embedded systems, classifications of embedded systems, fundamentals of embedded processor and microcontrollers, CISC vs. RISC, ASIC.

Module 2:

[9L]

Serial and parallel communication: devices and protocols, wireless communication: devices and protocols, parallel communication network using ISA, PCI, PCT-X, Internet embedded system network protocols, USB, Bluetooth.

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Module 3:**[5L]**

Program Modeling Concepts ; Fundamental issues in Hardware software co-design, Unified Modeling Language(UML), Hardware Software trade-offs DFG model, state machine programming model, model for multiprocessor system.

Module 4:**[5L]**

Real Time Operating Systems: Operating system basics, Tasks, Process and Threads, Multiprocessing and multitasking, task communication, task synchronization, qualities of good RTOS.

Module 5:**[12L]**

PIC microcontroller: introduction, architecture, comparison of PIC with other CISC and RISC based systems and microprocessors, assembly language programming, addressing modes, instruction set, Interfacing with various sensors and actuators using PIC microcontroller. Programming concepts and embedded programming, embedded architecture.

Text Books:

1. Introduction to Embedded Systems : Shibu K. V. (TMH)
2. Embedded System Design – A unified hardware and software introduction: F. Vahid (John Wiley)

Reference Book:

1. Embedded Systems : Rajkamal (TMH)
2. Embedded Systems : L. B. Das (Pearson)
3. Embedded System design : S. Heath (Elsevier)
4. Embedded microcontroller and processor design: G. Osborn (Pearson)

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	3	3	-	-
CO2	3	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	-	-	-	3	3	-	-	-	2	-	2
CO4	3	2	2	3	-	-	-	3	3	-	-	2	2	-	2
CO5	1	2	3	3	2	-	-	3	-	3	-	2	2	2	3

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B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Soft computing

Course Code: PEC-CS-D-602

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites:

Discrete Mathematics, Probability and Statistics

Course Outcome(s):

After completion of the course students will be able to

CO1	Understand and explain the basic concept of soft computing and hard computing and apply them in designing solution to engineering problem.
CO2	Identify and formulate learning rules for each of the architectures and learn several neural network paradigms and its applications to solving engineering and other problems
CO3	Explore relevant literature and apply fuzzy logic and reasoning to handle uncertainty and solving interdisciplinary engineering problems
CO4	Use genetic algorithms to combinatorial optimization problems and recognize the feasibility of applying a soft computing methodology for a particular problem
CO5	Implement the concept and techniques of designing of soft computing methods in real world problem.

Course Content:

Module-1: Introduction to Soft Computing:

8L

An Overview of Artificial Intelligence, Evolution of Computing - Soft Computing Constituents – From Conventional Artificial Intelligence to Computational Intelligence - Machine Learning Basics.

Soft Computing: Introduction of soft computing, soft computing vs. hard computing, various types of soft computing techniques, applications of soft computing

Module-2: Fuzzy sets and Fuzzy logic

7L

Introduction, Fuzzy sets versus crisp sets, operations on fuzzy sets, Extension principle, Fuzzy relations and relation equations, Fuzzy numbers, Linguistic variables, Fuzzy logic, Linguistic hedges, Applications, fuzzy controllers, fuzzy pattern recognition, fuzzy image processing, fuzzy database.

Module -3: Artificial Neural Networks

9L

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Artificial Neural Network: Introduction, basic models, Hebb's learning, Adeline, Perception, Multilayer feed forward network.

Back propagation, Different issues regarding convergence of Multilayer Perceptron, Competitive learning, Self-Organizing Feature Maps, Adaptive Resonance Theory, Associative Memories, Applications.

Module -4: Genetic Algorithms

7L

Evolutionary and Stochastic techniques: Genetic Algorithm (GA), different operators of Genetic Algorithm, Analysis of selection operations, Hypothesis of building Blocks, Schema theorem and convergence of Genetic Algorithm, Simulated annealing and Stochastic models, Boltzmann Machine, Applications.

Rough Set: Introduction, Imprecise Categories Approximations and Rough Sets, Reduction of Knowledge, Decision Tables, and Applications.

Module -5: Hybrid Systems

5L

Neural-Network-Based Fuzzy Systems, Fuzzy Logic-Based Neural Networks, Genetic Algorithm for Neural Network Design and Learning, Fuzzy Logic controlled Genetic Algorithm. Fuzzy Logic and Genetic Algorithm for Optimization, Applications.

Text book:

1. “Neural Networks, Fuzzy logic, and Genetic Algorithms”, S. Rajasekaran & G. A. V. Pai, PHI.
2. “Principles of Soft Computing”, S.N. Sivanandam, S.N. Deepa, Wiley publications.

Reference Books:

1. “Genetic Algorithms in Search, Optimization and Machine Learning”, David E. Goldberg, Addison Wesley, 1997.
2. “Intelligent Hybrid Systems”, D. Ruan, Kluwer Academic Publisher, 1997.
3. “Neural Networks”, S. Haykin, Pearson Education, 2ed, 2001.
4. “An Introduction to Genetic Algorithm”, Mitchell Melanie, Prentice Hall, 1998.

CO – PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	3	3	-	-
CO2	2	3	-	-	2	2	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	2	2	-	-	-	-	-	-	2	-	2
CO4	2	2	2	3	2		-	-	-	-	-	2	2	-	2
CO5	2	2	3	3	-	2	-	-	2	2	-	2	2	2	3

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Human Computer Interaction

Course Code: PCC-CS-A-602

Contacts:3:0:0

Total Contact Hours: 36

Credits:3

Prerequisites: Computer Programming Knowledge, Object Oriented Programming

Course Objectives:

The objectives of this course are to enable students to

1. learn the foundations of Human Computer Interaction.
2. become familiar with the design technologies for individuals and persons with disabilities.
3. be aware of mobile HCI.
4. learn the guidelines for user interface.

Course Outcome(s):

After completion of the course students will be able to

CO1	Understand the fundamental concepts of Human Computer Interaction to Explain or Illustrate and Identify problems where students can Apply the concept appropriately to Solve them.
CO2	Understand the fundamental concepts of Interactive Design so that they can Design user interface as per specification and Validate it.
CO3	Understand and Explain the fundamental concepts of HCI Models and Analyse their effectiveness and limitations.
CO4	Explain or Illustrate the fundamental design principles of HCI for mobile platforms and Use it appropriately to Design user interface as per specification.
CO5	Understand the fundamental design principles of Web-interface and Design web-interface as per specification and analyse the effectiveness as well as limitations of them making the students aware of its utilitarian importance for further explorations leading towards lifelong learning.

Course Contents:

Module-1: FOUNDATIONS OF HCI [9L]

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The Human: I/O channels – Memory – Reasoning and problem solving; The Computer: Devices – Memory – processing and networks; Interaction: Models – frameworks – Ergonomics – styles – elements – interactivity- Paradigms. – Case Studies

Module-2:DESIGN & SOFTWARE PROCESS [9L]

Interactive Design: Basics – process – scenarios – navigation – screen design – Iteration and prototyping. HCI in software process: Software life cycle – usability engineering – Prototyping in practice – design rationale. Design rules: principles, standards, guidelines, rules. Evaluation Techniques – Universal Design

Module-3:MODELS AND THEORIES[4L]

HCI Models: Cognitive models: Socio-Organizational issues and stakeholder requirements – Communication and collaboration models-Hypertext, Multimedia and WWW.

Module-4:MOBILE HCI[9L]

Mobile Ecosystem: Platforms, Application frameworks- Types of Mobile Applications: Widgets, Applications, Games- Mobile Information Architecture, Mobile 2.0, Mobile Design: Elements of Mobile Design, Tools. – Case Studies

Module-5:WEB INTERFACE DESIGN[5L]

Designing Web Interfaces – Drag & Drop, Direct Selection, Contextual Tools, Overlays, Inlays and Virtual Pages, Process Flow – Case Studies

Textbook:

1. Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale, —Human Computer Interaction, 3rd Edition, Pearson Education.
2. Preece J., Rogers Y.,Sharp H.,Baniyon D., Holland S. and Carey T. Human Computer Interaction,Addison-Wesley, 1994.

Reference Books:

3. Brian Fling, —Mobile Design and Development, First Edition, O'Reilly Media Inc., 2009
4. Bill Scott and Theresa Neil, —Designing Web Interfaces, First Edition, O'Reilly

CO-PO Mapping:

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COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	-	3	-	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	-	-	-	-	-	-	-	-	2	-	2
CO4	2	2	2	3	-	-	-	-	-	-	-	-	2	-	2
CO5	2	2	3	3	-	-	-	-	-	-	-	2	2	2	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Paper Name: Introduction to Internet of Things

Paper Code: OEC-CS-601A

Contact (Periods/Week): 3L/Week

Total Credit hours: 36

Credit Point: 3

Prerequisite:

1. Fundamental knowledge in computer networking.
2. Basic knowledge of Microcontroller fundamentals.

Course Objective(s):

Students will understand the concepts of Internet of Things and can able to build IoT applications.

Course Outcome(s):

After completion of the course students will be able to

CO1	Understand and differentiate the fundamental concepts of Internet of Things and the Internet.
CO2	Demonstrate the concepts of wireless sensor network, Analyze and Identify appropriate
CO3	Analyze and compare the basic protocols used in different OSI layer of wireless sensor network and IoT.
CO4	Describe IoT architecture and Machine to machine communication.
CO5	Design basic IoT applications and Solve different real life problems in different domains based upon the concept of IoT and sensor network.

Course Content

Module 1: [9L]

Wireless Sensor Network

Wireless sensor network, application of it, sensor nodes, Network & Communication aspects, Wireless medium access issues, MAC protocol, routing protocols, Sensor deployment & Node

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discovery, Data aggregation & dissemination.

Module 2: [4L]

Fundamental of IoT

The Internet of Things, Time for Convergence, Towards the IoT Universe, Internet of Things Vision, IoT Strategic Research and Innovation Directions, IoT Applications, Future Internet Technologies, Infrastructure, Networks and Communication, Design challenges, Development challenges, Security challenges, other challenges.

Module 3: [7L]

IoT and M2M

A Basic Perspective– Introduction, Some Definitions, M2M Value Chains, IoT Value Chains, An emerging industrial structure for IoT, The international driven global value chain and global information monopolies. M2M to IoT-An Architectural Overview– Building an architecture, Main design principles and needed capabilities, An IoT architecture outline, standards considerations.

Module 4: [7L]

IoT Architecture

Introduction, ArchitectureReference Model- Introduction, Reference Model and architecture, IoT reference Model, IoT Reference Architecture- Introduction, Functional View, Information View, Deployment and Operational View, Other Relevant architectural views.

Module 5: [5L]

IoT Applications for Value Creations

Introduction to Arduino and Raspberry Pi, Cloud Computing, Fog Computing, Connected Vehicles, Data Aggregation for the IoT in Smart Cities, Introduction, IoT applications for industry: Future Factory Concepts, Brownfield IoT, Smart Objects, Smart Applications, Four Aspects in your Business to Master IoT, Value Creation from Big Data and Serialization, IoT in health care, Value for Industry, smart home Management.

Module 6: [4L]

Internet of Things Privacy, Security and Governance

Introduction, Overview of Governance, Privacy and Security Issues, Trust in IoT-Data-Platforms for Smart Cities, First Steps Towards a Secure Platform, Smartie Approach. Data Aggregation for the IoT in smart cities, Security. Data analytics in IoT.

Text books:

1. Vijay Madiseti and Arshdeep Bahga, “Internet of Things (A Hands-on-Approach)”, 1st Edition, VPT, 2014.
2. Francis daCosta, “Rethinking the Internet of Things: A Scalable Approach to Connecting

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Everything”, 1st Edition, Apress Publications, 2013.

Reference books:

1.Cuno Pfister, Getting Started with the Internet of Things, O’Reilly Media, 2011, ISBN: 978-1-4493-9357-1

2.Waltenegus Dargie,ChristianPoellabauer, "Fundamentals of Wireless Sensor Networks: Theory and Practice"

Table 1: Mapping of Course Outcomes with POs (&PSOs)

CO #	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO8	PO9	PO1 0	PO1 1	PO1 2	PS O1	PS O2	PS O3
CO1	3	3	3	3	1	1	-	-	-	-	-	-	2	3	2
CO2	3	3	3	3	2	1	1	-	-	-	1	1	2	3	3
CO3	3	3	3	2	2	1	-	-	-	-	-	-	2	3	3
CO4	3	3	3	2	2	1	1	-	-	-	-	-	2	2	2
CO5	3	3	3	3	3	2	2		1	1	2	1	3	3	3

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Name of the Paper: Bio-informatics

Paper Code: OEC-CS- 601B

Contact (Periods/Week): L-T-P=3-0-0

Credit Point: 3

No. of Lectures: 35

Course Objective(s):

The objective of the course is to make the student able to:

1. Familiar with the basic concept of the Bioinformatics and Molecular Biology and also Familiar with a variety of currently available genomic and proteomic databases.
2. Search and retrieve information from genomic and proteomic databases (e.g. GenBank, Swiss-Prot), and to analyze their search results using software available on the internet (e.g. BLAST, ClustalW).
3. Familiar with the principles and applications of microarrays and locate consensus sequences, genes and open reading frames within biological sequences.
4. Learn how to compare and analyze biological sequences and how to interpret the results of their analyses and how to construct phylogenetic trees based on biological sequence data.
5. Perform elementary predictions of protein structure and function and use the scientific method of inquiry, through the acquisition of scientific knowledge.

Course Outcomes:

After completion of the course students will be able to

CO1	Acquire the knowledge of Bioinformatics technologies with the related concept of DNA, RNA and their implications
CO2	Develop idea in Molecular Biology
CO3	Understand the concept and techniques of different types of Data Organization and Sequence Databases with different types of Analysis Tools for Sequence Data Banks
CO4	Acquire the knowledge of the DNA Sequence Analysis
CO5	Analyze the performance of different types of Probabilistic models used in Computational Biology

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Course Content:

Module -1: Introduction to Molecular Biology [7L]

Concepts of Cell, tissue, types of cell, components of cell, organelle, Functions of different organelles;

Concepts of DNA: Basic Structure of DNA; Double Helix structure; Watson and crick model. Exons and Introns and Gene Concept;

Concepts of RNA: Basic structure, Difference between RNA and DNA. Types of RNA

Concept of Protein: Basic components and structure. Introduction to Central Dogma: Transcription and Translation, Introduction to Metabolic Pathways.

Introduction to Bioinformatics, Recent challenges in Bioinformatics.

Module 2: Introduction to Genomic and MSDN [10L]

Introduction to Genomic data, Data Organization and Sequence Databases: Sequence Data Banks - Introduction to sequence data banks - protein sequence data bank. Signal peptide data bank, Nucleic acid sequence data bank - GenBank, AIDS virus sequence data bank. RRNA data bank, structural data banks - protein Data Bank (PDB), The Cambridge Structural Database (CSD) : Genome data bank - Metabolic pathway data : Microbial and Cellular Data Banks.

Introduction to MSDN (Microbial Strain Data Network): Numerical Coding Systems of Microbes, Hibridoma Data Bank Structure, Virus Information System Cell line information system; Protein Sequence Databases, DNA sequence databases, sequence database search programs like BLAST and FASTA. NCBI different modules: GenBank; OMIM, Taxonomy browser, PubMed;

Module 3: DNA Sequence Analysis [8L]

DNA Mapping and Assembly: Size of Human DNA, Copying DNA: Polymerase Chain Reaction (PCR), Hybridization and Microarrays, Cutting DNA into Fragments, Sequencing

Secondary Structure predictions; Prediction algorithms; Chao-Fasman algorithm, Hidden-Markov model, Neural Networking

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Tertiary Structure predictions; Prediction algorithms; Chao-Fasman algorithm, Hidden-Markov model, Neural Networking

Module 4: Introduction Probabilistic models used in Computational Biology [10L]

Probabilistic Models; Gene Regulatory Method Application of HMM in Bioinformatics: Genefinding, profile searches, multiple sequence alignment and regulatory site identification.

Applications in Biotechnology: Protein classifications, Fold libraries, Protein structure prediction: Fold recognition (threading), Protein structure predictions: Comparative modeling (Homology), Advanced topics: Protein folding, Protein-ligand interactions, Molecular Modeling & Dynamics, Drug Designing.

Text Book:

1. Yi-Ping Phoebe Chen (Ed), “BioInformatics Technologies”, First Indian Reprint, Springer Verlag, 2007.

References Book:

1. Bryan Bergeron, “Bio Informatics Computing”, Second Edition, Pearson Education, 2003.
2. Arthur M Lesk, “Introduction to Bioinformatics”, Second Edition, Oxford University Press, 2005

CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-		1	1	-	-	-	-	-
CO2		1	2	1	-	-	-	-	-	1		1
CO3	1	2		2	2				1	-	-	-
CO4	2	-	-			2	2	-	-	1	1	-
CO5	-	3		1		3	-	1	-	-	2	-

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Paper Name: Robotics

Code: OEC-CS- 601C

Contacts: 3L

Credits: 3

Allotted hours: 35L

Prerequisite:

1. Microprocessor & Microcontroller
2. Computer Organization & Architecture

Course Objective(s):

- To study microcontroller operations for robotics.
- To study how different interfaces are actually implemented in a microcontroller.
- To learn how Microchip PIC micro PIC16F627 can be erased and reprogrammed
- To learn how different sensors, outputs, and peripherals can be wired to a microcontroller to work cooperatively and create a high-level control program.
- To design robots in a real time environment.

Course Outcome(s):

After the successful completion of this course, the student will be able to:

CO1	Understand the basic concepts of robotics exploring the characteristics of its various components, motion control, actuator and drive system and the functions of various sensors in robotics, and robot programming.
CO2	Apply the concepts of robotics for machine loading and their kinematics and analyze the kinematics of serial and parallel robots, motion control systems.
CO3	Illustrate concepts of Linear and angular velocity of links, Velocity propagation, Manipulator Jacobians for serial and parallel manipulators.

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CO4	Understand classical control concepts and use advanced topics in non-linear control of manipulators.
CO5	Develop algorithmic solutions and corresponding robot-programs for designing various robotic systems.

Course contents:

Module 1 [5L]

Brief history, types, classification and usage, Science and Technology of robots, Some useful websites, textbooks and research journals.

Position and orientation of a rigid body, Homogeneous transformations, Representation of joints, link representation using D-H parameters, Examples of D-H parameters and link transforms, different kinds of actuators – stepper, DC servo and brushless motors, model of a DC servo motor, Types of transmissions, Purpose of sensors, internal and external sensors, common sensors – encoders, tachometers, strain gauge based force-torque sensors, proximity and distance measuring sensors, and vision.

Module 2 [8L]

Introduction, Direct and inverse kinematics problems, Examples of kinematics of common serial manipulators, workspace of a serial robot, Inverse kinematics of constrained and redundant robots, Tractrix based approach for fixed and free robots and multi-body systems, simulations and experiments, Solution procedures using theory of elimination, Inverse kinematics solution for the general 6R serial manipulator.

Degrees-of-freedom of parallel mechanisms and manipulators, Active and passive joints, Constraint and loop-closure equations, Direct kinematics problem, Mobility of parallel manipulators, Closed-form and numerical solution, Inverse kinematics of parallel manipulators and mechanisms, Direct kinematics of Gough-Stewart platform.

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Module 3 [8L]

Linear and angular velocity of links, Velocity propagation, Manipulator Jacobians for serial and parallel manipulators, Velocity ellipse and ellipsoids, Singularity analysis for serial and parallel manipulators, Loss and gain of degree of freedom, Statics of serial and parallel manipulators, Statics and force transformation matrix of a Gough-Stewart platform, Singularity analysis and statics.

Mass and inertia of links, Lagrangian formulation for equations of motion for serial and parallel manipulators

Module 4 [9L]

Joint and Cartesian space trajectory planning and generation, Classical control concepts using the example of control of a single link, Independent joint PID control, Control of a multi-link manipulator, Non-linear model based control schemes, Simulation and experimental case studies on serial and parallel manipulators, Control of constrained manipulators, Cartesian control, Force control and hybrid position/force control, Advanced topics in non-linear control of manipulators.

Module 5 [5L]

Introduction and some well known wheeled mobile robots (WMR), two and three-wheeled WMR on flat surfaces, Slip and its modeling, WMR on uneven terrain, Design of slip-free motion on uneven terrain, Kinematics, dynamics and static stability of a three-wheeled WMR's on uneven terrain, Simulations using Matlab and ADAMS.

Introduction to chaos, Non-linear dynamics and chaos in robot equations, Simulations of planar 2 DOF manipulators, Analytical criterion for unforced motion. Gough-Stewart platform and its singularities, use of near singularity for fine motion for sensing, design of Gough-Stewart platform based sensors. Over-constrained mechanisms and deployable structures, Algorithm to obtain redundant links and joints, Kinematics and statics of deployable structures with pantographs or scissor-like elements (SLE's).

Textbooks:

1. Myke Predko, —Programming Robot Controllers— McGraw Hill, 1st edition, 2003.

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Reference books:

1. Michael Slater, —Microprocessor – based design: A comprehensive Guide to Effective Hardware Design, Prentice Hall, 1989.
2. Myke Predko, —Programming and customizing the 8051- micro-controller, Tata McGraw-Hill, New Delhi, 2000.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	1	-	-	-	-	-	2	-
CO2	2	3	-	1	-	-	-	-	-	-	-	-
CO3	2	3	3	-	-	-	-	-	-	-	-	-
CO4	2	2	2	1	1	-	-	-	1	-	-	3
CO5	3	3	3	3	3	2	2	-	2	-	2	2

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Practical

Course Name: Computer Networks Lab

Code: OEC-CS- 601C

Contact: L: T: P/0:0:3

Total Contact Hours: 36

Credits: 2

Course Outcome(s)

After completion of the course the student able to do

CO1	To design and implement small size network and to understand various networking commands.
CO2	To provide the knowledge of various networking tools and their related concepts.
CO3	To understand various application layer protocols for its implementation in client/server environment
CO4	Understand the TCP/IP configuration for Windows and Linux
CO5	Learn the major software and hardware technologies used on computer networks

Detail Syllabus:

Module 1:

Familiarization of UNIX or Linux environment, UNIX or Linux general Commands specially Network Commands. Familiarization of Internetworking - Network Cables - Color coding - Crimping. Internetworking Operating Systems - Configurations. [5L]

Module 2:

Implementation of flow control mechanisms [4L]

Module 3:

Socket Programming using TCP and UDP [15]

Module 4:

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Implementing routing protocols such as RIP, OSPF [4L]

Module 5:

Familiarization of advanced simulators like Packet Tracer, NS2/NS3, OMNET++, TinyOS.[4L]

Server Configuration: only web server [4L]

Text books:

1. B. A. Forouzan – “Data Communications and Networking (3rd Ed.) “ – TMH
2. S. Tanenbaum – “Computer Networks (4th Ed.)” – Pearson Education/PHI

Recommended books:

1. TCP sockets in C programs-Practical guide for Programmers By Micheal J Donahoo and Kenneth Calvert.
2. Socket Programming by rajkumar Buyaa.

CO-PO mapping

CO #	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
CO 1	1	-	-	-	-	-	-	-	-	-	-	-	-	-	3
CO 2	-	3	-	-	-	-	-	-	-	-	-	-	-	1	3
CO 3	-	-	-	-	-	-	-	-	3	2	3	2	3	-	-
CO 4	2	-	3	-	-	2	2	-	-	-	-	-	-	2	3
CO 5	1	-	-	-	-	-	-	-	-	-	3	-	-	-	2

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Software Engineering Lab

Course Code: PCC-CS 692

Contact: 3:0:0

Total Contact Hours: 36

Credits: 1.5

Prerequisites: Programming for Problem Solving

Course Objective(s)

- 1.To learn software development skill through various stages of software life cycle.
2. To ensure the quality of software through software development with various protocol based environment.
3. Graduates build upon and adapt knowledge of science, mathematics, and engineering to take on more expansive tasks that require an increased level of self-reliance, technical expertise, and leadership.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the basic concept of Software Engineering and mathematical knowledge and apply them in designing solution to engineering problem including he specification, design , implementation, and testing of software systems that meet specification, performance, maintenance and quality requirements
CO2	Analyze , elicit and specify software requirements through a productive working relationship with various stakeholders of the project
CO3	Design applicable solutions in one or more application domains using software engineering approaches that integrates ethical, social, legal and economic concerns.
CO4	Develop the code from the design and effectively apply relevant standards and perform testing, and quality management and practice team work.
CO5	Identify and Use modern engineering tools necessary for software project management time management and software reuse, and an ability to engage in life-long learning.

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Content:**Module-I: Introduction****6**

Preparation of requirement document for standard application problems in standard format. (e.g. Library Management System, Railway Reservation system, Hospital management System, University Admission system) .DFD of standard application problems.

Module-II: Software life cycle models**6**

Software Requirement Analysis: Describe the individual Phases/ modules of the project, Identify deliverables. Compute Process and Product Metrics (e.g Defect Density, Defect Age, Productivity, Cost etc.) Estimation of project size using Function Point(FP) for calculation.

Cost Estimation models.

Module-III: Software design**6**

Use Case diagram, Class Diagram, Sequence Diagram, Activity Diagram and prepare Software Design Document using tools like Rational Rose.(For standard application problems)

Module-IV: Software Coding& Testing**9**

Software Development, Coding Practice and Debugging, Design Test Script/Test Plan(both Black box and White Box approach)

Module-V: Software project management**9**

Software project management, Project planning and control, configuration control, cost estimation, project scheduling using PERT and GANTT charts, cost-time relations using standard tools.

Text Books:

1. Fundamentals of Software Engineering by Rajib Mall, –PHI-3rd Edition, 2009.
2. Software Engineering- Pankaj Jalote (Wiley-India)

Reference Books:

1. Software Engineering –Agarwal and Agarwal (PHI)

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2. Software Engineering, by Ian Sommerville, Pearson Education Inc., New Delhi, (2009).

3. Software Engineering: A Practitioner's Approach", by Roger S. Pressman, McGraw-Hill, (2005)

CO – PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO2	3	3	2	2	-	-	-	-	-	-	-	-	3	2	3
CO3	3	3	3	3	-	-	-	3	-	-	-	3	3	3	2
CO4	3	3	3	3	-	-	-	-	3	-	-	-	3	3	3
CO5	3	2	3	2	3	-	-	-	3	2	3	3	3	2	2

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B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Advanced Algorithms Lab

Course Code: PEC-CS-T-691

Contact: 0:0:2

Credits: 1.5

Prerequisites:

1. Programming knowledge
2. Knowledge of Design and Analysis of Algorithm

Objective of the course:

17. Design and implement efficient algorithms for a specified application.
18. Strengthen the ability to identify and apply the suitable algorithm for the given real-world problem.

Course Outcomes:

After completion of the course students will be able to

CO1	Introduce students to the advanced strategies of designing and analyzing algorithms.
CO2	The student should be able to prefer suitable algorithms and use it for a precise problem.
CO3	To familiarize students with basic paradigms and data structures used to solve advanced algorithmic problems.
CO4	Students should be able to understand different classes of problems concerning their computation difficulties.
CO5	To introduce the students to recent developments in the area of algorithmic design.

Course Content:

Write the following problems in any programming language. Programming Language used: C

1. **Divide and Conquer:** Implementation of finding Maximum and Minimum element from an array of integer, Quick Sort, Check the running time for different positions of pivot elements. Randomized version of quick sort using Divide and Conquer Method.
2. **Dynamic Programming:** Calculation of the minimum number of scalar multiplications needed for chain of Matrices Multiplication Technique, Implementation of Single Source shortest Path for a graph (Dijkstra and Bellman Ford Algorithm), Implement all pair Shortest path for a graph (FloydWarshall Algorithm)
3. **Greedy method:** Implementation of fractional Knapsack Problem, MST by Prim's algorithm, Implement MST by Kruskal's algorithm
4. **Graph Traversal Algorithm:** Implement Depth First Search (DFS), application of DFS (do topological sorting, identify strongly connected components)

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5. **String Matching:** Implement KMP algorithm
6. **Network Flow:** Implement Ford-Fulkerson algorithm to get maximum flow of a given flow network.
7. **Modulo Representation of integers/ polynomials:** Chinese Remainder Theorem
8. **Linear Programming:** Simplex Algorithm

CO – PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	3	3	2	2	3
CO2	3	3	3	3	-	-	-	-	-	-	3	3	2	3	3
CO3	3	3	3	3	-	-	-	-	-	-	3	3	3	3	3
CO4	3	3	3	3	-	-	-	-	-	-	3	3	3	2	3
CO5	3	3	3	3	-	-	-	-	-	-	3	3	3	3	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Advanced Operating Systems Lab

Course Code: PEC-CS-S-691

Contact: 0:0:3

Total Contact Hours: 36

Credits: 1.5

Prerequisites: Operating Systems

Course Objectives:

The objective of the course is to make the students able to -

- 1 . Understand and execute basic commands of shell script
- 2 Apply basic operations in shell scripts which are required for different applications.
- 3 Identify and understand concept of file systems in shell script
- 4 able to understand the concept of creating new process from parent process.
- 5 Able to understand concept of virtual file and execute basic commands on it

Course Outcomes:

After completion of the course students will be able to

CO1	Understand and implement basic services and functionalities of the operating system using system calls and able to Understand the benefits of thread over process and implement synchronized programs using multithreading concepts.
CO2	Analyze the design aspects and issues of distributed operating systems.
CO3	Analyze and simulate CPU Scheduling Algorithms like FCFS, Round Robin, SJF, and Priority.
CO4	Implement memory management schemes and page replacement schemes
CO5	Understand the concepts of deadlock in operating systems and implement them in Multiprogramming system.

Course Content:

Curriculum for B.Tech Under Autonomy
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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Preliminaries of Operating System: 6P

managing users, managing systems, file managements, useful commands, Shell scripting : shell syntax, executing shell scripts.

Process : 12P

creating new process, counting maximum number of processes a system can handle at a time, handling system calls; inter process communication through pipes and message passing, zombie process, orphan process.

Process Synchronization: 6P

handling threads and semaphores to achieve synchronization among processes using POSIX standard functions.

Signal : 6P

study of some POSIX signals (SIGINT, SIGILL, SIGFPE, SIGKILL, SIGHUP, SIGALRM, SIGABRT).

Text book:

1. Mukesh Singhal and Niranjana Shivaratri, Advanced Concepts in Operating Systems, McGraw-Hill.

Reference Books:

1. Nancy Lynch, Distributed Algorithms, Morgan Kaufmann.

CO – PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO2	3	3	3	3	-	-	-	-	-	-	-	-	3	2	3
CO3	3	3	3	3	-	-	-	-	-	-	-	3	3	3	2
CO4	3	3	3	3	-	-	-	-	-	-	-	-	3	3	2
CO5	3	3	3	3	-	-	-	-	3	3	-	-	3	2	2
CO	3	3	3	3	-	-	-	-	3	3	-	3	3	3	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Machine Learning Lab

Course Code: PEC-CS-D-691

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: Data Structure, Design and Analysis of Algorithms, Statistics, Artificial Intelligence, Python Programming

Course Objectives:

11. Comprehend and Implement the fundamental concepts of the evolving technologies in machine learning such as Supervised and Unsupervised Learning
12. Formulate an engineering problem within the scope of machine learning paradigm.
13. Implement the concepts of machine learning to solve problems of making automated decisions dealing with large scale data.
14. Develop and Implement ideas for proposing solutions to the challenging problems of machine learning
15. Analyze the effectiveness of various machine learning Frameworks using appropriate tools.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand and Implement the basics concepts of machine learning to Explain or Illustrate and Identify problems where students can Apply the concept appropriately to Solve them.
CO2	Understand and Implement the fundamental concepts of regression analysis so that they can propose models for predicting values based on exemplary data and Analyze their performances.
CO3	Understand and Implement the fundamental strategies of unsupervised machine learning paradigm to solve clustering problems and Analyze their performances.
CO4	Understand and Implement the concepts of Mining Frequent Patterns, Associations and Data Streams and Apply them to solve the relevant problems and Analyze their performances.

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CO5	Develop ideas to Propose solutions to the problems of supervised learning and Identify problems where students can Apply and Implement the concept appropriately with adequate documentation in collaborative environment for successfully carrying out projects on machine learning problems and investigate their effectiveness by analyzing the performances using proper techniques and tools and Assess the limitations of solutions underscoring utilitarian importance for further explorations leading towards lifelong learning.
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Course Content:

WEEK-1: Introduction to Machine Learning Programming Platform & Python Programming Basics

Introduction to Machine Learning Programming Platform and Python Programming Basics

WEEK-2: Data Exploration

Data Exploration: Data Types, Data Attributes, Statistical Description of Data, Data Visualization, Data Feature Vectors, Data Preprocessing: Data Cleaning, Data Transformation

WEEK -3: Regression

Implementation and Analysis of Linear and Nonlinear Regression Methods

WEEK -4: Classification

Implementation and Analysis of k-Nearest-Neighbor Classifier, Decision Tree Classifier, Naïve Bayes Classifier

WEEK -5: Classification

Implementation and Analysis of ANN-Backpropagation and SVM Based Classifier

WEEK-6: Clustering

Implementation and Analysis of k-Means and k-Medoids

WEEK -7: Association Analysis

Implementation and Analysis of Apriori Algorithm

WEEK -8: Mining Time-Series Data

WEEK -9: Discussion on Project Problems and Allocation (Problem Description Report Submission)

WEEK -11: Project Implementation, Verification and Documentation

Textbook:

- ### Reference Books:

- ### CO–PO Mapping:

[illegible]

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Name of the Paper: Web and Internet Technology

Paper Code: PEC-CS-A-601

Contact (Periods/Week): 3L/Week

Credit Point: 3

No. of Lectures: 36

Course Objective(s):

- To impart the design, development, and implementation of Static and Dynamic Web Pages.
- To develop programs for Web using Scripting Languages and .net framework.
- To give an overview of Server Side Programming in Web.

Course Outcomes:

After completion of the course students will be able to:

CO1	Understand networks, IP, DNS, routing and other related technologies used in internet and execute and solve problems related to them leading to engineering problems solutions
CO2	Understand different web based technologies like HTML, DHTML, CSS, XML and demonstrate their use in design of web based solutions leading to engineering problems
CO3	Comprehend and analyze different client and server side technologies like JavaScript, Servlet, CGI and design appropriate engineering solutions leading to life long learning
CO4	Understand and implement different types of technologies like JSP, JavaBean, JDBC and ODBC and evaluate their performances
CO5	Understand different web based applications and network security techniques and apply them to protect the network against different attacks and solve related problems preferably as a team

Course Contents:

Module 1: [6L]

Introduction (1L): Overview, Network of Networks, Intranet, Extranet, and Internet. World Wide Web (1L): Domain and Sub domain, Address Resolution, DNS, Telnet, FTP, HTTP. Review of TCP/IP (1L): Features, Segment, Three-Way Handshaking, Flow Control, Error Control, Congestion control, IP Datagram, IPv4 and IPv6. IP Subnetting and addressing (1L): Classful

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and Classless Addressing, Subnetting. NAT, IP masquerading, IP tables. Internet Routing Protocol (1L): Routing -Intra and Inter Domain Routing, Unicast and Multicast Routing, Broadcast. Electronic Mail (1L): POP3, SMTP, Clients - Servers Communication.

Module -2: [9L]

HTML, DHTML & CSS: Introduction, Elements, Attributes, Heading, Paragraph. Formatting[1L]; Link, Table, List, Block, Layout, Html Forms, and input [1L]; Iframe, Colors[1L], Image Maps and attributes of image area [1L]; Introduction to CSS, basic syntax and structure of CSS, different types internal, external and inline CSS [1L]; Basic Introduction of DHTML, Difference between HTML and DHTML, Documentary Object Model (DOM) [1L]. Extended Markup Language (XML) : Introduction, Difference between HTML & XML, XML-Tree [1L]; Syntax, Elements, Attributes, Validation and parsing, DTD [2L].

Module 3: [15L]

Java Scripts: Basic Introduction, Statements, comments, variable, operators, data types [1L]; condition, switch, loop, break [1L]; Java script functions, objects, and events[1L]. CGI Scripts: Introduction, Environment Variable, GET and POST Methods [1L]. Java Servlet: Servlet environment and role, Servlet life cycle [1L]; Servlet methods- Request, Response, Get and post [1L]; Cookies and Session [1L].

Java Server Page (JSP):

JSP Architecture [1L]; JSP Servers, JSP Life Cycle [1L]; Understanding the layout of JSP, JSP Scriptlet Tag [1L]; JSP implicit object (request and response) [1L]; Variable declaration, methods in JSP [1L]; JSP directive (Taglib and Include), JavaBean- inserting JavaBean in JSP [1L]; JSP Action tags (Forward & Include) [1L]; Creating ODBC data source name, Introduction to JDBC, prepared statement and callable statement [1L].

Module-4: [6L]

Threats[1L]: Malicious code-viruses, Trojan horses, worms; eavesdropping, spoofing, modification, denial of service attacks. Network security techniques: Password and Authentication; VPN, IP Security [1L], security in electronic transaction, Secure Socket Layer (SSL), Secure Shell (SSH)[1L]. Firewall (1L): Introduction, Packet filtering, Stateful, Application layer, Proxy. Search Engine and Web Crawler: Definition, Meta data, Web Crawler [1L], Indexing, Page rank, overview of SEO[1L].

Textbooks:

1. “Web Technology: A Developer's Perspective”, N.P. Gopalan and J. Akilandeswari, PHI Learning, Delhi, 2013. (Topics covered: html, CSS, imagemap, xml)
2. “Learning PHP, MySQL & JavaScript”, Robin Nixon, O'Reilly Publication. (Topics covered: Java Script)
3. “Head First Servlet's & JSP”, Bryan Basham, Kathy Sterra, Bert Bates, O'Reilly Publication. (Topics covered: Servlet, JSP)

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4.Cryptography and Network Security by William Stallings Publisher: Pearson Education India(Topics covered: Threats, Security techniques, Firewall)

Recommended books:

1. "Programming the World Wide Web", Robert. W. Sebesta, Fourth Edition, Pearson Education, 2007.
2. “Core Web Programming”- Second Edition-Volume I and II, Marty Hall and Larry Brown, Pearson Education, 2001

CO-PO & PSO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	-	-	-	-	-	-	-	2	3	3
CO2	3	3	3	3	2	-	-	-	-	-	-	-	3	3	3
CO3	3	3	3	3	2	-	-	-	-	-	-	3	3	3	3
CO4	3	3	3	3	2	-	-	-	-	-	-	-	3	3	3
CO5	3	3	3	3	2	-	-	-	3	-	2	-	3	3	3

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MANDATORY COURSE

Course Name: Intellectual Property Right

Course Code: MC 601

Contacts: 2:0:0

Total Contact Hours: 24

Credit: 0

Prerequisite: None

Course Outcome(s):

On successful completion of the learning sessions of the course, the learner will be able to:

After completion of the course students will be able to:

CO1	Explain fundamental aspects of Intellectual property Rights to students
CO2	To disseminate knowledge on patents, patent regime in India and abroad and registration aspects
CO3	To disseminate knowledge on copyrights and its related rights and registration aspects
CO4	To aware about current trends in IPR and Govt. steps in fostering IPR
CO5	To disseminate knowledge on Design, Geographical Indication (GI), Plant Variety and Layout Design Protection and their registration aspects

Course Content:

Module 1: [4L]

Overview of the IPR: Introduction and the need for intellectual property right (IPR) - Kinds of Intellectual Property Rights: Patent, Copyright, Trade Mark, Design, Geographical Indication, Plant Varieties and Layout Design – Genetic Resources and Traditional Knowledge – Trade Secret - IPR in India : Genesis and development – IPR in abroad - International organizations. Agencies and treaties,

Module 2:[4L]

Patents- Trips Definition, kind of inventions protected by patent-Patentable and Non patentable inventions. Elements of Patentability: Novelty , Non Obviousness (Inventive Steps), Legal requirements for patents — Granting of patent - Rights of a patent-exclusive right. Patent application process: Searching a patent- Drawing of a patent- Filing of a patent- Types of patent applications- Patent document: specification and Claims.

[illegible]

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B.Tech. – Computer Science & Technology

L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

CO2	3	3	3	3	2	-	-	-	-	-	-	-	3	3	3
CO3	3	3	3	3	2	-	-	-	-	-	-	3	3	3	3
CO4	3	3	3	3	2	-	-	-	-	-	-	-	3	3	3
CO5	3	3	3	3	2	-	-	-	3	-	2	-	3	3	3

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4 th Year 1 st Semester: 7 th Semester								
Sl No	Course Code	Paper Code	Theory	Contact Hours /Week				Credit Points
				L	T	P	Total	
A. THEORY								
1	Professional Elective courses	PEC-CS-T-701	Information Theory and Coding	3	0	0	3	3
		PEC-CS-S-701	Ad-Hoc and Sensor Networks					
		PEC-CS-D-701	Data Mining and Data Warehousing					
		PEC-CS-A-701	Cloud Computing					
2	Professional Elective courses	PEC-CS-T-702	Quantum Computing	3	0	0	3	3
		PEC-CS-S-702	Mobile Computing					
		PEC-CS-D-702	Natural Language Processing					
		PEC-CS-A-702	Cryptography and Network Security					
3	Open Elective courses	OEC-CS-701A	High Performance Computing	3	0	0	3	3
		OEC-CS-701B	VLSI					
		OEC-CS-701C	Optimization Techniques					
4	Open Elective courses	OEC-CS-702A	Cyber Law and Ethics	3	0	0	3	3
		OEC-CS-702B	Soft Skills and Interpersonal Communication					

		OEC-CS-702C	Foreign Language					
B. PRACTICAL								
5	Professional Elective courses	PEC-CS-T-791	Information Theory and Coding Lab	0	0	0	3	1.5
		PEC-CS-S-791	Ad-Hoc and Sensor Networks Lab					
		PEC-CS-D-791	Data Mining and Data Warehousing Lab					
		PEC-CS-A-791	Cloud Computing Lab					
6	Open Elective courses	OEC-CS-791A	High Performance Computing Lab	0	0	3	3	1.5
		OEC-CS-791B	VLSI Lab					
		OEC-CS-791C	Optimization Techniques Lab					
7	PROJECT	PR 791	Major Project-I	0	0	0	4	2
8	PROJECT	PR 792*	Industrial Training / Internship	0	0	0	0	1
9	PROJECT	PR 793	Skill Development VII: Seminar & Group Discussion	1	0	0	1	0.5
C. MANDATORY ACTIVITIES / COURSES								
10	MC	MC 781	Entrepreneurship & Innovation Skill	3	0	0	3	3 Units
TOTAL CREDIT WITHOUT MOOCS COURSES								18.5
D.MOOCS COURSES**								
11	MOOCS COURSES	HM701	MOOCS COURSE-V	3	1	0	4	4
TOTAL CREDIT WITH MOOCS COURSES								22.5

Curriculum for B.Tech Under Autonomy

B.Tech. – Computer Science & Technology

L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Information Theory and Coding

Course Code: PEC-CS-T-701

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: Probability & Statistics

Course Objective:

The objective of the course is to make the students able to

1. Illustrate the basic concept of information and apply this knowledge in designing solution.
2. Illustrate the basic concept of coding theory and use this knowledge to design and solve problem.
3. Interpret the concept of channel models to determine the mutual information in the channels.
4. Compare the existing error detection techniques and design a model for building a new solution.
5. Understand convolutional theory and develop a new approach.

Course Outcomes:

After completion of the course students will be able to

CO1	Illustrate the basic concept of information and apply this knowledge to design solution for real life engineering problem.
CO2	Illustrate the basic concept of coding theory and use this knowledge to design and solve mathematical and engineering problem leading to lifelong learning.
CO3	Interpret the concept of channel models to find amount of mutual information in the channels.
CO4	Compare the existing error detection techniques and design a model for building a new solution as a professional engineering practice as a team.

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B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

CO5	Understand how convolutional theory works and develop an approach by means of existing and new methods as a team work.
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Course Content:

Module 1: Information Theory [4]

Introduction, Measure of Information, Average Information Content (Entropy) of a Zero Memory Source, Extension of Zero Memory Source, Entropy of a Source with Memory.

Module 2: Source Coding [9]

Introduction, Types of Codes, Prefix Codes, Source Coding Theorem, Shannon's Encoding Theorem, Huffman Coding, Arithmetic Coding, Lempel-Ziv Algorithm, Run Length Encoding, An Overview on Speech and Image Compression.

Module 3: Information Channels[4]

Introduction, Channel Models, System Entropies, Mutual Information (Trans information), Channel Capacity, Capacity of Channels, Continuous Channels.

Module 4: Error Control Coding [8]

Introduction, need for Error Control Coding, Types of Codes, Coding Gain, Linear Block Codes, The Hamming Codes, Probability of an Undetected Error Pattern for an LBC over a BSC, Equivalent Codes, Cyclic Codes, Golay Codes, Shortened Cyclic Codes.

Module 5: Burst Error Correcting Codes [6]

Introduction, Burst Errors, Interleaved Codes, Product Codes, Fire Codes, BCH Codes, Non-Binary BCH Codes and Reed-Solomon Codes.

Module 6: Convolution Codes[5]

Introduction, Convolution Encoder, Representation of Convolution Code, Transfer Function of a Convolution Code, Distance Properties of Convolution Codes, Decoding of Convolution Codes, Stack Algorithm, Known Good Convolution Codes.

Textbook:

1. Information theory, coding and cryptography - Ranjan Bose; TMH.

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2. Information and Coding - N Abramson; McGraw Hill.

Reference Books:

1. Introduction to Information Theory - M Mansurpur; McGraw Hill.
2. Information Theory - R B Ash; Prentice Hall.
3. Error Control Coding - Shu Lin and D J Costello Jr; Prentice Hall.

CO–PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO2	3	3	3	3	-	-	-	-	-	-	-	3	3	2	3
CO3	3	3	3	3	-	-	-	-	-	-	-	-	3	3	1
CO4	3	3	3	3	-	-	-	-	3	-	-	-	3	2	3
CO5	3	3	3	3	-	-	-	-	3	2	-	-	3	2	3
CO	3	3	3	3	-	-	-	-	3	2	-	3	3	3	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Name of the Paper: Ad-Hoc and Sensor Networks

Paper Code: PEC-CS-S-701

Contact :=3L/Week

Credit Point: 3

No. of Lectures: 35

Prerequisite:

4. Basic concept of computer network and communication engineering
5. Basic programming knowledge

Course Objective(s)

The objective of the course is to make the students able to –

- To understand and illustrate the principles of mobile ad-hoc networks and sensor networks, and their impact on protocol design
- To design and develop MAC and routing protocols for mobile ad-hoc and sensor networks
- To develop efficient protocols for mobile ad-hoc and sensor networks
- To analyze and develop information dissemination protocols for mobile ad-hoc and sensor networks

Course Outcome(s)

After completion of the course students will be able to

CO1	Understand and explain the Fundamental Concepts and applications of ad- hoc and wireless sensor networks
CO2	Describe and analyze the MAC protocol issues of ad-hoc networks
CO3	Design and develop routing protocols for ad-hoc wireless networks with respect to TCP design issues
CO4	Explain the concepts of network architecture and MAC layer protocol for WSN
CO5	Develop and analyze the WSN routing issues by considering QoS measurements

Course Content:

Module I: Introduction [5L]:

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Fundamentals of Wireless Communication Technology -The Electromagnetic Spectrum - Radio propagation Mechanisms - Characteristics of the Wireless channel mobile ad-hoc networks (MANETs) - Wireless Sensor Networks (WSNs): concepts and architectures - Applications of Ad-Hoc and Sensor Networks - Design Challenges in Ad-hoc and Sensor Networks.

Module II: MAC Protocols for Ad Hoc Wireless Networks [7L]

Issues in designing a MAC Protocol - Issues in Designing a MAC Protocol for Ad-Hoc Wireless Networks - Design Goals of a MAC Protocol for Ad-Hoc Wireless Networks - Classification of MAC Protocols -Contention based protocols - Contention based protocols with Reservation Mechanisms - Contention based protocols with Scheduling Mechanisms - Multi channel MAC - IEEE 802.11.

Module III: Routing Protocols and Transport Layer in Ad Hoc Wireless Networks [10L]

Routing Protocol: Issues in designing a routing protocol for Ad-hoc networks - Classification- proactive routing - reactive routing (on-demand) - hybrid routing - Transport Layer protocol for Ad-hoc networks - Design Goals of a Transport Layer Protocol for Ad-Hoc Wireless Networks -Classification of Transport Layer solutions-TCP over Ad-hoc wireless.

Network Security - Security in Ad-Hoc Wireless Networks – Network Security Requirements.

Module IV: Wireless Sensor Networks (WSNs) and MAC Protocols [5L]

Single node architecture: hardware and software components of a sensor node -WSN Network architecture: typical network architectures -data relaying and aggregation strategies -MAC layer protocols: self-organizing - Hybrid TDMA/FDMA and CSMA based MAC -IEEE 802.15.4.

Module V: WSN Routing, Localization & QOS [8L]

Issues in WSN routing –OLSR - Localization –Indoor and Sensor Network Localization - absolute and relative localization - triangulation - QOS in WSN - Energy Efficient Design – Synchronization.

Text books:

- 1) C. Siva Ram Murthy, and B. S. Manoj, "Ad Hoc Wireless Networks: Architectures and Protocols ", Pearson Education, 2008.
- 2) Labiod. H, "Wireless Adhoc and Sensor Networks", Wiley, 2008.
- 3) Li, X, "Wireless ad -hoc and sensor Networks: theory and applications", Cambridge University Press, 2008.

Reference books:

1. Carlos De Moraes Cordeiro, Dharma Prakash Agrawal "Ad Hoc & Sensor Networks: Theory and Applications", World Scientific Publishing Company, 2nd edition, 2011.

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2. Feng Zhao and Leonides Guibas, "Wireless Sensor Networks", Elsevier Publication
3. Holger Karl and Andreas Willig "Protocols and Architectures for Wireless Sensor Networks", Wiley, 2005
4. Kazem Sohraby, Daniel Minoli, & Taieb Znati, "Wireless Sensor Networks Technology, Protocols, and Applications", John Wiley, 2007.
5. Anna Hac, "Wireless Sensor Network Designs", John Wiley, 2003.

CO-PO Mapping

CO #	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	2	2	2	2	1	1	-	-	3	2	3	2	2	2
CO2	3	3	3	3	2	1	1	-	-	3	2	3	2	2	2
CO3	3	3	2	3	2	1	1	-	-	3	2	3	2	2	2
CO4	3	3	2	2	2	1	1	-	-	3	2	3	2	2	2
CO5	3	3	3	3	2	1	1	-	-	3	2	3	2	2	2
CO	3	3	2	3	2	1	1	-	-	3	2	3	2	2	2

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Course Name: Data Mining and Data Warehousing

Course Code: PEC-CS-D-701

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: Data Structure, Design and Analysis of Algorithms, Database Management Systems, Statistics, Artificial Intelligence

Course Objectives:

- 1 Comprehend the fundamental concepts of the evolving technologies in Data Mining (such as Mining Frequent Patterns and Data Streams, Associations, Supervised and Unsupervised Learning, Graph Mining, Web Mining etc.) and Data Warehousing (such as Data Cube and OLAP)
- 2 Formulate an engineering problem within the scope of Data Mining and Data Warehousing paradigm.
- 3 Apply the concepts of Data Mining and Data Warehousing to solve problems of making automated decisions dealing with large scale data.
- 4 Develop and Implement ideas for proposing solutions to the challenging problems of Data Mining and Data Warehousing.
- 5 Analyze the effectiveness of various Data Mining and Data Warehousing Frameworks.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand and explain the fundamental concepts of the evolving technologies in Data Mining (such as Mining Frequent Patterns and Data Streams, Associations, Supervised and Unsupervised Learning, Graph Mining, Web Mining etc.) and Data Warehousing (such as Data Cube and OLAP) recognizing their utilitarian importance in current technological context for further exploration leading towards lifelong learning.
CO2	Identify and formulate an engineering problem within the scope of Data Mining and Data Warehousing paradigm.
CO3	Explore relevant literature and apply the concepts of Data Mining and Data Warehousing to solve problems of making automated decisions dealing with large scale data.

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CO4	Develop ideas for proposing solutions to the challenging problems of Data Mining and Data Warehousing.
CO5	Implement ideas of Data Mining and Data Warehousing through developing feasible algorithms or frameworks and investigate their effectiveness in solving the relevant problems by analyzing the performances using proper techniques.

Course Content:

MODULE-1: Introduction to Data Mining [5L]

Basic Concepts, Data Exploration: Data Types, Data Attributes, Statistical Description of Data, Data Visualization, Data Similarity Measures; Data Preprocessing: Data Cleaning, Data Integration, Data Reduction, Data Transformation & Discretization.

MODULE-2: Introduction to Data Warehousing [6L]

Basic Concepts, Data Warehouse Modeling: Data Cube and OLAP (OnLine Analytical Processing); Data Warehouse Design, Usage, and Implementation; Data Generalization by Attribute-Oriented Induction.

MODULE-3: Mining Frequent Patterns, Associations and Correlation Analysis [5L]

Basic Concepts, Frequent Itemset Mining Methods: The Apriori Algorithm, Mining Frequent Item Sets without Candidate Generation, Mining Frequent Item Sets Using Vertical Data Format, Correlation Analysis; Pattern Mining in Multilevel and Multidimensional Space.

MODULE-4: Classification and Regression [6L]

Basic Concepts, k-Nearest-Neighbor Classifier, Decision Tree Classifier, Naïve Bayes Classifier; ANN-Backpropagation Based Classifier, Support Vector Machine Based Classifier, Linear and Nonlinear Regression Methods.

MODULE-5: Clustering and Outlier Analysis [5L]

Basic Concepts, Partitioning Methods: k-Means and k-Medoids, Hierarchical Methods: Agglomerative and Divisive Hierarchical Clustering, Density-Based Methods: DBSCAN: Density-Based Clustering Based on Connected Regions with High Density, Frequent Pattern–Based Clustering Method; Outlier Analysis.

MODULE-6: Mining Data Stream, Time-Series, and Sequence Data [3L]

Basic Concepts of Data Stream Mining; Mining Time Series Data; Mining Sequence Patterns in Biological Data.

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MODULE-7: Introduction to Graph Mining, Social Network Analysis, Multi-relational Data Mining, Text Mining and World Wide Web (WWW) Mining [6L]

Graph Mining: Methods for Mining Frequent Subgraphs (Apriori-based Approach & Pattern Growth Approach); Basic Concepts of Social Network Analysis and Multi-relational Data Mining; Basic Concepts of Text Mining; Basic Concepts of World Wide Web (WWW) Mining.

Textbook:

14. Han J & Kamber M, “Data Mining: Concepts and Techniques”, Morgan Kaufmann Publishers, Third Edition.
15. Parteeek Bhatia, “Data Mining and Data Warehousing: Principles and Practical Techniques”, Cambridge University Press.

ReferenceBooks:

15. Pang-Ning Tan, Vipin Kumar, Michael Steinbach, “Introduction to Data Mining”, Pearson Education.
16. Robert Layton, “Learning Data Mining with Python”, Packt Publishing

CO–PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	3	3	-	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	-	-	-	-	-	-	-	-	2	-	2
CO4	2	2	2	3	-	-	-	-	-	-	-	2	2	-	2
CO5	2	2	3	3	2	2	2	-	-	-	-	2	2	2	3

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B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Cloud Computing

Course Code: PEC-CS-A-701

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisite:

1. The student must have basic knowledge in Computer Network and Distributed System

Course Objectives(s):

1. To provide students with the fundamentals and essentials of Cloud Computing.
2. Understand the importance of virtualization in distributed computing and how this has enabled the development of Cloud Computing.
3. Understand the importance of protocols and standards in computing.

Course Outcome(s)

After completion of the course the student able to do

CO1	Identify the appropriate cloud services for a given application
CO2	Assess the comparative advantages and disadvantages of Virtualization technology
CO3	Analyze authentication, confidentiality and privacy issues in cloud computing
CO4	Identify security implications in cloud computing.
CO5	Understand the importance of protocols and standards in management for cloud services.

Detail Syllabus:

Module 1: Definition of Cloud Computing and its Basics [8L]

Definition of Cloud Computing: Defining a Cloud, Cloud Types – NIST model, Cloud Cube model, Deployment models (Public, Private, Hybrid and Community Clouds), Service models – Infrastructure as a Service, Platform as a Service, Software as a Service with examples of services/ service providers, Cloud Reference model, Characteristics of Cloud Computing – a shift in paradigm Benefits and advantages of Cloud Computing [3]

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Cloud Architecture: Cloud Infrastructure, Architecture of each components, Virtualization versus Traditional Approach, Virtualization Model for Cloud Computing. [2]

Services and Applications by Type [3]

IaaS – Basic concept, Workload, partitioning of virtual private server instances, Pods, aggregations, silos.

PaaS – Basic concept, tools and development environment with examples

SaaS - Basic concept and characteristics, Open SaaS and SOA, examples of SaaS platform

Identity as a Service (IDaaS) Compliance as a Service (CaaS)

Module 2: Use of Platforms in Cloud Computing [6L]

Concepts of Abstraction and Virtualization [2L]

Virtualization technologies: Types of virtualization, Load Balancing and Virtualization: Basic Concepts, Network resources for load balancing; Classification of Virtualization Environment: Scheduling-based Environment, Load-Distribution-Based Environment, Energy Aware-Based Environment, Operational-Based Environment, Distributed Pattern-Based Environment, Transactional-Based Environment

Mention of The Google Cloud as an example of use of load balancing Hypervisors: Virtual machine technology and types, VMware vSphere Machine imaging (including mention of Open Virtualization Format – OVF) [2L]

Porting of applications in the Cloud: The simple Cloud API and AppZero Virtual Application appliance

Concepts of Platform as a Service [2L]

Definition of services, Distinction between SaaS and PaaS (knowledge of Salesforce.com and Force.com), Application development. Use of PaaS Application frameworks.

Module 3: Cloud Service Models [6L]

Use of Google Web Services [2L]

Discussion of Google Applications Portfolio – Indexed search, Dark Web, Aggregation and disintermediation, Productivity applications and service, Adwords, Google Analytics, Google Translate, a brief discussion on Google Toolkit (including introduction of Google APIs in brief), major features of Google App Engine service.

Use of Amazon Web Services [2L]

Amazon Web Service components and services: Amazon Elastic Cloud, Amazon Simple Storage system, Amazon Elastic Block Store, Amazon SimpleDB and Relational Database Service

Use of Microsoft Cloud Services [2L]

Windows Azure platform: Microsoft's approach, architecture, and main elements, overview of Windows Azure AppFabric, Content Delivery Network, SQL Azure, and Windows Live services

Module 4: Cloud Infrastructure [10L]

Types of services required in implementation – Consulting, Configuration, Customization and Support

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Cloud Management [3L]

An overview of the features of network management systems and a brief introduction of related products from large cloud vendors, Monitoring of an entire cloud computing deployment stack – an overview with mention of some products, Lifecycle management of cloud services (six stages of lifecycle)

Live Migration of Virtual Machines: [2L]

Need of Live Migration of Virtual Machine, A Designing Process of Live Migration, and Security Issues during live migration.

Concepts of Cloud Security [3L]

Infrastructure Security, Infrastructure Security: The Network Level, The Host Level, The Application Level, Data Security and Storage, Aspects of Data Security, Data Security Mitigation Provider Data and Its Security, Identity and Access Management.

Auditing and Compliance in Cloud Environment: [2L]

Data Security in Cloud Computing Environment, Need for Auditing in Cloud Computing Environment, Third Party Service Provider, Cloud Auditing Outsourcing Lifecycle Phases, Auditing Classification.

Module 5: Concepts of Services and Applications [6L]

Service Oriented Architecture: Basic concepts of message-based transactions, Protocol stack for an SOA architecture, Event-driven SOA, Enterprise Service Bus, Service catalogs [6]

Applications in the Cloud: Concepts of cloud transactions, functionality mapping, Application attributes, Cloud service attributes, System abstraction and Cloud Bursting, Applications and Cloud APIs [2]

Cloud-based Storage: Cloud storage definition – Manned and Unmanned. [1]

Webmail Services: Cloud mail services including Google Gmail, Mail2Web, Windows Live Hotmail, Yahoo mail, concepts of Syndication services [1]

Text Book

1. Kai Hwang, Geoffrey C Fox, Jack J Dongarra: Distributed and Cloud Computing – From Parallel Processing to the Internet of Things, Morgan Kaufmann Publishers – 2012.
2. Barrie Sosinsky, “Cloud Computing Bible”, Wiley India Edition.

Reference books

1. Anthony Velte, tobyVelte, Robert Elsenpeter, “Cloud Computing – A Practical Approach”, Tata McGraw-Hill Edition.
2. Alex Amies, Harm Sluiman, QiangGuo Tong and Guo Ning Liu: Developing and Hosting Applications on the cloud, IBM Press, 2012.
3. George Reese: Cloud Application Architectures: Building Applications and Infrastructure in the Cloud (Theory in Practice), O’Reilly Publications, 2009

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Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Quantum Computing

Course Code: PEC-CS-T-702

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: Discrete Structures

Objective of the course:

The objective of the course is to make the students able to –

1. Understand the basic idea of quantum computing including background of mathematics and physics.
2. Understand and explain the concept of quantum circuits using single and multiple qubit gates and also designing of quantum circuits.
3. Compare between classical and quantum information theory and explain and apply Bell states, Quantum teleportation, Quantum Cryptography and no cloning theorem.
4. Understand, explain and apply different quantum algorithms including classical computation on quantum computers like Deutsch's algorithm, Deutsch's-Jozsa algorithm, Shor factorization, Grover search and also relate between quantum and classical complexity classes.
5. Understand noise and error correction including graph states and codes, quantum error correction, fault-tolerant.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the basic idea of quantum computing including background of mathematics and physics required for developing and solving complex engineering problem in the domain of quantum computing possibly using modern engineering tools.
CO2	Understand and explain the concept of quantum circuits using single and multiple qubit gates and also designing of quantum circuits for solving engineering problem including societal and environmental issues.
CO3	Compare between classical and quantum information theory and explain and apply Bell states, Quantum teleportation, Quantum Cryptography and no cloning theorem in solving engineering problem possibly in a team maintain proper ethics of professional collaboration.
CO4	Understand, explain and apply different quantum algorithms including classical computation on quantum computers like Deutsch's algorithm, Deutsch's-Jozsa algorithm, Shor factorization, Grover search and also relate between quantum and classical complexity classes for solving engineering problem.

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CO5	Understand noise and error correction including graph states and codes, quantum error correction, fault-tolerant computation and apply it in designing and solving complex engineering problems leading to their lifelong learning.
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Course Content:

Module1: Introduction to Quantum Computation: 8L

Quantum bits, Bloch sphere representation of a qubit, multiple qubits. Background Mathematics and Physics: Hilber space, Probabilities and measurements, entanglement, density operators and correlation, basics of quantum mechanics, Measurements in bases other than computational basis.

Module2: Quantum Circuits: 6L

Single qubit gates, multiple qubit gates, design of quantum circuits.

Module3: Quantum Information and Cryptography: 6L

Comparison between classical and quantum information theory. Bell states. Quantum teleportation. Quantum Cryptography, no cloning theorem.

Module4: Quantum Algorithms: 8L

Classical computation on quantum computers. Relationship between quantum and classical complexity classes. Deutsch's algorithm, Deutsch's-Jozsa algorithm, Shor factorization, Grover search.

Module5: Noise and error correction: 8L

Graph states and codes, Quantum error correction, fault-tolerant computation.

Text book:

1. Nielsen M. A., Quantum Computation and Quantum Information, Cambridge University Press.
2. Benenti G., Casati G. and Strini G., Principles of Quantum Computation and Information, Vol. I: Basic Concepts, Vol II: Basic Tools and Special Topics, World Scientific.
3. Pittenger A. O., An Introduction to Quantum Computing Algorithms

Reference Books:

1. P Kaye, R Laflamme and M Mosca, An Introduction to Quantum Computing.
2. Eleanor G. Rieffel , Wolfgang H. Polak , "Quantum Computing - A Gentle Introduction"

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(Scientific and Engineering Computation)

3. Yanofsky's and Mannucci, Quantum Computing for Computer Scientists.
4. Riley Tipton Perry, “Quantum Computing from the Ground Up”, World Scientific Publishing Ltd.
5. Scott Aaronson, “Quantum Computing since Democritus”, Cambridge.
6. P. Kok, B. Lovett, “Introduction to Optical Quantum Information Processing”, Cambridge.

CO – PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	-	-	-	-	-	-	2	3	1	1
CO2	3	3	3	3	-	2	2	-	-	-	-	2	3	1	3
CO3	3	3	3	3	-	-	-	2	2	-	-	2	3	3	1
CO4	3	3	3	3	-	-	-	-	-	-	-	2	3	3	1
CO5	3	3	3	3	-	-	-	-	-	-	-	3	3	3	3

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Name of the Paper: Mobile Computing

Paper Code: PEC-CS-S-702

Contact :=3L/Week

Credit Point: 3

No. of Lectures: 35

Prerequisite:

6. Basic concept of computer network and communication engineering
7. Basic programming knowledge

Course Objective(s)

The objective of the course is to make the students able to –

- Understand and illustrate the basic concepts and principles in mobile computing
- Understand and demonstrate the various routing algorithms for both infrastructure based and ad hoc networks.
- Identify and develop mobility and bandwidth management in cellular network
- Design and build an energy efficient and secure mobile computing environment using heterogeneous wireless technologies
- Predict and explain the technical issues related to recent mobile computing environment

Course Outcome(s)

After completion of the course students will be able to

CO1	Illustrate the concepts and working of modern communication technologies.
CO2	Demonstrate the various routing algorithms for both infrastructure based and ad hoc networks.
CO3	Develop mobility and bandwidth management in cellular network
CO4	Design and build an energy efficient and secure mobile computing environment using heterogeneous wireless technologies
CO5	Predict the technical issues related to recent mobile computing

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	environment.
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Course Content:

Module I: Introduction [6L]:

Evolution of different types of wireless communication devices; Effects of mobility of devices; Cellular mobile networks – mobility management (call setup, handoff, interoperability and internetworking), bandwidth management, energy management, security; Brief introduction about different generations of wireless communication technology – 1G, 2G, 3G, 4G, 5G.

Module II: Mobile Data Communication [5L]

Mobile Data Communication, WLANs (Wireless LANs) IEEE 802.11 standard, Bluetooth technology, Bluetooth Protocols, Ad hoc networks initialization, leader election, location identification, communication protocols, energy and security.

Module III: Mobility Management in Cellular Networks [4L]

Call setup in PLMN (location update, paging), GPRS, Call setup in mobile IP networks; Handoff management; Mobility models- random walk, random waypoint, Brownian, map-based, group-based.

Module IV: Bandwidth Management in Cellular Mobile networks [3L]

Mathematical formulation of the channel assignment problem (CAP); CAP and generalized graph coloring; Benchmark instances; Lower bound on bandwidth, Genetic algorithms for channel assignment- concept of critical block in a hexagonal cellular network, coalesced CAP, fast near-minimal channel assignment algorithm.

Module V: Localization of Nodes in a Mobile Network [4L]

Different approaches, Indoor and outdoor localizations, LOS and NLOS signals, Outdoor localization techniques – triangulation (TOA-based, AOA- based), errors due to inaccuracies in coordinates of beacon nodes and in measurements, selection of beacon nodes; Location region identification- computational geometric technique.

Module VI: Message Communication in Ad Hoc Networks [6L]

Collision avoidance mechanism (different schemes for a deterministic transmission schedule), collision resolution mechanism – successive partitioning approach; Time slot assignment based on location information, Point-to-point routing in ad hoc networks – proactive, reactive and hybrid approaches, different protocols - DSDV, DSR, AODV, TORA, ZRP

Module VII: Energy-efficient Communication [3L]

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Energy efficiency at various layers - Physical layer, MAC layer, Network layer, Application layer, performance analysis in noisy channel environment.

Module VIII: Secure Wireless Communication [4L]

Introduction-different types of attacks, internal attacks, external attacks; measures against attacks (authentication, intrusion detection, encryption); RC4 algorithm, Lightweight cryptographic algorithms; antijamming techniques.

Text books:

- 1) K. Sinha, S.Ghosh and B. P. Sinha, Wireless Networks and Mobile Computing, CRC Press : New York, 2015.
- 2) J. Schiller, Mobile Communication, Pearson
- 3) Yi-Bing Lin & Imrich Chlamtac, Wireless and Mobile Networks Architectures, John Wiley & Sons, 2001
- 4) Raj Pandya, Mobile and Personal Communication systems and services, Prentice Hall of India, 2001
- 5) 5. XiangYang Li, Wireless Adhoc and Sensor Networks, Cambridge University Press.

Recommended books:

- 1) Research articles published on secure wireless communication (authentication, mitigation of DoS, DDoS, eavesdropping) published in leading journals.
- 2) Mark Ciampa, Guide to Designing and Implementing wireless LANs, Thomson learning, Vikas Publishing House, 2001.
- 3) P.Stavronlakis, Third Generation Mobile Telecommunication systems, Springer Publishers.

CO-PO Mapping

CO #	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	2	2	1	1	-	-	3	2	3	2	2	2
CO2	3	3	3	3	2	1	1	-	-	3	2	3	2	2	2
CO3	3	3	2	3	2	1	1	-	-	3	2	3	2	2	2
CO4	3	3	2	2	2	1	1	-	-	3	2	3	2	2	2
CO5	3	3	3	3	2	1	1	-	-	3	2	3	2	2	2

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Natural Language Processing

Course Code: PEC-CS-D-702

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites:

Statistics, Automata, Compiler Design

Objective of the course:

19. To learn the basics and details of NLP algorithms, principles & application, different NLP techniques and different tools and their uses.
20. To familiarize the concepts and techniques of Natural language Processing for analyzing words based on Morphology and CORPUS.
21. To learn the fundamental strategies of Language Modelling and Word Sense Disambiguation acquiring enough knowledge to Propose models for Word Prediction & Disambiguation.
22. To learn the concepts of Markov Model for POS Tagging and Probabilistic Context Free Grammars and Probabilistic parsing.
23. To learn the techniques of Syntax & Semantics Analysis for Machine Translation and Identify problems where students can Apply the concept appropriately.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the basic concepts of NLP to Explain or Illustrate and Identify problems where students can Apply the concept appropriately to Solve them.
CO2	Understand the fundamental concepts of Text Preprocessing and Morphology so that they can Apply the concept to Analyze their CORPUS.
CO3	Explain or Illustrate the fundamental strategies of Language Modelling and Word Sense Disambiguation acquiring enough knowledge to Propose models for Word Prediction & Disambiguation and Evaluate their performances.
CO4	Explain or Illustrate the concepts of Markov Model for POS Tagging and Probabilistic Context Free Grammars and Probabilistic parsing so that they can Apply them to solve the relevant problems and Analyze their performances.

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CO5	Develop ideas to Propose solutions to the problems of Syntax & Semantics Analysis for Machine Translation and Identify problems where students can Apply the concept appropriately and Analyze the effectiveness as well as limitations of solutions underscoring the utilitarian importance for further exploration of NLP issues leading towards lifelong learning.
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CourseContent:

Module-I: Introduction to NLP [4L]

Introduction to NLP - Various stages of NLP –The Ambiguity of Language: Why NLP Is Difficult Parts of Speech: Nouns and Pronouns, Words: Determiners and adjectives, verbs, Phrase Structure. Statistics Essential Information Theory: Entropy, perplexity, The relation to language, Cross entropy.

Module-II: Text Preprocessing and Morphology [5L]

Character Encoding, Word Segmentation, Sentence Segmentation, Introduction to Corpora, Corpora Analysis. Inflectional and Derivation Morphology, Morphological analysis and generation using Finite State Automata and Finite State transducer.

Module-III: Language Modeling [4L]

Introduction to N-grams, Chain Rule, Smoothing – Add-One Smoothing, Witten-Bell Discounting; Backoff, Deleted Interpolation, N-grams for Spelling and Word Prediction, Evaluation of language models.

Module-IV: Word Sense Disambiguation [5L]

Methodological Preliminaries, Supervised Disambiguation: Bayesian classification, Aninformation theoretic approach, Dictionary-Based Disambiguation: Disambiguation based on sense, Thesaurus based disambiguation, Disambiguation based on translations in a second-language corpus.

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Module-V: Markov Model and POS Tagging [5L]

Markov Model: Hidden Markov model, Fundamentals, Probability of properties, Parameter estimation, Variants, Multiple input observation. The Information Sources in Tagging: Markov model taggers, Viterbi algorithm, Applying HMMs to POS tagging, Applications of Tagging.

Module-VI: Probabilistic Context Free Grammars and Probabilistic parsing [5L]

The Probability of a String, Problems with the Inside-Outside Algorithm, Parsing for disambiguation, Treebanks, Parsing models vs. language models, Phrase structure grammars and dependency, Lexicalized models using derivational histories, Dependency-based models.

Module-VII: Syntax & Semantics Analysis and Machine Translation [8L]

Shallow Parsing and Chunking, Shallow Parsing with Conditional Random Fields (CRF), Lexical Semantics, WordNet, Thematic Roles, Semantic Role Labelling with CRFs. Statistical Alignment and Machine Translation, Text alignment, Word alignment, Information extraction, Text mining, Information Retrieval, NL interfaces, Sentimental Analysis, Question Answering Systems, Social network analysis.

Textbook:

3. Speech and Language Processing, Jurafsky and Martin, Pearson Education
4. Foundation of Statistical Natural Language Processing, Manning and Schutze, MIT Press
5. Multilingual Natural Language Processing Applications from Theory to Practice: Bikel, Pearson.
6. Ela Kumar, “Natural Language Processing”, Wiley

Reference Books:

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

1. Allen, James. 1995. – “Natural Language Understanding”. Benjamin/Cummings, 2ed. 2.
2. Bharathi, A., Vineet Chaitanya and Rajeev Sangal. 1995. Natural Language Processing- “A Paninian Perspective”. Prentice Hall India, Eastern Economy Edition.
3. Hobson lane, Cole Howard, Hannes Hapke, “Natural language processing in action” MANNING Publications, 2019

CO–PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	3	2	3	2	3
CO2	3	2	2	3	-	-	-	-	-	-	2	3	2	2	3
CO3	3	2	2	3	-	-	-	-	-	-	3	-	2	3	3
CO4	3	3	3	3	-	-	-	-	-	-	3	3	2	3	3
CO5	3	3	3	3	-	-	-	-	-	-	3	-	2	3	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Cryptography and Network Security

Course Code: PEC-CS-A-702

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisite:

1. The student must have basic knowledge about Computer Network and mathematics.

Course Objective(s):

1. To provide introduction to the concept of Network Security Model and Cryptography systems.
2. To give the knowledge of Digital Signature and other Security Measures available.
3. To familiarize with the various techniques like PGP and S/MIME.
4. To showcase IP Security Architecture & Transport Layer Security to identify the vulnerability of the Internet systems and recognize the mechanisms of the attacks.
5. To explain the firewall design principles and various intrusion detection system.

Course Outcome(s)

CO1	Understand cryptography and network security concepts and application.
CO2	Apply security principles to system design.
CO3	Identify and investigate network security threat
CO4	Analyze and design network security protocols
CO5	Conduct research in network security

Curriculum for B.Tech Under Autonomy

B.Tech. – Computer Science & Technology

L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Contents :

Module-1 [7L]

Introduction - Services, Mechanisms, and Attacks, OSI security architecture, Network security model[1L]
 Classical Encryption techniques (Symmetric cipher model, substitution techniques, transposition techniques, steganography) [3L]
 Finite Fields and Number Theory: Groups, Rings, Fields, Modular arithmetic, Euclid's algorithm[1L]
 Polynomial Arithmetic, Prime numbers, Fermat's and Euler's theorem[1L]
 Testing for primality -The Chinese remainder theorem - Discrete logarithms [1L]

Module-2 [9L]

Data Encryption Standard- Block cipher principles, block cipher modes of operation[2L]
 Advanced Encryption Standard (AES), Triple DES, Blowfish, RC5 algorithm[3L]
 Public key cryptography: Principles of public key cryptosystems, The RSA algorithm[2L]
 Key management - Diffie Hellman Key exchange, Elliptic curve arithmetic, Elliptic curve cryptography [2L]

Module-3 [6L]

Authentication requirement, Authentication function, MAC, Hash function [2L]
 Security of hash function and MAC, MD5, SHA, HMAC, CMAC [2L]
 Digital signature and authentication protocols, DSS, ElGamal, Schnorr [2L]

Module-4 [7L]

Authentication applications, Kerberos, X.509 [1L]
 Internet Firewalls for Trusted System: Roles of Firewalls, Firewall related terminology- Types of Firewalls, Firewall designs principles [1L]
 SET for E-Commerce Transactions [1L]
 Intruder, Intrusion detection system [1L]
 Virus and related threats, Countermeasures [1L]
 Trusted systems, Practical implementation of cryptography and security [2L]

Module-5 [7L]

E-mail Security: Security Services for E-mail-attacks possible through E-mail, Establishing keys privacy, authentication of the source [1L]
 Message Integrity, Non-repudiation, Pretty Good Privacy, S/MIME [2L]
 IP Security: Overview of IPSec, IPv4 and IPv6-Authentication Header, Encapsulation Security Payload (ESP) [1L]
 Internet Key Exchange (Phases of IKE, ISAKMP/IKE Encoding) [1L]
 Web Security: SSL/TLS Basic Protocol, computing the keys, client authentication [1L]
 PKI as deployed by SSL Attacks fixed in v3, Exportability, Encoding, Secure Electronic Transaction [1L]

Textbooks

- [1] Kahate, A. (2013). Cryptography and network security. Tata McGraw-Hill Education.
- [2] Forouzan, B. A., & Mukhopadhyay, D. (2015). Cryptography and network security. New York, NY: Mc

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Graw Hill Education (India) Private Limited.

Reference Books

[1] Stallings, W. (2006). Cryptography and network security, 4/E. Pearson Education India.

[2] Daras, N. J., & Rassias, M. T. (Eds.). (2015). Computation, cryptography, and network security (pp. 253-287). Springer.

[3] Kumar, A., & Bose, S. (2017). Cryptography and network security. Pearson Education India.

CO-PO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	2	-	2	-	-	-	-	2	-	3	3	2	3
CO2	1	1	2	-	2	-	-	-	-	2	-	2	2	2	3
CO3	3	2	3	3	-	-		-	-	-	-	1	2	3	3
CO4	3	2	3	-		-		-	-	-	-	-	2	3	3
CO5	3	2	3	2		-		2	-	2	-	1	2	3	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: High Performance Computing

Course Code: OEC-CS-701A

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites:

Computer Architecture Lab, Operating System Lab, Compiler design Lab

Course Objectives:

The objective of the course is to make the students able to -

1. Explain the basic knowledge of Parallel Processing.
2. Understand the development and design of parallel system with Compute Unified Device Architecture (CUDA).
3. Differentiate and analyze different Design Issues in Parallel Computing.
4. Understand the limitation of Parallel Computing.
5. Understand and distinguish different Power-Aware Computing and Communication system and also different elements of cloud computing services.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the basic knowledge of Parallel Processing and apply it in solving Complex Engineering Problem.
CO2	Develop and design parallel system with Compute Unified Device Architecture (CUDA) for engineering problem leading to lifelong learning.
CO3	Differentiate and analyze different Design Issues in Parallel Computing and apply the knowledge in solving Complex Engineering Problem including the problem in societal and environmental contexts.
CO4	Understand the limitation of Parallel Computing and apply the knowledge to create and select appropriate techniques, resources, and modern engineering and IT tools to complex engineering problem.

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

CO5	Understand and distinguish different Power-Aware Computing and Communication system and also different elements of cloud computing services leading to developing new computing system and analyzing of existing one.
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Course Content:

MODULE-1: Parallel Processing Concepts (Quick Overview) 10L

Levels of parallelism (instruction, transaction, task, thread, memory, function); Models (SIMD, MIMD, SIMT, SPMD, Dataflow Models, Demand-driven Computation etc); Architectures: N-wide superscalar architectures, multi-core, multi-threaded.

MODULE-2: Parallel Programming with CUDA (Compute Unified Device Architecture) 8L

Processor Architecture, Interconnect, Communication, Memory Organization, and Programming Models in high performance computing architectures: (Examples: IBM CELL BE, Nvidia Tesla GPU, Intel Larrabee Microarchitecture and Intel Nehalem microarchitecture); Memory hierarchy and transaction specific memory design; Thread Organization.

MODULE-3: Fundamental Design Issues in Parallel Computing 6L

Synchronization; Scheduling; Job Allocation; Job Partitioning; Dependency Analysis; Mapping Parallel Algorithms onto Parallel Architectures; Performance Analysis of Parallel Algorithms

MODULE-4: Fundamental Limitations Facing Parallel Computing 4L

Bandwidth Limitations; Latency Limitations; Latency Hiding/Tolerating Techniques and their limitations

MODULE-5: Power-Aware Computing and Communication 6L

Power-aware Processing Techniques; Power-aware Memory Design; Power-aware Interconnect Design; Software Power Management

MODULE-6: Cloud computing services 2L

Infrastructure as a service; Platform as a service; Software as a service

Text book:

1. High Performance Cluster Computing, Volume 1, Architecture and Systems, Rajkumar Buyya, Pearson Education.
2. Berman, Fox and Hey, Grid Computing – Making the Global Infrastructure a Reality, Wiley India.
3. Hurwitz, Bllor, Kaufman, Halper, Cloud Computing for Dummies, Wiley India.

Reference Books:

1. Ronald Krutz, Cloud Security, Wiley India.
2. Cloud Computing, A Practical Approach, Anthony Velte, Toby Velte, Robert Elsenpeter, McGrawHill.

[illegible]

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: VLSI
Course Code: OEC-CS-701B
Contacts: 3:0:0
Credits: 3
Total Contact Hours: 36

Prerequisite: Concept of courses Solid State Devices; Analog Electronic Circuit; Digital Electronic and Circuit

Course Objective(s):

Objective of the course is:

1. To understand the basic concepts of designing combinational and sequential circuits and the design of VLSI ICs
2. To motivate students to design VLSI circuits in the area of digital , analog
3. To encourage for the design of IC with low power and high speed.
4. To study various programmable logic devices like PLDs and FPGA.

Course Outcome(s):

After completion of the course students will be able to

CO1	Understand basic CMOS circuits and properties of CMOS transistors and able to draw stick diagram and layout of CMOS circuits.
CO2	Apply CMOS realization for combinational logic design and analyze the delay models for combinational circuits and understand power dissipation and low power design principles in CMOS circuits.
CO3	Describe fabrication steps of IC and construct stick diagram & layout of CMOS inverter and basic gates based on Layout design rules
CO4	Understand different architectures for address and analyze the speed and area trade off and also understand accumulators, multipliers, dividers and barrel shifters.
CO5	Understand the techniques of chip design using programmable devices like VHDL or Verilog Combinational & Sequential Logic circuit Design

Course Contents:

Module –1: Introduction to VLSI Design [9L]

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Historical perspective development of VLSI from discrete electronic circuit to VLSI IC, MSI, LSI, Microelectronics & VLSI

Types of VLSI Chips (General purpose, ASIC, PLA, FPGA), photo-resist Basic CMOS Technology – (Steps in fabricating CMOS), Basic n-well CMOS proc VLSI Design Concepts, Moor's Law, Scale of Integration (SSI, MSI, LSI, VLSI, ULSI – basic idea only), Types of VLSI Chips (Analog & Digital VLSI chips, General purpose, ASIC, PLA, FPGA), Design principles (Digital VLSI – Concept of Regularity, Granularity etc), Design Domains (Behavioural, Structural, Physical), Y-Chart, Digital VLSI Design Steps.

Module-2: MOS structure [2L]

E-MOS & D-MOS, Charge inversion in E-MOS, Threshold voltage, Flat band voltage, Potential balance & Charge balance, Inversion, MOS capacitances. Three Terminal MOS Structure: Body effect. Four Terminal MOS Transistor: Drain current, I-V characteristics. Current-voltage equations (simple derivation). Scaling in MOSFET, General scaling, Constant Voltage & Field scaling.] CMOS: CMOS inverter, Simple Combinational Gates - NAND gate and NOR Gate using CMOS.

Module-3: Micro-electronic Processes for VLSI Fabrication [10L]

Silicon Semiconductor Technology- An Overview, Wafer processing, Oxidation, Epitaxial deposition, Ion-implantation & Diffusion, Cleaning, Etching, Photo-lithography – Positive & Negative ess, p-well CMOS process, Twin tub process, Silicon on insulator Layout Design Rule: Stick diagram with examples, Layout rules.

Module –4: Hardware Description Language [6L]

VHDL or Verilog Combinational & Sequential Logic circuit Design.

Text Books:

1. Digital Integrated Circuit , J.M.Rabaey, Chandrakasan, Nicolic, Pearson Education
2. CMOS Digital Integrated Circuits Analysis and Design , S.M.Kang & Y.Leblebici,TMH.

Reference Books:

1. Microelectronic Circuits , Sedra & Smith , Oxford
2. Introduction to VLSI Circuits and System , Uyemura , Wiley

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2	2	-	-	-	-	-	-	2	2

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

C02	3	3	3	3	2	-	-	-	-	-	2	2
C03	2	2	3	2	-	-	-	-	-	-	2	2
C04	3	2	2	3	-	-	-	-	-	1	2	2
C05	3	2	2	3	-	1	1	1	-	1	2	2

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Optimization Techniques

Course Code: OEC-CS-701C

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisite:

1. Linear algebra
2. Probability

Course Objective(s):

- To build an understanding on the basics of optimization techniques.
- To introduce basics of linear programming and meta- heuristic search techniques)

Course Outcome(s):

After completion of the course students will be able to

CO1	Understand Decision making procedure and its applications - Explain or Illustrate and Identify queuing model and simulation in real life scenario.
CO2	Understand the essential features and scope of optimization techniques - Learn and Analyze the properties of objective function and formalization of optimization problem.
CO3	Learn numerical methods to find optimum point and value of a function - Learn to solve the LPP.
CO4	Explain or Illustrate transportation problems and assignment problems. - Apply in real life situations.
CO5	Learn applications of network models and analyse the model – Learn to use Tabu Search methods in various fields.

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Contents:**Module I :** [7L]

Decision -making procedure under certainty and under uncertainty – Operations Research – Probability and decision – making – Queuing or Waiting line theory – Simulation and Monte – Carlo Technique – Nature and organization of optimization problems – Scope and hierarchy of optimization – Typical applications of optimization.

Module II : [6L]

Essential features of optimization problems – Objective function – Formulation of optimization problems – Continuous functions – Discrete functions – Unimodal functions – Convex and concave functions, Investment costs and operating costs in objective function – Optimizing profitably constraints – Internal and external constraints.

Module III : [6L]

Necessary and sufficient conditions for optimum of unconstrained functions – Numerical methods for unconstrained functions – One – dimensional search – Gradient – free search with fixed step size. Linear Programming – Basic concepts of linear programming – Graphical interpretation – Simplex method – Apparent difficulties in the Simplex method.

Module IV : [6L]

Transportation Problem, Loops in transportation table, Methods of finding initial basic feasible solution, Tests for optimality. Assignment Problem, Mathematical form of assignment problem, methods of solution.

Module V : [6L]

Network analysis by linear programming and shortest route, maximal flow problem. Introduction to Non – traditional optimization, Computational Complexity – NP – Hard, NP – Complete. Tabu Search – Basic Tabu search, Neighborhood, Candidate list, Short term and Long term memory.

Module VI : [5L]

Genetic Algorithms – Basic concepts, Encoding, Selection, Crossover, Mutation. Simulated Annealing – Acceptance probability, Cooling, Neighborhoods, Cost function Application of GA and Simulated Annealing in solving sequencing and scheduling problems and Travelling salesman problem.

Text Books:

1. Rao S.S., Optimization Theory and Applications, Wiley Eastern.
2. Hamdy A. Taha, Operations Research – An introduction, Prentice – Hall India.

Curriculum for B.Tech Under Autonomy
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Reference books:

1. Gass S. I., Introduction to Linear Programming, Tata McGraw Hill.
2. Reeves C., Modern heuristic techniques for combinatorial problems, Orient Longman.
3. Goldberg, Genetic algorithms in Search, optimization and Machine Learning, Addison Wesley.
4. K. Deb, Optimization for engineering design – algorithms and examples, Prentice Hall of India.

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	-	3	-	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	-	-	-	-	-	-	-	-	2	-	2
CO4	2	2	2	3	-	-	-	-	-	-	-	-	2	-	2
CO5	2	2	3	3	-	-	-	-	-	-	-	2	2	2	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Cyber Law and Ethics

Course Code: OEC-CS-702A

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisite:

1. Familiarity in computer Networking.
2. Basic concepts about network security.

Course Objective(s) :

- To understand, explore and acquire acritical understanding of Cyber Law.
- To learn the basics of a Cyber security

Course Outcome(s):

After completion of the course students will be able to

CO1	To understand the importance of professional practice, Law and Ethics in their personal lives and professional careers.
CO2	To acquire in depth knowledge of information technology act, security policies, and legal framework of right to privacy, data security and data protection
CO3	To develop the understanding of relationship between commerce and cyberspace
CO4	To be familiar with network security threats and countermeasures
CO5	To develop competencies for dealing with frauds and deceptions (Confidence Tricks, Scams

Course Contents:

Module – 1: Introduction of Cybercrime [7]

Cybercrime, Forgery, Hacking, Software Piracy, Computer Network intrusion

Criminals plan attacks, passive attack, Active attacks, cybers talking.

Security challenges in mobile devices, cryptographic security for mobile devices, Attacks on mobile/cell phones, Theft, Virus, Hacking. Bluetooth; Different viruses on laptop.

Proxy servers, Password checking, Random checking, Trojan Horses and Backdoors; DOS & DDOS attacks; SQL injection: Buffer over flow Attacks, Scripts Kiddies and Packaged Defense.

Phishing methods, ID Theft; Online identity method Legal aspects, Indian laws, IT act, Public key certificate, Design of Cyber Security Policy of an Organization ,Unictrnal Model Law.

The Importance of Cyber Law, Significance of cyber Ethics, Need for Cyber regulations and Ethics. Ethics in Information society, Introduction to Artificial Intelligence Ethics: Ethical Issues in AI and core Principles, Introduction to Block chain Ethics.

1. Cyber security by Nina Gobole&SunitBelapune; Pub: Wiley India.
2. Chris Reed & John Angel, Computer Law, OUP, New York, (2007).
3. Justice Yatindra Singh, Cyber Laws, Universal Law Publishing Co, New Delhi, (2012).
4. Verma S, K, Mittal Raman, Legal Dimensions of Cyber Space, Indian Law Institute, New Delhi, (2004)

1. Kenneth J. Knapp, “Cyber Security and Global Information Assurance: Threat Analysis and Response Solutions”, IGI Global, 2009.
2. Jonathan Rosenoer, “Cyber law: the Law of the Internet”, Springer-Verlag, 1997
3. Sudhir Naib, The Information Technology Act, 2005: A Handbook, OUP, New York,
4. Vasu Deva, Cyber Crimes and Law Enforcement, Commonwealth Publishers, New Delhi, (2003) .

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Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

CO2	2	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	-	-	-	-	-	-	-	-	2	-	2
CO4	2	-	2	3	-	-	-	-	-	-	-	-	2	-	2
CO5	2	2	3	3	-	-	-	-	-	-	-	2	2	2	3

Course Name: Soft Skills and Interpersonal Communication

Course Code: OEC-CS-702B

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Foreign Language

Course Code: OEC-CS-702C

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

French (Open Elective)

Pre-requisites: Basic high school level reading, writing and communication skills in English.

Course Outcome(s):

By the end of the course the students will be able to –

CO1	Read basic French and interpret the meaning
CO2	Construct simple sentences in French
CO3	Interact with others and hold simple conversations in French
CO4	Demonstrate a basic knowledge of French culture, manners, geography and world view
CO5	To understand Vocabulaire, Grammaire and Français Interactif

Course Content:

Module 1:[6L]

Vocabulaire

- L' alphabet français (The Alphabets)
- Les nombres (cardinaux et ordinaux) (Numbers)
- Les mois de l'année (The Months of the Year)
- Les saisons (The Seasons)
- Les jours de la semaine (The Days of the Week)

Curriculum for B.Tech Under Autonomy

B.Tech. – Computer Science & Technology

L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

- Les couleurs (The Colours)
- La famille (The Family)
- Les nationalités (The Nationalities)

Grammaire

- Les Verbes—*être, avoir et aller*
- Nouns—Gender and Number
- Les articles (définis, indéfinis, contracté et partitif)
- Les adjectifs—possessifs et démonstratifs

Français Interactif (Listening and Speaking)

- Les salutations
- Les forms de politesse
- Présentez-vous (About Yourself)

Module2:[6L]

Vocabulaire

- L'heure (the time)
- La maison (the house)
- Le corps (the body)
- Les vêtements (clothes)
- Les professions (professions)
- Les loisirs (pastimes)
- Le sport (Sports)

Grammaire

- Les Verbes—*voir, savoir, venir, aller, sortir, connaître, partir.*
- Les négations
- Le futur
- Les interrogatifs

Français Interactif (Listening and Speaking)

- Décrivez les images
- La dictée
- Lisez le journal

Module3 : [6L]

Vocabulaire

- La nourriture (Food)
- Les repas (Meals)
- Les légumes (Vegetables)
- Les fruits (Fruits)

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

- Les fleurs (Flowers)
- Les animaux (Animals)
- Les oiseaux (Birds)

Grammaire

- Les adverbes
- Les adjectifs
- Les prépositions

Français Interactif (Listening and Speaking)

- Ecoutez la radio/la télévision
- Dialogues—À la médecin, au café, a la gare

Module4: [6L]

Vocabulaire

- Le jardin (The Garden)
- Le temps (the weather)
- Les voyages (Travel)
- La ville (the City)
- Les vacances (Holidays)

Grammaire

- Pronoms interrogatifs
- Mood—subjonctif et l'impératif

Français Interactif (Listening and Speaking)

- Se présenter (expressing ideas/opinions on general topics)
- Ecoutez le programme sur la radio/la télévision

Module 5: [6L]

Vocabulaire

- Les modes de transport (Transport)
- L'Ecole (the School)
- À la Campagne (in the Country)
- À la restaurant (at the Restaurant)
- Le Cinema (at the Cinema)
- La Marché (at the Market)

Grammaire

- Passé Composé
- Passé Récent

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Ecrivez en Français (Writing)

- Decrivez votre ville
- Decrivez votre maison/appartement
- Qu'est que son métier?

Module 6:[6L]

En France

- Les villes de France (the Cities of France)
- Les montagnes et rivières de France (Mountains and Rivers)
- La géographie de France (Geography of France)
- La gastronomie française (French Food and Gastronomy)
- Le fêtes (Festivals of France)

Grammaire

- Les Verbes Pronominaux
- Les Pronoms Personnels

Ecrivez en Français (Writing)

- Decrivez votre vacation à l'mer
- Quels sont les loisirs préférés?
- Les magasins de supermarché

Recommended Texts:

1. *Le Nouveau Sans Frontières-1* (Paris: CLE International, 1999)
2. Dondo, *Modern French Course* (1930, Oxford:Oxford UP, 1999)
3. Dictionnaire Larousse

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	3	-	-	-	-	-	-	-	-	3	-	3
CO2	2	3	2	-	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	-	-	-	-	-	-	-	-	2	2	2
CO4	2	-	2	3	-	-	-	-	-	-	-	-	2	-	2
CO5	2	2	3	3	-	-	-	-	-	-	-	2	2	2	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Practical

Course Name: Information Theory and Coding Lab

Course Code: PEC-CS-T-791

Contact: 0:0:3

Credits: 1.5

Prerequisites: Knowledge of C programming and MATLAB

Course Objective:

The objective of the course is to make the students able to

1. Illustrate and apply proper code in appropriate platform using suitable syntax to solve problems.
2. Understand the concept of variables, constants, data type, operator, expression, statements, loops, vector, matrix, array, function, file handling and apply this knowledge to design the problem
3. Apply systematic approach to design the programs for solving problems.
4. Solve and analyse engineering-related computational problems by applying a variety of common numeric techniques
5. Interpret the result of the experiments, prepare laboratory reports based on observed output and analyse it.

Course Outcomes:

After completion of the course students will be able to

CO1	Illustrate and apply proper code in appropriate platform using suitable syntax for developing program to solve problems related to Mathematics and Engineering field leading to lifelong learning.
CO2	Understand the concept of variables, constants, data type, operator, expression, statements, loops, vector, matrix, array, function, file handling and apply this knowledge to design the problem using modern tools for solving complex engineering problems.
CO3	Apply systematic approach to design the programs for solving problems as a professional engineering practice.
CO4	Solve and analyse engineering-related computational problems by applying a variety of common numeric techniques
CO5	Interpret the result of the experiments, prepare laboratory reports based on observed output and analyse it to validate professional ethics and responsibilities and norms of the engineering practice.

Curriculum for B.Tech Under Autonomy
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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Content:

Module-1: Revision on programming using C language. Familiarization with MATLAB environment setup, syntax, variables, commands, data types, operators, decisions, loops, vectors, matrix, arrays, functions, and advanced part, creating and editing basic MATLAB program in an editor, compilation and execution of MATLAB program.

Module-2: Determination of various entropies and mutual information using C/MATLAB of the following channels

- a. Noise free channel
- b. Noisy channel

Module-3: Generation and evaluation of following variable source coding using C/MATLAB

- a. Shannon – Fano coding
- b. Huffman Coding and Decoding
- c. Lempel Ziv Coding and Decoding

Module-4: Coding & Decoding of the following codes using C/MATLAB

- a. Linear block codes
- b. Cyclic codes
- c. Convolutional codes

Module-5: Coding & Decoding of the following codes using C/MATLAB

- a. BCH code
- b. RS code

Module-6: Problem based on

- a. Coded and uncoded communication system (Calculate the error probability) using C/MATLAB.
- b. Source coding and channel coding for transmitting a text file using C/MATLAB.

Textbook:

1. Information theory, coding and cryptography - Ranjan Bose; TMH.
2. Process Control – A First Course with MATLAB - Pao C. Chau; Cambridge University Press

Reference Books:

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

- 1.Introduction to Information Theory - M Mansurpur; McGraw Hill.
2. Information Theory - R B Ash; Prentice Hall.
3. Error Control Coding - Shu Lin and D J Costello Jr; Prentice Hall.
- 4.Information and Coding - N Abramson; McGraw Hill.

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-	-	-	-	-	-	3	3	3	3
CO2	3	3	3	3	3	-	-	-	-	-	-	-	3	3	3
CO3	3	3	3	3	-	-	-	3	-	-	-	-	3	3	3
CO4	3	3	3	3	-	-	-	3	3	-	-	-	3	3	3
CO5	-	-	-	-	-	-	-	3	-	3	-	-	3	2	3
CO	3	3	3	3	3	-	-	3	3	3	-	3	3	3	3

Curriculum for B.Tech Under Autonomy
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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Name of the Paper: Ad-Hoc and Sensor Networks Lab

Paper Code: PEC-CS-S-791

Contact :=3P/Week

Credit Point: 1.5

Prerequisite:

8. Basic concept of computer network and communication engineering
9. Basic programming knowledge

Course Objective(s)

The objective of the course is to make the students able to –

- To understand and illustrate the principles of mobile ad-hoc networks and sensor networks, and their impact on protocol design
- To design and develop MAC and routing protocols for mobile ad-hoc and sensor networks
- To develop efficient protocols for mobile ad-hoc and sensor networks
- To analyze and develop information dissemination protocols for mobile ad-hoc and sensor networks

Course Outcome(s)

After completion of the course students will be able to

CO1	Understand and explain the Fundamental Concepts and applications of ad-hoc and wireless sensor networks
CO2	Describe and analyze the challenges of designing MAC protocol in mobile ad-hoc networks
CO3	Describe and analyze the challenges in designing routing and transport protocols for mobile ad-hoc networks.
CO4	Describe and analyze the challenges of designing MAC protocol in sensor networks
CO5	Describe and analyze the challenges in designing routing and transport protocols for sensor networks.

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Course Content:

All the Experiments may be Conducted using Network Simulation software like NS-2/ NSG2.1/ ONE Simulator/ Wire SHARK/ SDR etc.

- 1) Introduction of Ad-hoc network and sensor network applications and its simulation
- 2) Network Simulator installation for ad-hoc network.
- 3) Network Simulator installation for wireless sensor network.
- 4) Evaluation of the performance of various LAN Topologies
- 5) Evaluation of the performance of various routing protocols of ad-hoc network
- 6) Evaluation of the performance of TCP and UDP Protocols
- 7) Evaluation of the performance of AODV, DSR and DSDV routing protocols
- 8) Evaluation of the performance of IEEE 802.11 and IEEE 802.15.4
- 9) Capturing and Analysis of TCP and IP Packets
- 10) Analysis of CDMA Downlink

CO-PO Mapping

CO #	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	2	2	1	1	-	-	3	2	3	2	2	2
CO2	3	3	3	3	2	1	1	-	-	3	2	3	2	2	2
CO3	3	3	2	3	2	1	1	-	-	3	2	3	2	2	2
CO4	3	3	2	2	2	1	1	-	-	3	2	3	2	2	2
CO5	3	3	3	3	2	1	1	-	-	3	2	3	2	2	2

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Data Mining and Data Warehousing Lab

Course Code: PEC-CS-D-791

Contact: 3:0:0

Credits: 1.5

Prerequisites: Data Structure & Programming, Design and Analysis of Algorithms, Database Management Systems, Statistics, Artificial Intelligence, Python Programming

Course Objectives:

The objective of the course is to enable students to

6. Comprehend the practical aspects of Data Mining and Data Warehousing through implementation of ideas using proper techniques and tools.
7. Formulate a problem to fit Data Mining and Data Warehousing frameworks by exploring the contextual data and its characteristics.
8. Explore the well-known strategies of Data Mining and Data Warehousing by implementing their respective algorithmic solutions to large scale data using proper techniques and tools to solve contextual problems.
9. Develop ideas and propose technical solutions to the challenging problems of Data Mining and Data Warehousing.
10. Analyze the effectiveness of a Data Mining Model and Data Warehousing Frameworks in offering solutions to the respective problem.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the practical aspects of Data Mining and Data Warehousing through implementation of ideas using proper techniques and tools to recognize their utilitarian importance in current technological context for further exploration leading towards lifelong learning.
CO2	Identify and formulate an engineering problem by exploring contextual data and its characteristics within the scope of Data Mining and Data Warehousing paradigm.
CO3	Explore relevant literature and apply the concepts of Data Mining and Data Warehousing by implementing well-known algorithmic solutions to large scale data using proper techniques and tools to solve contextual problems.
CO4	Develop ideas and propose technical solutions to the challenging problems of Data Mining and Data Warehousing.

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CO5	Plan and Implement Data Mining based ideas as executable programs (preferably termed as models) by developing suitable algorithms with adequate documentation in collaborative environment for successfully carrying out projects on Data Mining and Data Warehousing and investigate their effectiveness by analyzing the performances using proper techniques and tools.
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Course Content:

WEEK-1: Introduction to Data Mining Programming Platform

Introduction to Data Mining Programming Platform and Data Exploration: Data Types, Data Attributes, Statistical Description of Data, Data Visualization, Data Feature Vectors, Data Preprocessing: Data Cleaning, Data Transformation

WEEK -2: Affinity Analysis

Implementation and Analysis of Recommending Engine

WEEK -3: Association Analysis

Implementation and Analysis of Apriori Algorithm

WEEK -4: Regression

Implementation and Analysis of Linear and Nonlinear Regression Methods

WEEK -5: Classification

Implementation and Analysis of k-Nearest-Neighbor Classifier, Decision Tree Classifier, Naïve Bayes Classifier

WEEK -6: Classification

Implementation and Analysis of ANN-Backpropagation and SVM Based Classifier

WEEK-7: Clustering

Implementation and Analysis of k-Means and k-Medoids

WEEK -8: Mining Time-Series Data

Implementation and Analysis of Time-Series Data Mining Models

WEEK -9: Discussion on Project Problems and Allocation (Problem Description Report Submission)

WEEK -10: Designing Solution Model and Proposal Report Submission

WEEK -12: Project Demonstration and Project Report Review

16. Han J &Kamber M, “Data Mining: Concepts and Techniques”, Morgan Kaufmann Publishers, Third Edition.
17. Parteek Bhatia, “Data Mining and Data Warehousing: Principles and Practical Techniques”, Cambridge University Press.

17. Pang-Ning Tan, Vipin Kumar, Michael Steinbach, “Introduction to Data Mining”, Pearson Education.
18. Robert Layton, “Learning Data Mining with Python”, Packt Publishing

[illegible]

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Cloud Computing Lab

Course Code: PEC-CS-A-791

Contact: 3:0:0

Credits: 1.5

Prerequisite:

2. The student must have basic knowledge in Computer Network and Distributed System

Course Objectives(s):

4. To provide students with the fundamentals and essentials of VM.
5. Understand the importance of virtualization in distributed computing and how this has enabled the development of Cloud Computing.
6. Understand the importance of protocols and standards in computing.

Course Outcome(s)

After completion of the course the student able to do

CO1	Understand the concept of virtual machines in cloud computing
CO2	Implementation of virtual machines on different platforms
CO3	<i>To provide the knowledge of various cloud computing tools and their related concepts.</i>
CO4	To work on tools used in cloud computing online
CO5	Exploring different cloud computing and analysis there capability

Students are advised to do laboratory/practical practice not limited to, but including the following types of problems:

1. Create virtual machines that access different programs on same platform. [4L]
2. Create virtual machines that access different programs on different platforms. [4L]
3. Working on tools used in cloud computing online [16L]
 - a. Storage
 - b. Sharing of data
 - c. manage your calendar, to-do lists
 - d. a document editing tool
4. Exploring Google cloud [4L]
5. Exploring Microsoft cloud [4L]
6. Exploring Amazon cloud [4L]

LIST OF EXPERIMENTS

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

1. Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of windows7 or 8.
2. Install a C compiler in the virtual machine created using virtual box and execute Simple Programs
3. Install Google App Engine. Create hello world app and other simple web applications using python/java.
4. Use GAE launcher to launch the web applications.
5. Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim.
6. Find a procedure to transfer the files from one virtual machine to another virtual machine.
7. Find a procedure to launch virtual machine using trystack (Online Openstack Demo Version)
8. Install Hadoop single node cluster and run simple applications like wordcount.
9. Hands on virtualization using XenServer
10. Hands on containerisation using Docker 3
11. Deployment and Configuration options in Amazon (AWS)
12. Deployment and Configuration options in Google Cloud
13. Deployment and Configuration options in Microsoft Azure
14. Building a 'HelloWorld' app for the cloud
15. Deploying the 'HelloWorld' app for the cloud

CO-PO mapping

CO #	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
CO 1	1	-	-	-	-	-	-	-	-	-	-	-	-	-	3
CO 2	-	3	-	-	-	-	-	-	-	-	-	-	-	1	3
CO 3	-	-	-	-	-	-	-	-	3	2	3	2	3	-	-
CO 4	2	-	3	-	-	2	2	-	-	-	-	-	-	2	3
CO 5	1	-	-	-	-	-	-	-	-	-	3	-	-	-	2

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: High Performance Computing Lab

Course Code: OEC-CS-791A

Contact: 0:0:3

Total Contact Hours: 36

Credits: 1.5

Prerequisites: Computer Architecture Lab, Operating System Lab, Compiler design Lab

Objective of the course:

The objective of the course is to make the students able to -

1. Understand, design and develop effective programs applying device Query, Vector Addition, different Matrix Multiplication, different Image processing algorithms using Xeon Phi Programming.
2. Understand, design and develop effective programs applying device Query, Vector Addition, different Matrix Multiplication, different Image processing algorithms using Open MPI programming.
3. Design and develop effective programs to solve problems like DAXPY, Matrix Multiply, Calculation of pi using work sharing and reduction, Producer consumer problem, Molecular dynamics simulation using Open MPI programming.
4. Implement and analyze program to solve problems like DAXPY, Calculation of π - MPI Bcast and MPI Reduce, Ocean Kernel, and also for different Large Matrices using MPI programming.
5. Prepare laboratory reports on interpretation of experimental results and analyze it for validating the same.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand, design and develop effective programs for engineering and mathematical problems applying device Query, Vector Addition, different Matrix Multiplication, different Image processing algorithms using Xeon Phi Programming or modern programming tools leading to lifelong learning.
CO2	Understand, design and develop effective programs for engineering and mathematical problems applying device Query, Vector Addition, different Matrix Multiplication, different Image processing algorithms using OpenMPI programming or modern programming tools leading to lifelong learning.
CO3	Design and develop effective programs for engineering and mathematical problems like DAXPY, Matrix Multiply, Calculation of pi using work sharing and reduction, Producer consumer problem, Molecular dynamics simulation using Open MPI programming possibly as a team maintaining proper ethics of collaboration.

Curriculum for B.Tech Under Autonomy
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CO4	Implement and analyze program for engineering and mathematical problems like DAXPY, Calculation of π - MPI Bcast and MPI Reduce, Ocean Kernel, and also for different Large Matrices using MPI programming leading to lifelong learning.
CO5	Prepare laboratory reports on interpretation of experimental results and analyze it for validating the same maintaining proper ethics of collaboration.

Course Content:

Device Query, Vector Addition, Matrix Multiplication, Tiled Matrix Multiplication, Picture Scaling, Image Blur, Image Grayscale. 1D, 2D, and 3D Stencil Operations. Histogramming, Convolution, Scan, Reduction using Xeon Phi Programming (3 Labs)

Vector Addition, Matrix Multiplication, Tiled Matrix Multiplication, Picture Scaling, Image Blur, Image Grayscale. 1D, 2D, and 3D Stencil Operations. Histogramming, Convolution, Scan, Reduction using OpenMPI programming (3 Labs)

DAXPY, Matrix Multiply, Calculation of pi using worksharing and reduction, Producer consumer problem, Molecular dynamics simulation problem using Open MPI programming (3 Labs)

DAXPY, Calculation of π - MPI Bcast and MPI Reduce, Ocean Kernel, Reduction example, Collective Communication - Scatter – Gather, MPI Derived Datatypes, Matrix Multiplication on a Cartesian Grid (2D Mesh) using Cannon's Algorithm, Matrix Multiplication using Cannon's Algorithm for Large Matrices using MPI programming. (3 Labs)

Text book:

1. High Performance Cluster Computing, Volume 1, Architecture and Systems, Rajkumar Buyya, Pearson Education.
2. Berman, Fox and Hey, Grid Computing – Making the Global Infrastructure a Reality, Wiley India.
3. Hurwitz, Bllor, Kaufman, Halper, Cloud Computing for Dummies, Wiley India.

Reference Books:

1. Ronald Krutz, Cloud Security, Wiley India.
2. Cloud Computing, A Practical Approach, Anthony Velte, Toby Velte, Robert Elsenpeter, McGrawHill.

CO – PO Mapping:

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
C01	3	3	3	3	3	-	-	-	-	-	-	3	3	3	3
C02	3	3	3	3	3	-	-	-	-	-	-	3	3	3	3
C03	3	3	3	3	2	-	-	2	2	-	-	-	1	1	3
C04	3	3	3	3	-	-	-	-	-	-	-	3	3	1	3
C05	-	-	-	-	-	-	-	3	-	3	-	-	3	2	3

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: VLSI Lab
Course Code: OEC-CS-791B
Contact: 3:0:0
Total Contact Hours: 36
Credits: 3

Prerequisite: Concept of courses Solid State Devices; Analog Electronic Circuit; Digital Electronic and Circuit

Course Objective:

The objective of the course is to make the students able to

- 1 Plan a sequence of processing steps to fabricate a solid state device to meet geometric, electrical, and/or processing parameters.
- 2 Design VLSI circuits by keeping technological process constraints in mind.
- 3 Design code converters, parity generator and comparator using various styles of modeling.

Course Outcomes:

After completion of the course students will be able to

CO1	Design basic and universal gates, adder and subtractor using VHDL.
CO2	Design multiplexers, demultiplexers, encoders and decoders
CO3	Design code converters, parity generator and comparator using various styles of modeling.
CO4	Design sequential circuits like counters, shift registers and flip-flops.
CO5	Implement digital circuit on FPGA kit.

Course Content:

Perform the following experiments using VHDL:

Experiment 1. Design of gates: AND, OR, NOT, NAND and NOR.

Experiment 2. Design of Half-Adder and Full Adder.

Experiment 3. Design of Half Subtractor and Full Subtractor.

Experiment 4. Design of 4:1 MUX.

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L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Experiment 5. Design of 1:8 DEMUX.

Experiment 6. Design of 3:8 Decoder.

Experiment 7. Design of 8:3 Priority Encoder.

Experiment 8. Design of 4 Bit Binary to Grey code Converter.

Experiment 9. Design of 4 Bit Binary to BCD Converter using sequential statement.

Experiment 10. Design of 4-Bit Binary to Excess-3 converter using sequential statement.

Experiment 11. Design of 9-Bit parity generator using structural style.

Experiment 12. Design of 4-bit comparator.

Experiment 13. Design of all type of Flip-Flops using sequential statements.

Experiment 14. Design of 8-Bit Shift Register.

Text Book:

1. Microelectronic Circuits , Sedra & Smith , Oxford
2. Introduction to VLSI Circuits and System , Uyemura , Wiley

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2	2	-	-	-	-	-	-	2	2
CO2	3	3	3	3	2	-	-	-	-	-	2	2
CO3	2	2	3	2	-	-	-	-	-	-	2	2
CO4	3	2	2	3	-	-	-	-	-	1	2	2
CO5	3	2	2	3	-	1	1	1	-	1	2	2

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

Course Name: Optimization Techniques Lab

Course Code: OEC-CS-791C

Contact: 0:0:3

Credits: 1.5

Course Objective(s):

- To introduce the optimization techniques using both linear and non-linear programming.
- To focus on the convex optimization though some techniques will be covered for non-convex function optimization.
- To frame engineering minima maxima problems in the framework of optimization problems.

Course Outcome(s):

After the completion of the course, the students will be able to

CO1	Demonstrate the basic principles and concepts of Python
CO2	Explore the applicability of programming skills in Python
CO3	Summarize various optimization techniques like LPP models.
CO4	Analyse the transportation, inventory and assignment problems.
CO5	Evaluate the concepts of sequencing, game theory and dynamic programming.

List of Experiments (Includes but Not Limited to)

1. Matrix Operations
2. Minimum Cost Path
3. Finding Maximum Number in An Array
4. Array Sorting
5. Linear Programming Problem
6. Queuing Problem
7. Sequencing Problem

- ### Recommended Books :

- ### CO-PO Mapping:

[illegible]

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

MANDATORY COURSE

Course Name: Entrepreneurship & Innovation Skill

Course Code: MC 781

Credits: 0

Total Lectures: 36

Credit: 0

Prerequisite: None

Course Outcome(s):

After the completion of the course, the students will be able to:

CO1	Comprehend the role of bounded rationality, framing, causation and effectuation in entrepreneurial decision making.
CO2	Demonstrate an ability to design a business model canvas.
CO3	Evaluate the various sources of raising finance for startup ventures.
CO4	Explain the fundamentals of developing and presenting business pitching to potential investors.
CO5	Implement case study of entrepreneurial venture in your nearby area

Course Content:

Module 1:[4L]

Introduction to Entrepreneurship: Entrepreneurs; entrepreneurial personality and intentions - characteristics, traits and behavioral; entrepreneurial challenges. Entrepreneurial Opportunities: Opportunities. discovery/ creation, Pattern identification and recognition for venture creation: prototype and exemplar model, reverse engineering.

Module 2: [4L]

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Entrepreneurial Process and Decision Making: Entrepreneurial ecosystem, Ideation, development and exploitation of opportunities; Negotiation, decision making process and approaches, Effectuation and Causation; Advantage and Limitations of Entrepreneurship; Process of Entrepreneurship.

Module 3:[4L]

Crafting business models and Lean Start-ups: Introduction to business models; Creating value propositions-conventional industry logic, value innovation logic; customer focused innovation; building and analyzing business models; Business model canvas, Introduction to lean startups, Business Pitching.

Module 4: [4 L]

Organizing Business and Entrepreneurial Finance: Forms of business organizations; organizational structures; Evolution of Organisation, sources and selection of venture finance options and its managerial implications. Policy Initiatives and focus; role of institutions in promoting entrepreneurship.

Module 5: [4L]

Entrepreneurs as problem solvers: Innovations and Entrepreneurial Ventures – Global and Indian; Role of Technology – E-commerce and social media; Social Entrepreneurship – Concept; Entrepreneurship – The Indian Scenario

Module 6: [4 L]

Project/Case Study: (Any One)

1. Visit of the District Industries Centre and prepare a report of activities and programs undertaken by them
2. Conduct a case study of any entrepreneurial venture in your nearby area.
3. Field Visit: Visit any business firm near your locality; interact with the owner of the business firm and prepare a field report on parameters like: type of business, scale of business, product/service dealing in, target customer, problems faced and measures to solve the faced challenges.
4. Know your State Handicraft and Handlooms as a means of economic activity

Text Books:

1. Bessant, J. (2003) High Involvement Innovation: Building and Sustaining Competitive Advantage Through Continuous Change. Chicester: John Wiley & Sons.
2. Bygrave, W and Zackarakis, A (2013) Entrepreneurship, 3rd Edition, John Wiley and Co.
- Drucker, P. (1999) Innovation and Entrepreneurship, Butterworth Heinemann, Oxford.

Reference Books:

1. Fagerberg, J, Mowery, DC and Nelson, RR (2005) The Oxford Handbook of Innovation, Oxford University Press, NY.

- ### CO-PO Mapping:

[illegible]

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

4th Year 2nd Semester: 8th Semester

Sl No	Course Code	Paper Code	Theory	Contact Hours /Week			Credit Points	
				L	T	P		Total
A. THEORY								
1	Professional Elective courses	PEC-CS-T-801	Advance Graph Algorithms	3	0	0	3	3
		PEC-CS-S-801	Real Time Systems					
		PEC-CS-D-801	Data Analytics					
		PEC-CS-A-801	VLSI					
2	Open Elective courses	OEC-CS-801A	Human Resource Development and Organizational Behavior	3	0	0	3	3
		OEC-CS-801B	Block Chain					
		OEC-CS-801C	Simulation and Modeling					
3	Open Elective courses	OEC-CS-802A	Values and Ethics in Profession	3	0	0	3	3
		OEC-CS-802B	History of Science					
		OEC-CS-802C	Economic Policies in India					
B. PRACTICAL								
4	PROJECT	PR 891	Major Project-II	0	0	0	12	6
5	PROJECT	PR 892	Grand Viva	0	0	0	0	1
C. MANDATORY ACTIVITIES / COURSES								
6	MC	MC 801	Essence of Indian Knowledge Tradition	0	0	3	3	3 Units

Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
 L – Lecture; T- Tutorial; P- Practical [1L=1Cr, 1T=1Cr, 1P =0.5 Cr]

TOTAL CREDIT	16
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Course Name: Advance Graph Algorithms

Course Code: PEC-CS-T-801

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: Discrete Mathematics, Graph Theory, Data Structures and Algorithms

Course Objectives:

1. Use of a graph algorithm and apply them in designing solution to Computer Science and engineering domain problem.
2. Understand the basic concept of graph implementation programming and use of data structure to represent graph for developing the solution to the graph based problems
3. Understand of how to apply the graph theory principles and concepts to design an efficient algorithm and use suitable data structure for problem solving.
4. Understand the design better algorithm in compare to some existing algorithm and create new efficient data structures for solving many graph based real life problem in Computer science domain.
5. Discuss the application of graph concepts in the development of computer algorithms and applications.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the basic concept of Graph theory and Graph Algorithm and apply them in designing solution to Computer Science and engineering domain problem.
CO2	Demonstrate the basic concept of graph implementation programming and use of data structure to represent graph for developing the solution to the graph based problems in computer science domain
CO3	Formulate graph algorithms and programs in solving Computer science and engineering problem leading to lifelong learning and Analyze the efficiency of algorithms using time and space complexity.

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CO4	Apply the graph theory principles and concepts to design an efficient algorithm and use suitable data structure for problem solving.
CO5	Design and develop better algorithm in compare to some existing algorithm and create new efficient data structures for solving many graph based real life problem in Computer science domain.

Course Content:

Module-I: Introduction of Graph

6

Fundamental concept of Graph :Definition of Graphs, Representations of graphs, degree sequences, distance in graphs, complete, regular and bipartite graphs, basic properties of Paths, Cycles, and Trails, Connection, Bipartite Graphs, Eulerian Circuits, Vertex Degrees and Counting, Degree-sum formula.

Module-II: Graph algorithms

5

Graph traversal algorithm-Introduction to Graphs & its Applications:Breadth-first search and tree, Depth-first search and tree , Topological sort, Strongly connected components

Module-III: Tree

5

Trees and connectivity –Properties of trees, vertex and edge,connectivity, Mengers theorem, Minimum Spanning Trees, Growing a minimum spanning tree, The algorithms of Kruskal and Prim

Module-IV: Shortest Paths problem

5

Shortest Paths problem: Single-source shortest paths in directed acyclic graphs, Dijkstra's algorithm, The Bellman-Ford algorithm, All-Pairs Shortest Paths, The Floyd-Warshall algorithm

Module-V: Flow networks problem

7

Flow networks problem: The Ford-Fulkerson method, Maximum bipartite matching, Cuts and Connectivity, k-Connected Graphs, Max-Flow Min-cut Theorem, Menger's Proof using Max-Flow Min-Cut Theorem.

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Module-VI: Planar Graphs

4

Planar Graphs, Characterization of Planar Graphs, Kuratowski's Theorem, Wagner's Theorem, Graph-coloring, Hamiltonian Graph, Traveling Salesman Problem

Module-VII: Matchings and Cover

4

Matchings and Covers: Hall's Condition, Min-Max Theorem, Independent Sets, Covers and Maximum Bipartite Matching, Augmenting Path Algorithm, Weighted Bipartite Matching, Hungarian Algorithm, Stable Matchings and Faster Bipartite Matching, Factors & Perfect Matching in General

Text Books:

1. Graph Theory, by J. A. Bondy and U. S. R. Murthy, Springer Verlag (2008.)
2. Introduction to Graph Theory, by D. B. West, PHI, 2004.
3. T. H. Cormen, C. E. Leiserson, R. L. Rivest and C. Stein, "Introduction to Algorithms"

Reference Books:

- 1 Graph Theory, by R. Diestel : Springer Verlag (Free Download available).(2003)
- N. Deo, Graph Theory, Prentice Hall of India, 1974.
2. E.M.Reingold, J.Nievergelt and N.Deo "Combinational Algorithms Theory and Practice", Prentice Hall.

CO – PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	3	2	2	-	-	-	-	-	-	-	3	3	3
CO2	3	3	2	3	-	-	-	-	-	-	-	-	3	2	3
CO3	3	3	3	2	-	-	-	-	2	2	-	3	3	3	2
CO4	3	3	3	3	-	-	-	-	-	-	-	-	3	3	2
CO5	3	3	3	3	-	-	-	-	3	2	-	-	3	2	2

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Course Name: Real Time Systems

Course Code: PEC-CS-S-801

Total Contact Hours: 36

Credit:3

Prerequisites:

1. Concepts of Operating systems and Algorithm.
2. Knowledge of Distributed System basics.

Course Objective(s):

- To understand the real-time systems
- Obtain a broad understanding of the technologies and applications for emerging and exciting domain of real-time systems.
- Get in-depth hands-on experience in designing and developing a real time system.

Course Outcome(s):

After completion of the course students will be able to

CO1	Understand the concepts of Real-Time systems
CO2	Recognize the characteristics of a real-time system
CO3	Understand and develop document on an architectural design of a real-time system.
CO4	Develop and document Task scheduling, resource management, real-time operating systems and fault tolerance applications of real-time systems.
CO5	Apply the basics of RTOS in interpretation of real time systems.

Course Contents:

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Module-1: Introduction [8L]

Definition, Typical Real Time Applications: Digital control, High Level Controls, Signal processing etc. , Release Times, Deadline period and time constraints, Hard and soft real time systems, Reference models for RTOS: Processors and Resources, Temporal parameters of Real-time workload, Periodic Task Model, Precedence Constraints and Data Dependency.

Module-2: Real Time Scheduling. [8L]

Common Approaches to Real Time Scheduling: Clock Driven Approach, Weighted Round Robin Approach, Priority Driven Approach, Dynamic Versus Static Systems, Optimality of Effective-Deadline-First (EDF) and Least-Stack-Time-First (LST) algorithms, Rate Monotonic algorithm, Offline versus Online Scheduling.

Module-3: Resources Sharing. [8L]

Effect of Resource Contention and Resource Access Control (RAC), Non-pre-emptive Critical Sections, Basic Priority- Inheritance and Priority-Ceiling Protocols, Stack based Priority Ceiling Protocol, Use of Priority Ceiling Protocol in Dynamic priority systems, Pre-emption Ceiling Protocol, Access control in Multiple Module Resources, Controlling Concurrent Accesses to Data Objects.

Module-4: Real Time Communication. [6L]

Basic Concepts of Real time Communication, Soft and Hard real-time Communication systems, Model of Real-time Communication, Priority based service and Weighted Round Robin Service disciplines for switched Networks, Medium Access control protocols for broadcast networks, Internet and resource reservation protocols

Module-5: Real Time Operating Systems and Databases. [6L]

Features of RTOS, Time Services, UNIX as RTOS, POSIX Issues, Characteristic of temporal data, temporal consistency, on-currency Control, and Overview of Commercial Real Time databases.

Text Books

1. Real Time Systems – Jane W. S. Liu, Pearson Education Publication

Reference Books

1. Real Time Systems – Mall Rajiv, Pearson Education
2. Real-Time Systems: Scheduling, Analysis, and Verification – Albert M. K. Cheng, Wiley.

CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
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Course Name: Data Analytics

Course Code: PEC-CS-D-801

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: Data Structure, Design and Analysis of Algorithms, Database Management Systems, Statistics, Artificial Intelligence, Programming skills of Python

Course Objectives:

2. Comprehend the fundamental concepts of the Big Data Analytics exploring machine learning strategies such as Supervised and Unsupervised Learning etc. for analyzing various types of large scale structured as well as unstructured data distributed across multiple locations (Map Reduce, Hadoop and NoSQL Framework).
3. Formulate an engineering problem of analyzing large scale data distributed across multiple locations to make automated meaningful decisions
4. Apply the concepts of Big Data Analytics to solve problems of making automated decisions dealing with large scale structured as well as unstructured data distributed across multiple locations.
5. Excogitate and Implement ideas to address the challenging issues of Big Data Analytics.
6. Analyze the effectiveness of various Big Data Analytics Frameworks.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand and explain the fundamental concepts of the Big Data Analytics which are primarily explored for making automated decisions using machine learning strategies on analyzing large scale structured as well as unstructured data distributed across multiple locations (Map Reduce, Hadoop and NoSQL Framework) underscoring the utilitarian importance in current technological context for further exploration leading towards lifelong learning.
CO2	Identify and formulate an engineering problem of analyzing large scale data distributed across multiple locations to make automated meaningful decisions within the scope of Big Data Analytics Frameworks.
CO3	Explore relevant literature and apply the concepts of Big Data Analytics to solve problems of making automated decisions dealing with large scale structured as well as unstructured data using Map Reduce, Hadoop and advanced SQL Frameworks.
CO4	Excogitate ideas for proposing solutions to the challenging problems of Big Data Analytics.

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CO5	Implement ideas of Big Data Analytics through developing feasible algorithms or frameworks and investigate their effectiveness in solving the relevant problems by analyzing the performances using proper techniques.
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Course Content:

Module – 1: Introduction to Basic Analytics [10L]

Introduction: Big data overview, Analyst's perspective on data repositories, Current analytical architecture, Drivers of big data, Examples of big data analytics.

Life Cycle of Data Analytics: Phase 1: Discovery, Phase 2: Data preparation, Phase 3: Model planning, Phase 4: Model building, Phase 5: Communication of results, Phase 6: Making operational.

Basic Analytic Methods: Visualization, Dirty data, Data exploration versus presentation, Statistical methods for evaluation – hypothesis testing, difference of means, rank sum test, type I and type II errors, ANOVA.

Module - 2: Advanced Analytic Methods I [8L]

Clustering: Overview, K-means, Determining the number of clusters, Diagnostics.

Association Rules: Overview, Apriori algorithm, Evaluation of candidate rules, Application of association rules, Validation and testing, Diagnostics.

Regression: Linear regression - model description, Logistic regression – model description, Other regression models.

Classification: Decision trees – overview, General algorithm, Decision tree algorithms, Evaluating a decision tree, Naïve Bayes – Bayes theorem, Naïve Bayes classifier, Diagnostics of classifiers.

Module – 3: Advanced Analytic Methods II [8L]

Time Series Analysis: Overview, Box-Jenkins methodology, Autocorrelation function (ACF), Autoregressive model, Moving average model, ARMA and ARIMA model, Building and evaluating an ARIMA model.

Text Analysis: Steps in text analysis, Collecting raw text, Representing text, Term Frequency-Inverse Document Frequency (TFIDF), Categorizing documents by types, Determining sentiments.

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Map Reduce and Hadoop: Analytics for unstructured data – map reduce, Apache Hadoop, Hadoop Ecosystem – Pig, Hive, Hbase, Mahout.

Module – 4: Advanced Analytic Methods III [10L]

Technology and Tools: SQL essentials - Join, Set, Grouping extensions, Advanced SQL – Window functions, User-defined functions, Ordered aggregates, MADlib, NoSQL.

Integration of Techniques: Communicating and operationalizing an analytic project.

Creating final deliverables – Developing core materials, project goals, Main findings, Approach, Model description and model details, Recommendations, Providing technical specifications and code.

Data visualization basics - Key points, evolution of a graph, common representation methods, how to clean up a graphic.

Textbook:

1. EMC Education Services (Editor), Data Science and Big Data Analytics. John Wiley & Sons, 2015.
2. Mike Barlow, Real-Time Big Data Analytics: Emerging Architecture. O'Reilly, 2013.

ReferenceBooks:

3. Nathan Marz and James Warren, Big Data: Principles and Best Practices for Scalable Real-time Data Systems. Manning Publications, 2015.
4. Venkat Ankam, Big Data Analytics. Packt Publishing Ltd., UK, 2016.

CO–PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	3	3	-	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	-	-	-	-	-	-	-	-	2	-	2
CO4	2	2	2	3	-	-	-	-	-	-	-	2	2	-	2
CO5	2	2	3	3	2	2	2	-	-	-	-	2	2	2	3

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Course Name: Image Processing

Course Code: PEC-CS-A-801

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites: Design and Analysis of Algorithms, UG Level Mathematics

Course Objectives:

16. Understand the basic concepts of digital image processing and identify problems where students can apply the concept appropriately.
17. Understand the fundamental concepts of image enhancement strategies and identify the scope of enhancement where students can apply the appropriate strategy and analyze the performance.
18. Illustrate the fundamental image restoration strategies and apply them appropriately to eliminate noise in the image.
19. Illustrate various Image Compression Techniques and Analyze their performances.
20. Understand the ideas of Morphological Image Processing and Image Segmentation to propose solutions to the related problems and analyze the effectiveness as well as limitations of solutions underscoring its utilitarian importance for further explorations leading towards lifelong learning.

Course Outcomes:

After completion of the course students will be able to

CO1	Understand the basic concepts of digital image processing to Explain or Illustrate and Identify problems where students can Apply the concept appropriately to Solve them.
CO2	Understand the fundamental concepts of image enhancement strategies and Identify the scope of enhancement where students can Apply the appropriate strategy and Analyze the performance.
CO3	Illustrate the fundamental image restoration strategies and Apply them appropriately to eliminate noise in the image.
CO4	Illustrate various Image Compression Techniques and Apply them to compress the images and Analyze their performances.

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CO5	Understand and Develop ideas to Propose solutions to the problems of Morphological Image Processing and Image Segmentation and Analyze the effectiveness as well as limitations of solutions underscoring its utilitarian importance for further explorations leading towards lifelong learning.
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Course Content:

MODULE-1: Introduction to Digital Image Processing [3L]

Applications of digital image processing, fundamental steps in digital image processing, components of image processing system. Digital Image Fundamentals: A simple image formation model, image sampling and quantization, Some Basic Relationships Between Pixels-Neighbors and Connectivity of pixels in image, Color Image Models.

MODULE-2: Image Enhancement [10L]

Image Enhancement in The Spatial Domain: Some Basic Gray Level Transformations, Histogram Processing, Basics of Spatial Filtering, Smoothing Spatial Filters, Sharpening Spatial Filters.

Image Enhancement in Frequency Domain:

Introduction, Fourier Transform, Discrete Fourier Transform (DFT) and its relation with image characterization, fundamental steps of image enhancement in Frequency Domain, Smoothing and Sharpening frequency domain filters – Ideal, Butterworth and Gaussian filters.

MODULE-3: Image Restoration [5L]

Basics of Image restoration and Noise characterization, Estimating the degradation function, Noise removal using spatial and frequency domain filtering, Image Restoration techniques.

MODULE-4: Morphological Image Processing [5L]

Basic Concepts, Erosion, Dilation, Opening, Closing, Skeletonization, Hole filling, Connected components, Boundary Detection.

MODULE-5: Image Compression [5L]

Basic Concepts – Types of redundancy, Types of coding techniques, Lossless Compression: Run-Length Encoding, Huffman Coding, Lossy Compression: Vector Quantization, Sequential DCT-based Compression (JPEG Baseline Algorithm).

MODULE-6: Image Segmentation [8L]

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Detection of Points, lines and Edges (Sobel and Canny); Edge Linking, Image Thresholding (Otsu's method), Region based segmentation, color-feature based segmentation in color images.

Textbook:

18. Digital Image Processing, Rafael C. Gonzales, Richard E. Woods, Third Edition, Pearson Education, 2010.
19. Digital Image Processing, S. Sridhar, Oxford University Press, 2nd Ed, 2016.

Reference Books:

19. Fundamentals of Digital Image Processing- Anil K. Jain, 2nd Edition, Prentice Hall of India.
20. Image Processing, analysis and Machine Vision, Milan Sonka, Thomson Press India Ltd, Fourth Edition.

CO–PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	-	3	-	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	-	-	-	-	-	-	-	-	2	-	2
CO4	2	2	2	3	-	-	-	-	-	-	-	-	2	-	2
CO5	2	2	3	3	-	-	-	-	-	-	-	2	2	2	3

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Paper Name: Human Resource Development and Organizational Behaviour

Paper Code: OEC-CS-801A

Credits: 3

No. of lectures: 36

Course Objective(s):

- To develop an understanding of the nature, functioning and design of organisation as social collectivises.
- The basic concepts and theories underlying individual behaviour besides developing better insights into one's own self.
- To gain insight into the organisational learning processes, how they can be fostered and enhanced.
- Individual behaviour in groups, dynamics of groups and team building besides developing a better awareness of how they can be better facilitators for building effective teams as leaders themselves.

Course Outcome(s):

At the end of the course students are able to:

After completion of the course students will be able to

CO1	To understand key functions in management as applied in practice.
CO2	To identify and analyse major practices associated with HRD in modern work and organisations;
CO3	To evaluate the connections between the HRD process and the contemporary performance management concerns of organisations
CO4	To assess the potential effects of organisational-level factors (such as structure, culture and change) on organisational behaviour.

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CO5	To evaluate the potential effects of important developments in the external environment (such as globalisation and advances in technology) on organisational behaviour
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Course Contents

Module – 1 HRD-Macro Perspective: HRD Concept, Origin and Need, HRD as a Total System; Approaches to HRD; Human Development and HRD; HRD at Macro and Micro Climate. **3L**

Module -2 HRD–Micro Perspective: Areas of HRD; HRD Interventions Performance Appraisal, Potential Appraisal, Feedback and Performance Coaching, Training, Career Planning, OD or Systems Development, Rewards, Employee Welfare and Quality of Work Life and Human Resource Information; Staffing for HRD: Roles of HR Developer; Physical and Financial Resources for HRD; HR Accounting; HRD Audit, Strategic HRD **6L**

Module – 3 Instructional Technology for HRD: Learning and HRD; Models and Curriculum; Principles of Learning; Group and Individual Learning; Transactional Analysis; Assessment Centre; Behaviour Modeling and Self Directed Learning; Evaluating the HRD.**5L**

Module – 4 Human Resource Training and Development: Concept and Importance; Assessing Training Needs; Designing and Evaluating T&D Programmes; Role, Responsibilities and challenges to Training Managers.**4L**

Module – 5 Organisational Effectiveness (OE): Concept; Approaches to O E; Adoptive Coping Cycle for Effectiveness; Achieving OE; Organisational Climate: Concept, Determinants of Organisational Climate.**3L**

Module-6 Organization Theory: Classical Theory; Neo-Classical Theory, Modern Behavioural Theories, contingency theory, system theory, modern structural models; Organizational Culture; Creating and Sustaining Culture; Work Culture.**6L**

Module –7 Motivation: Types of Motives; Theories of Maslow; Herzberg, McGregor, Alderfers, Porter and Lawler’s Model; Job Enlargement, Job Enrichment, Behaviour Modification. **3L**

Module– 8(a) Group & Group Dynamics - concept, importance, classification of groups, reason for group, formation, group cohesiveness. (b) Team work: meaning, concept, types, creating, and an effective team. (c) Leadership: Concept, Leader vs. Manager; Classical Studies on Leadership; Trait Theories; Behavioral Theories; Group and Exchange Theories; Contingency Theory of Leadership; Leadership Styles.**6L**

- 1) Rao, T.V and Pareek, Udai: Designing and Managing Human Resource Systems, Oxford IBH Pub. Pvt.Ltd., New Delhi , 2005
- 2) Viramani, B.R and Seth, Parmila: Evaluating Management Development, Vision Books, New Delhi.
- 3) Rao, T.V: Human Resource Development, Sage Publications, New Delhi.
- 4) Luthans, Fred: Organisational Behaviour, Tata McGraw-Hill Co. New Delhi, 2004.
- 5) Stephen, P. Robins: Organisational Behaviour, Prentice-Hall of India Pvt., Ltd., 2004.
- 6) John, W. Mewstrom & Davis, Keith :Organisational Behavior (Human Behavior at Work), Tata McGraw-Hill, New Delhi , 2002

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Course Name: Block Chain
Course Code: OEC-CS-801B
Contact: 3:0:0
Total Contact Hours: 36
Credits: 3

Course Objective:

- The students should be able to understand a broad overview of the essential concepts of blockchain technology.
- To familiarize students with Bitcoin protocol followed by the Ethereum protocol – to lay the foundation necessary for developing applications and programming.
- Students should be able to learn about different types of blockchain and consensus algorithms.

Course Outcome(s):

After completion of the course students will be able to

CO1	Understand blockchain terminologies and its properties and the emerging models for blockchain technology
CO2	Familiarize with the functional/operational aspects of crypto currency ecosystem
CO3	Design, code, deploy and execute a smart contract – the computational element of the blockchain technology using Solidity and Remix IDE
CO4	Build private - permissioned blockchain -based applications for enterprises and businesses
CO5	Explore the blockchain decentralization and cryptography concepts.

Course Contents:

Module 1 [6 Lectures]

Basics: The Double-Spend Problem, Byzantine Generals' Computing Problems, Public-Key Cryptography, Hashing, Distributed Systems, Distributed Consensus.

Module 2 [10 Lectures]

Technology Stack: Blockchain, Protocol, Currency.

Bitcoin Blockchain: Structure, Operations, Features, Consensus Model, Incentive Model.

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Module 3 [10 Lectures]

Ethereum Blockchain: Smart Contracts, Ethereum Structure, Operations, Consensus Model, Incentive Model.

Module 4 [6 Lectures]

Tiers of Blockchain Technology: Blockchain 1.0, Blockchain 2.0, Blockchain 3.0, Types of Blockchain: Public Blockchain, Private Blockchain, Semi-Private Blockchain, Sidechains.

Module 5 [10 Lectures]

Types of Consensus Algorithms: Proof of Stake, Proof of Work, Delegated Proof of Stake, Proof Elapsed Time, Deposit-Based Consensus, Proof of Importance, Federated Consensus or Federated Byzantine Consensus, Practical Byzantine Fault Tolerance. Blockchain Use Case: Supply Chain Management.

Text Books/References:

1. Kirankalyan Kulkarni, Essentials of Bitcoin and Blockchain, Packt Publishing.
2. Anshul Kaushik, Block Chain & Crypto Currencies, Khanna Publishing House.
3. Tiana Laurence, Blockchain for Dummies, 2nd Edition 2019, John Wiley & Sons.
4. Mastering Blockchain: Deeper insights into decentralization, cryptography, Bitcoin, and popular Blockchain frameworks by Imran Bashir, Packt Publishing (2017).
5. Blockchain: Blueprint for a New Economy by Melanie Swan, Shroff Publisher O'Reilly Publisher Media; 1st edition (2015).
6. Mastering Bitcoin: Programming the Open Blockchain by Andreas Antonopoulos.

Corresponding Online Resources:

1. <https://www.coursera.org/specializations/blockchain>.
2. <https://nptel.ac.in/courses/106105184/>
3. Introduction to Blockchain Technology and Applications, https://swayam.gov.in/nd1_noc20_cs01/preview

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	2	-	-	-	-	-	-	-	-	3	3	2
CO2	2	3	-	-	-	-	-	-	-	-	-	-	-	3	2
CO3	2	2	3	2	-	-	-	-	-	-	-	-	2	3	2
CO4	2	2	2	3	-	-	-	-	-	-	-	-	2	3	2
CO5	2	2	3	3	-	-	-	-	-	-	-	2	2	2	3

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Course Name: Simulation and Modelling

Course Code: OEC-CS-801C

Contact: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisites:

1. Programming and Data Structures
2. Discrete Mathematics and Probability theory
3. Game theory
4. Numerical Analysis

Course Objective(s):

- To understand the Models and Simulation of Continuous and Discrete Systems.
- To enable students to analyze Continuous Uniformly Distributed Random Numbers
- To assess the strengths and weaknesses of various methods and to analyze their behavior.

Course Outcome(s):

On completion of the course students will be able to

CO1	Student will be able to summarize the issues in Modeling and Simulation and to explain the System Dynamics & Probability concepts in Simulation.
CO2	Student will be able to solve the Simulation of Queuing Systems
CO3	Student will be able to analyze the Simulation output.
CO4	Student will be able to identify the application area of Modeling and Simulation, and apply them.
CO5	Student will be able to implement new model by applying their knowledge

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Course Contents:

Module-1: Introduction to Modeling and Simulation [7L]

Nature of Simulation. Systems , Models and Simulation, Continuous and Discrete Systems, system modeling, Components of a simulation study, Introduction to Static and Dynamic System simulation , Application areas, Advantages ,Disadvantages and pitfalls of Simulation.

Module –2: System Dynamics & Probability concepts in Simulation [10L]

Exponential growth and decay models, Generalization of growth models, Discrete and Continuous probability functions, Continuous Uniformly Distributed Random Numbers, Generation of a Random numbers, Generating Discrete distributions, Non-Uniform Continuously Distributed Random Numbers, Rejection Method.

Module-3: Simulation of Queuing Systems and Discrete System Simulation [14L]

Poisson arrival patterns, Exponential distribution, Service times, Normal Distribution Queuing Disciplines, Simulation of single and two server queue. Application of queuing theory in computer system. Discrete Events ,Generation of arrival patterns ,Simulation programming tasks

Gathering statistics, Measuring occupancy and Utilization , Recording Distributions and Transit times .

Module-4 : Analysis of Simulation output [5L]

Sensitivity Analysis, Validation of Model Results

Text Books:

1. Jerry Banks, John Carson, B.L.Nelson and D.M.Nicol — Discrete Event System Simulation, Fifth Edition, Pearson.
2. NarsinghDeo, 1979, System Simulation with Digital Computers, PHI.

Reference Books:

1. Averill M. Law and W.DavidKelton, —Simulation Modeling and Analysis, Third Edition, McGraw Hill 5. J. N. Kapoor.. Mathematical Modeling, Wiley eastern Limited
2. Geoffrey Gordon, —System Simulation, PHI.

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
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Curriculum for B.Tech Under Autonomy
B.Tech. – Computer Science & Technology
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Course Name: Values and Ethics in Profession

Course Code: OEC-CS-802A

Contact: 3L:0T: 0P

Credit: 3

No. of Lectures: 34

Prerequisites: Basic knowledge of engineering and management.

Course Outcome(s):

On Completion of this course student will be able to

CO1	Understand the core values that shape the ethical behaviour of an engineer and Exposed awareness on professional ethics and human values.
CO2	Understand the concept of profession, professional ethics, and various moral issues.
CO3	Understand various social issues, industrial standards, code of ethics and role of professional ethics in engineering field.
CO4	Aware of responsibilities of an engineer for safety and risk benefit analysis, professional rights and responsibilities of an engineer
CO5	Acquire knowledge about various roles of engineers in variety of global issues and able to apply ethical principles to resolve situations that arise in their professional lives.

Course Content:

Module: 1: Introduction

6L

Basic Concepts, Governing Ethics, Personal & Professional Ethics, Ethical Dilemmas, Life Skills, Emotional Intelligence, Thoughts of Ethics, Value Education, Dimensions of Ethics, Profession and professionalism, Professional Associations, Professional Risks, Professional Accountabilities, Professional Success, Ethics and Profession.

Module 2: Theories of Self Development

6L

Emotional Intelligence (EI): Concept, Johari Window, Importance and Measurement, Concept of Motivation, Maslow's theory, Kohlberg's theory, Gilligan's theory – towards a comprehensive approach to moral behaviour – truth – approach to knowledge in technology.

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Module 3: Moral and Ethical Concerns**4L**

Variety of Moral Issues, Moral Dilemmas, Nature of values, Value Crisis in contemporary society, Value Spectrum of a good life, Steven Covey's Pursuit of Excellence.

Module 4: Professional Practices in Engineering**8L**

Professions and Norms of Professional Conduct, Norms of Professional Conduct vs. Profession; Responsibilities, Obligations and Moral Values in Professional Ethics, Professional codes of ethics, the limits of predictability and responsibilities of the engineering profession, Ethical and Unethical practices – case studies, Whistle blowing and beyond and Case studies.

Module 5: Global issues in Professional Ethics**10L**

Introduction – Current Scenario, Technology Globalization of MNCs, International Trade, World Summits, Issues, Business Ethics and Corporate Governance, Sustainable Development Ecosystem, Energy Concerns, Ozone Deflection, Pollution, Ethics in Manufacturing and Marketing, Media Ethics; Bio Ethics, Intellectual Property Rights.

Reference Books

1. Govindarajan M, Natarajan S, Senthil Kumar V S, "Engineering Ethics" PHI India, 2004
2. P Aarne Vesblind, Alastair S Gunn, "Engineering Ethics and the Environment"
3. Edmund G Seebauer, Robert L Barry, "Fundamentals of Ethics for scientists and engineers" Oxford University Press 2001
4. Mike W Martin, Roland Schinzinger, "Ethics in Engineering", Tata McGraw -Hill, 2003
5. Professional Ethics: R. Subramanian, Oxford University Press
6. Engineering Ethics, Concepts Cases: Charles E Harris Jr., Michael S Pritchard, Michael J Rabins, 4e, Cengage learning, 2015
7. Business Ethics concepts & Cases: Manuel G Velasquez, 6e, PHI, 2008.

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	-	-	-	-	-	-	-	2	3	2
CO2	3	3	3	3	3	-	-	-	-	-	-	-	2	3	2
CO3	3	3	3	3	3	-	-	-	-	-	-	3	2	3	2
CO4	3	3	3	3	3	-	-	-	-	-	-	-	2	3	2
CO5	3	3	3	3	3	-	-	-	3	-	2	-	2	3	2
CO	3	3	3	3	3	-	-	-	3	-	2	3	2	3	2

Curriculum for B.Tech Under Autonomy
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Course Name: History of Science and Technology in India

Course Code: OEC-CS-802B

Contact: 3:0:0

Total Contact Hours: 36

Credit: 3

Prerequisites: None

Course Outcome(s):

After completion of the course students will be able to

CO1	To understand the development of science and technology in ancient India.
CO2	To understand that development is not solo pursuit rather an interactive and collective process.
CO3	To familiarize with the evolution of scientific ideas and technical solution and their linkages with the socio-culture necessities
CO4	To analyse the socio-cultural and philosophical context in which the various scientific
CO5	To apply technological ideas got developed in India. and thereby help in repositioning India's contributions in science and technology

Course Content:

Module 1: Historical Perspective [5L]

The nature of science and technology, Roots of science and technology in India, Science and society, Scientists and society, Science and Faith and the rise of applied sciences.

Module 2: Policies and Plans after Independence [7L]

Nehru's vision of science for independent India, Science and technology developments in the new era science and technology developments during the Five Year Plan Periods and science and technology policy resolutions.

Module 3: Research and Development (R&D) in India [7L]

Expenditure in R&D, Science and Technology Education, Research activities and promotion of technology development, Technology mission, Programs aimed at technological self-reliance, activities of council of scientific and industrial research (CSIR).

Module 4: Science and Technological Developments in Major Areas [11L]

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Space – Objectives of space programs, Geostationary Satellite Services – INSAT system and INSAT services remote sensing applications, Launch Vehicle Technology
 Ocean Development – Objectives of ocean development, Biological and mineral resources, Marine research and capacity building
 Defence Research – Spin-off technologies for civilian use,
 Biotechnology – Applications of biotechnology in medicine, Biocatalysts, Agriculture, Food, Fuel and Fodder, Development of biosensors and animal husbandry
 Energy – Research and development in conservation of energy, India's nuclear energy program, technology spin-offs.

Module 5:Nexus between Technology Transfer and Development [6L]

Transfer of Technology – Types, Methods, Mechanisms, Process, Channels and Techniques, Appropriate technology, Technology assessment, Technological forecasting, Technological innovations and barriers of technological change.

Textbooks:

1. Kalpana Rajaram, Science and Technology in India, Published and Distributed by Spectrum Books (P) Ltd., New Delhi – 58.
2. Srinivasan, M., Management of Science and Technology (Problems & Prospects), East-West Press (P) Ltd., New Delhi.

Reference Books:

1. Ramasamy, K.A., and Seshagiri Rao, K., (Eds), Science, Technology and education for Development, K., Nayudamma Memorial Science Foundation, Chennai – 8.
2. Kohili, G.R., The Role and Impact of Science and Technology in the Development of India, Surjeet Publications.
3. Government of India, Five Year Plans, Planning Commission, New Delhi.
4. Sharma K.D., and Quresh M.A., Science, Technology and Development, Sterling Publications (P) Ltd., New Delhi.

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	-	-	-	-	-	-	-	2	3	2
CO2	3	3	3	3	3	-	-	-	-	-	-	-	2	3	2
CO3	3	3	3	3	3	-	-	-	-	-	-	3	2	3	2
CO4	3	3	3	3	3	-	-	-	-	-	-	-	2	3	2
CO5	3	3	3	3	3	-	-	-	3	-	2	-	2	3	2

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Course Name: Economic Policies in India
Course Code: OEC-CS-802C
Contact: 3L:0T:0P
Total Contact Hours: 36
Credit: 3

Course Objective(s):

This course reviews major trends in Indian Economic Policies towards the attainment of economic and social development in the post-independence period.

Course Outcome(s):

CO1	To understand the basic characteristics of Indian economy, it's potential
CO2	To understand the importance, causes and impact of population growth and its distribution, translate and relate them with economic development
CO3	To understand the importance of planning undertaken by the government of India, have knowledge on the various objectives, failures and achievements as the foundation of the ongoing planning and economic reforms taken by the government
CO4	To analyse the progress and changing nature of different sectors and their contribution to the economy as a whole.
CO5	To analyse the developmental trends of Indian economy.

Course Content

Module 1: Introduction:[4L]

Indian Economy on the eve of Independence, British rule and its impact on Indian Economy, Emergence and development of Planning exercise in India – historical debates, plan models and shift in focus over time

Module 2: Issues in Growth, Development and Sustainability[4L]

Output (National Income) and Employment Structure of Indian Economy; Composition and relative rates of growth of agriculture, industry and services sectors; Sub-sectoral analysis.

Module 3: Factors in development: Capital formation (physical and human)[12L]

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Trends and patterns in structure of population over time – growth rate, gender, rural-urban, literacy, regional; Structure and trends of Poverty and Inequality (interpersonal and regional); Inflation – trends, structure and causes; Unemployment – trends, structure and types.

Module 4: Trends in Agricultural Production and Productivity [6L]

Firm size and productivity, Land Reforms – Genesis, Progress and current status, new agricultural strategies.

Module 5: Trends and Patterns of Industrial Sector[6L]

Changes in the structure of Indian Industry; Small Scale Industries – Growth, Structure and its contribution in national economy; Public Sector – Growth, Structure, Historical role, Evolution and Dilution.

Module-6: Trends in Exports and Imports[4L]

Composition and Direction of Foreign Trade; Balance of Payments – Current Status

References:

- 1) R Dutta and K P M Sundaram: Indian Economy, S Chand
- 2) A.N.Agarwal: Indian Economy, Problems of Development and Planning, New Age.
- 3) Mishra and Puri: Indian Economy, Himalaya.
- 4) Planning Commission: Eleventh Five Year Plan, Vol I, II and III, Academic Foundation.
- 5) Government of India: Economic Survey (latest issue)

CO-PO Mapping:

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CO2	3	3	-	3	3	-	-	-	-	-	-	-	2	3	2
CO3	3	3	3	3	3	-	-	-	-	-	-	3	2	3	2
CO4	3	-	3	3	2	-	-	-	-	-	-	-	2	3	2
CO5	3	3	3	3	3	-	-	-	3	-	2	-	2	3	2
CO	2	3	3	3	3	-	-	-	3	-	2	3	2	3	2

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MANDATORY COURSE

Course Name: Essence of Indian Knowledge Tradition

Course Code: MC 801

Contact: 3L:0T:0P

Total Contact Hours: 36

Prerequisite: None

Course Outcome(s) :

At the end of the course, student will be able to:

CO1	Identify the concept of Traditional knowledge and its importance
CO2	Explain the need and importance of protecting traditional knowledge.
CO3	Interpret the concepts of Intellectual property to protect the traditional knowledge.
CO4	Illustrate the various enactments related to the protection of traditional knowledge.
CO5	Explain the importance of Traditional knowledge in Agriculture and Medicine.

Course Content:

Module 1:[7L]

Introduction to traditional knowledge: Define traditional knowledge, nature and characteristics, scope and importance, kinds of traditional knowledge, Indigenous Knowledge (IK), characteristics, traditional knowledge vis-a-vis indigenous knowledge, traditional knowledge Vs western knowledge traditional knowledge.

Module 2:[8L]

Protection of traditional knowledge: The need for protecting traditional knowledge Significance of traditional knowledge Protection, value of TK in global economy, Role of Government to harness traditional knowledge.

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Module 3:[6L]

Legal framework and TK: The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006, Plant Varieties Protection and Farmer's Rights Act, 2001 (PPVFR Act); The Biological Diversity Act 2002 and Rules 2004, the protection of traditional knowledge bill, 2016.

Module 4:[7L]

Traditional knowledge and intellectual property: Systems of traditional knowledge protection, Legal concepts for the protection of traditional knowledge, Patents and traditional knowledge, Strategies to increase protection of traditional knowledge.

Module 5:[8L]

Traditional Knowledge in Different Sectors: Traditional knowledge and engineering, Traditional medicine system, TK in agriculture, Traditional societies depend on it for their food and healthcare needs, Importance of conservation and sustainable development of environment, Management of biodiversity, Food security of the country and protection of traditional knowledge.

Text Books:

1. Traditional Knowledge System in India, by Amit Jha, 2009.

Reference Books:

1. Traditional Knowledge System in India by Amit Jha Atlantic publishers, 2002.
2. "Knowledge Traditions and Practices of India" Kapil Kapoor1, Michel Danino2.

Web Links:

- 1.<https://www.youtube.com/watch?v=LZP1StpYEPM>
- 2.<http://nptel.ac.in/courses/121106003/>

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CO3	3	3	3	3	3	-	-	-	-	-	2	3	2	3	2
CO4	3	2	3	3	-	-	-	-	-	-	-	-	2	3	2
CO5	3	3	3	3	3	-	-	-	3	-	2	-	2	3	2
CO	3	3	3	3	3	-	-	-	3	-	2	3	2	3	2

Curriculum for B.Tech Under Autonomy

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